S-122

IHO Marine Protected Areas (MPA) - Annex A Data Classification and Encoding Guide

Edition 2.0.0 – March 2026

IHO

International Hydrographic Organization

Published by the
International Hydrographic Organization
4b quai Antoine 1er
Principauté de Monaco
Tel: (377) 93.10.81.00
Fax: (377) 93.10.81.40
info@iho.int
\www.iho.int

© Copyright International Hydrographic Organization 2026

This work is copyright. Apart from any use permitted in accordance with the Berne Convention for the Protection of Literary and Artistic Works (1886), and except in the circumstances described below, no part may be translated, reproduced by any process, adapted, communicated or commercially exploited without prior written permission from the International Hydrographic Organization (IHO). Copyright in some of the material in this publication may be owned by another party and permission for the translation and/or reproduction of that material must be obtained from the owner.

This document or partial material from this document may be translated, reproduced or distributed for general information, on no more than a cost recovery basis. Copies may not be sold or distributed for profit or gain without prior written agreement of the IHO Secretariat and any other copyright holders.

In the event that this document or partial material from this document is reproduced, translated or distributed under the terms described above, the following statements are to be included:

> "Material from IHO publication [reference to extract: Title, Edition] is reproduced with the permission of the IHO Secretariat (Permission No ……./…) acting for the International Hydrographic Organization (IHO), which does not accept responsibility for the correctness of the material as reproduced: in case of doubt, the IHO's authentic text shall prevail. The incorporation of material sourced from IHO shall not be construed as constituting an endorsement by IHO of this product."

> "This [document/publication] is a translation of IHO [document/publication] [name]. The IHO has not checked this translation and therefore takes no responsibility for its accuracy. In case of doubt the source version of [name] in [language] should be consulted."

The IHO Logo or other identifiers shall not be used in any derived product without prior written permission from the IHO Secretariat.

Contents

1 Overview....1
1.1 Introduction....1
1.2 Document Metadata....1
1.3 Terms and definitions....1
1.4 Abbreviated terms....1
1.5 Use of language....1
1.6 Maintenance....2
2 General....2
2.1 Introduction....2
2.2 Descriptive characteristics....2
2.3 Spatial characteristics....3
2.4 Attributes....3
2.5 Associations....14
2.6 Datasets....17
2.7 Geographic names....19
2.8 Scale policy....20
2.9 Masking....22
2.10 Linear surface features....23
3 Description of table format for feature and information types....25
4 Meta-Features....27
4.1 Introduction....27
4.2 Mandatory meta features....27
4.3 Data Coverage....27
4.4 Quality of Non-Bathymetric Data....30
5 Abstract Geo Features....33
5.1 Introduction....33
5.2 Feature Type....33
6 Domain Features....37
6.1 General....37
6.2 Information Area....37
6.3 Marine Protected Area....39
6.4 Restricted Area....44
6.5 Vessel Traffic Service Area....47
7 Cartographic Features....49
7.1 Introduction....49
7.2 Text Placement....49
8 Abstract Information Types....51
8.1 Information Type....51
8.2 AbstractRxN....53
9 Textual Regulations....57
9.1 Introduction....57
9.2 Regulations, etc., for specific locations....59
9.3 Regulations applying only to vessels with specific characteristics or cargoes....59
9.4 Regulations....59
9.5 Restrictions....60
9.6 Recommendations....62
9.7 Nautical Information....63
10 Services, Organisations and Schedules....65
10.1 Work schedules and holidays....65
10.2 Contact information....65
10.3 Authority....66
10.4 Contact Details....67
10.5 Service Hours....70
10.6 Non-Standard Working Day....71
11 Limitations....73
11.1 Introduction....73
11.2 Defining subsets of vessels by dimensions, cargo, and other characteristics....73
11.3 Characterizing the relationship between the vessel set and the feature or regulation....75

11.4 Production hints and recommended practices (informative) 76
11.5 Applicability 77
12 Spatial Quality 81
12.1 Introduction 81
12.2 Spatial Quality 81
13 Feature Associations 87
13.1 Text Association 87
14 Information Associations 89
14.1 Additional Information 89
14.2 Authority Contact 89
14.3 Authority Hours 89
14.4 Associated RxN 89
14.5 Exceptional Workday 89
14.6 Protected Area Authority 89
14.7 Related Organisation 90
14.8 Inclusion Type 90
14.9 Permission Type 90
14.10 Service Control 90
14.11 Spatial Association 91
15 Association Roles 91
15.1 Authority 91
15.2 Authority Service Hours 91
15.3 Contact Details 91
15.4 Control Authority 91
15.5 Is Applicable To 91
15.6 Organisation-Related RxN 92
15.7 Permission 92
15.8 Partial Working Day 92
15.9 Responsible Authority 92
15.10 Service Hours (reference) 92
15.11 The Information 92
15.12 The Organisation 92
15.13 The Quality Information 93
15.14 The RxN 93
15.15 The Applicable RxN 93
15.16 The Cartographic Text 93
15.17 The Position Provider 93
15.18 The service hours for a non-standard workday 93
16 Simple Attributes 95
16.1 Administrative Division 95
16.2 Application Profile 95
16.3 Call Name 95
16.4 Call Sign 95
16.5 Cardinal Direction 95
16.6 Category of Authority 96
16.7 Category of Cargo 97
16.8 Category of Communication Preference 98
16.9 Category Of Dangerous Or Hazardous Cargo 98
16.10 Category of Relationship 99
16.11 Category of Restricted Area 100
16.12 Category of Schedule 101
16.13 Category of Temporal Variation 101
16.14 Category of Text 102
16.15 Category of Vessel Registry 102
16.16 City Name 102
16.17 Communication Channel 102
16.18 Comparison Operator 103
16.19 Contact Instructions 103
16.20 Country Name 103
16.21 Date End 103

16.22 Date Fixed....104
16.23 Date Start....104
16.24 Date Variable....104
16.25 Day of Week....104
16.26 Day of Week is Range....105
16.27 Delivery Point....105
16.28 Designation Identifier....105
16.29 Designation Scheme....105
16.30 Destination....106
16.31 Distance....106
16.32 File Locator....106
16.33 File Reference....106
16.34 Frequency Shore Station Receives....106
16.35 Frequency Shore Station Transmits....107
16.36 Headline....107
16.37 Horizontal Distance Uncertainty....107
16.38 In Ballast....107
16.39 Interoperability Identifier....108
16.40 Jurisdiction....108
16.41 Language....108
16.42 Linkage....108
16.43 Logical Connectives....109
16.44 Maximum Display Scale....109
16.45 Membership....109
16.46 Minimum Display Scale....110
16.47 MMSI Code....110
16.48 Name....110
16.49 Name of Resource....110
16.50 Name Usage....110
16.51 Online Function....111
16.52 Online Resource Description....111
16.53 Optimum Display Scale....111
16.54 Orientation Uncertainty....112
16.55 Orientation Value....112
16.56 Pictorial Representation....112
16.57 Picture Caption....112
16.58 Picture Information....113
16.59 Postal Code....113
16.60 Protocol....113
16.61 Protocol Request....113
16.62 Quality of Horizontal Measurement....113
16.63 Reported Date....114
16.64 Restriction....114
16.65 Scale Minimum....116
16.66 Source....116
16.67 Source Date....117
16.68 Source Type....117
16.69 Status....117
16.70 Telecommunication Carrier....118
16.71 Telecommunication Identifier....118
16.72 Telecommunication Service....118
16.73 Text....119
16.74 Text Offset Bearing....119
16.75 Text Offset Distance....120
16.76 Text Rotation....120
16.77 Text Type....120
16.78 Thickness of Ice Capability....120
16.79 Time of Day End....120
16.80 Time of Day Start....121
16.81 Uncertainty Fixed....121

16.82 Uncertainty Variable Factor 121
16.83 Vessel Performance 121
16.84 Vessels Characteristics 122
16.85 Vessels Characteristics Unit 122
16.86 Vessels Characteristics Value 123
16.87 Action or Activity 124
16.88 Category of Marine Protected Area 125
16.89 Category of RxN 125
16.90 Category of Vessel 126
17 Complex Attributes 129
17.1 Bearing Information 129
17.2 Contact Address 129
17.3 Designation 129
17.4 Feature Name 130
17.5 Fixed Date Range 130
17.6 Frequency Pair 130
17.7 Graphic 131
17.8 Horizontal Position Uncertainty 131
17.9 Information 131
17.10 Online Resource 132
17.11 Orientation 132
17.12 Periodic Date Range 133
17.13 RxN Code 133
17.14 Schedule by Day of Week 133
17.15 Spatial Accuracy 133
17.16 Source Indication 134
17.17 Survey Date Range 134
17.18 Telecommunications 134
17.19 Text Content 135
17.20 Time Intervals by Day of Week 135
17.21 Vessel Measurements Specification 136

1 Overview

1.1 Introduction

The “Data Classification and Encoding Guide” has been developed to provide consistent, standardized instructions for encoding S-100 compliant Marine Protected Areas (MPA) (S-122) data.

The purpose of the Data Classification and Encoding Guide is to facilitate S-122 encoding to meet IHO standards for the proper display of Marine Protected Areas information in an ECDIS and other electronic charting displays. This document describes how to encode information that the modeller considers relevant to a Marine Protected Areas data product. The content of a dataset is at the discretion of the producing authority provided that the conventions described within this document are followed. A “producing authority” is a Hydrographic Office (HO) or other organization authorized by a government, to produce definitive nautical information. The entire S-100 Universal Hydrographic Data Model, including the S-122 MPA Product Specification, is available at the following web site, https://iho.int.

1.2 Document Metadata

NOTE: This information uniquely identifies this Data Classification and Encoding Guide to the Product Specification and provides information about its creation and maintenance.

Table 1-1 — Document metadata

Metadata	Content
Title:	Marine Protected Areas, Data Classification and Encoding Guide
Version:	2.0.0
Date:	23 March 2026
Language:	English
Classification:	Unclassified
Contact:	International Hydrographic Organization 4 Quai Antoine 1er B.P. 445 MC 98011 MONACO CEDEX Telephone: +377 93 10 81 00 Fax: +377 93 10 81 40 URL: https://iho.int
Identifier:	S-122 Annex A Data Classification and Encoding Guide
Maintenance:	Changes to S-122 Annex A; Data Classification and Encoding Guide are coordinated by the IHO Nautical Information Provision Working Group (NIPWG) and must be made available via the IHO web site.

1.3 Terms and definitions

For terms and definitions, see the Marine Protected Areas Product Specification, Clause 1.4.2.

1.4 Abbreviated terms

For a list of abbreviations, see the Marine Protected Areas Product Specification, Clause 1.4.3.

1.5 Use of language

Within this document:

- “Must” indicates a mandatory requirement;
- “Should” indicates an optional requirement, that is the recommended process to be followed, but is not mandatory;
- “May” means “allowed to” or “could possibly”, and is not mandatory, or recommended.

1.6 Maintenance

Changes to the Data Classification and Encoding Guide must occur in accordance with the IHO Resolution 2/2007 as amended.

2 General

2.1 Introduction

This Data Classification and Encoding Guide (DCEG) contains rules and guidance for converting data describing the real world into data products that conform to the S-122 specification.

The S-122 specification contains an application schema (UML model) describing the conceptual domain model in terms of classes and relationships, and a Feature Catalogue (see S-122 Annex C) that specifies the data model, i.e., specifies the data model types and associations corresponding to the various classes and relationships in the application schema.

To simplify the DCEG text, the various data model types will be provided without the suffixes “class”, “type” or “instance”; e.g. the term “feature” should be understood as “feature class” or “feature type” or “feature instance” as best fits the immediate context in which it is used (and where there might be confusion, it is written out in full as feature class/type/instance).The model defines real world entities as a combination of descriptive and spatial characteristics (S-122 Product Specification clause 6).

This clause of the DCEG contains general information needed to understand the encoding rules and describes fundamental common rules and constraints. It also describes datasets and metadata. The data model object types used within S-122 and their encoding rules and guidelines are defined in detail in subsequent clauses of this document.

Within this document the features, information types, associations, and attributes appear in **bold text** or *italic text*, to distinguish them from surrounding words.

2.2 Descriptive characteristics

2.2.1 Feature

A feature contains descriptive attributes that characterize real world entities.

The word ‘feature’ as used in the ISO 191xx series and in S-100 based product specifications has two distinct but related senses – ‘feature type’ and ‘feature instance’. A feature instance is a single occurrence of the feature and is represented as an object in a dataset. The location of a feature instance on the Earth’s surface is indicated by a relationship to one or more spatial primitive instances. A feature instance may exist without referencing a spatial primitive instance.

2.2.1.1 Geographic feature class

Geographic (Geo) feature types carry the descriptive characteristics of a real world entity which is provided by a spatial primitive instance.

2.2.1.2 Meta feature class

A meta feature type contains information about other features.

2.2.1.3 Charted background feature

The data product would mostly be visualized as an overlay of an ENC or other GIS applications. Consequently, all necessary descriptive and spatial characteristics to provide a charted background should be provided by the underlying application.

2.2.1.4 Information type

An information type has no geometry and therefore is not associated to any spatial primitives to indicate its location.

An information type may have attributes and can be associated with features or other information types in order to carry information particular to these associated features or information types.

2.3 Spatial characteristics

2.3.1 Spatial primitives

The allowable spatial primitive for each feature is defined in the Feature Catalogue. Allowable spatial primitives are point, curve, and surface.

Within this document, allowable spatial primitives are included in the description of each feature. For easy reference, Table 2-1 below summarises the allowable spatial primitives for each feature. In the table, abbreviations are as follows: point (P), curve (C), surface (S), and none (N). Abstract features are excluded from this table since they cannot have feature instances in datasets.

Table 2-1 — Features and their spatial primitives

Feature	P	C	S	N
InformationArea			S	
MarineProtectedArea		C	S	
RestrictedArea			S	
VesselTrafficServiceArea			S	
DataCoverage			S	
QualityOfNonBathymetricData			S	
TextPlacement	P			

2.3.2 Capture density guideline

Coordinate density can have a significant impact on file size and system performance. A rule of thumb is to limit the coordinate density to 0.3 mm at maximum permitted display scale. For a scaleless product, the producer should keep in mind the expected scale range for typical use and the density of coordinates needed to suit the needs of the product.

The capture density will follow the recommendation of the S-101 (ENC) DCEG, which states curves and surface boundaries should not be encoded at a point density greater than 0.3 mm at permitted display scale.

A curve consists of one or more curve segments. Each curve segment is defined as a loxodromic line on WGS84, or as an arc or circle. Long lines may need to have additional coordinates inserted to cater for the effects of projection change.

The presentation of line styles may be affected by curve length. Therefore, the encoder must be aware that splitting a curve into numerous small curves may result in poor symbolization.

2.4 Attributes

Attributes may be simple type or complex type. Complex (C) attributes are aggregates of other attributes that can be simple type or complex type attributes. Simple (S) attributes are assigned to one of the types collected in clause 2.4.1.

The binding of attributes to a feature, the binding of attributes to attributes to construct complex attributes, and attribute multiplicity are all defined in the Feature Catalogue.

Within this document, the allowable attributes are included in the description of each feature, as well as the allowable values for enumeration type attributes.

2.4.1 Simple attribute types

Each simple attribute is assigned one of the attribute datatypes in Table 2-2:

Table 2-2 — Simple attribute types

Abbreviation	Attribute type	Description
BO	Boolean	A value representing binary logic. The value can be either *true* or *false*. The default state for Boolean type attributes (i.e. where the attribute is not populated for the feature) is *false*. NB: The XML schema specification states that a boolean data type can have the following legal literals: true, false, 1, 0.
CL	Code List	A type of flexible enumeration (see “EN” below). A code list type is a list of literals which may be extended only in conformance with specified rules. Attributes of a code list type may take values from the list or other values which are defined according to the rules. The rules should be part of the specification of the individual codelist type. A code list could either be closed (fixed) or open (extensible). A code list type has the following properties: 1. A description of the code list type, 2. The URI where the list could be found, and 3. An encoding instruction.
DA	Date	A date provides values for year, month and day according to the Gregorian Calendar. Example (XML/GML): 1998-09-18 (YYYY-MM-DD) S-122 uses only XML-based formats (including GML) and therefore the ISO “basic” format described in S-100 is not used.
DT	Date and Time	A DateTime is a combination of a date and a time type. Example (XML/GML): 1985-04-12T10:15:30 (YYYY-MM-DDThh:mm:ss) S-122 uses only XML-based formats (including GML) and therefore the ISO “basic” format described in S-100 is not used.
EN	Enumeration	A fixed list of valid identifiers of named literal values. Attributes of an enumerated type may only take values from this list.
IN	Integer	A signed integer number. The representation of an integer is encapsulation and usage dependent. Integer attribute values must not be padded by non-significant zeroes. For example, for a number of 19, the value populated for the attribute must be 19 and not 019. Examples: 29, -65547
RE	Real	A signed real (floating point) number consisting of a mantissa and an exponent. The representation of a real is encapsulation and usage dependent. Real attribute values must not be padded by non-significant zeroes. For example, for a signal period of 2.5 seconds, the value populated for the attribute signal period must be 2.5 and not 02.50. Examples: 23.501, -0.0001234, -23.0, 3.141296
TD	Truncated Date	One or more significant components of the modelling date are omitted. Example: A GML dataset would use a GML built-in type and encode it as `<gMonth>--02<gMonth>` S-122 uses only XML-based formats (including GML) and therefore the ISO “basic” format described in S-100 is not used.
TE	Free text	An arbitrary-length sequence of characters including accents and special characters from a repertoire of one of the adopted character sets.
TI	Time	A time is given by an hour, minute, and second. Time zone according to UTC is optional. Character encoding of a time is a string that follows the local time.

Abbreviation	Attribute type	Description
		Examples (XML/GML): 18:30:59Z; 18:30:59+01:00; 18:30:59
URL	URL	A uniform resource locator (URL) is a URI that provides a means of locating the resource by describing its primary access mechanism (RFC 3986). Example: https://registry.iho.int
URN	URN	A persistent, location-independent, resource identifier that follows the syntax and semantics for URNs specified in RFC 2141. Example: urn:mrn:iho:S-122:1:0:0:Regulations

2.4.2 Mandatory attributes

Some attributes are mandatory and must be populated for a given feature. There are some reasons why attribute values may be considered mandatory:

- They are fundamental to the definition of a feature;
- They are required to support the correct portrayal of a feature instance;
- Certain features make no logical sense without specific attributes;
- Some attributes are required for safety of navigation.

Within this document, mandatory attributes are those with a multiplicity of 1,1 or 1,n (n>1) or 1,*. The attribute multiplicity is identified in the description of each feature class.

2.4.3 Conditional attributes

The feature classes or information types do not contain conditional attributes.

Complex attributes which are assigned to feature classes or information types have at least one sub-attribute which is mandatory (or conditionally mandatory). Where the sub-attribute of a complex attribute is conditional, this is indicated in the Remarks sub-clause for the relevant feature class entries.

2.4.4 Missing attribute values

Where a value of a mandatory attribute is not known, the attribute must be populated with an empty (null) value, except in the bindings listed in Table 2-3 below. Nilling the attributes in Table 2-3 must be treated as an error in the dataset.

Table 2-3 — Attributes which cannot be nilled

Feature / Information type / Complex attribute	Un-nillable attribute(s)
InformationArea	categoryOfRelationship actionOrActivity
TextPlacement	textOffsetBearing textOffsetDistance textType
onlineResource	linkage
featureName	language name
graphic	pictorialRepresentation
periodicDateRange	dateStart dateEnd
vesselMeasurementsSpecification	comparisonOperator vesselsCharacteristics vesselsCharacteristicsValue
DataCoverage	maximumDisplayScale
PermissionType	categoryOfRelationship
InclusionType	membership

Where the value of a non-mandatory attribute is not known, the attribute must not be included in the dataset.

2.4.5 Multiplicity

In order to control the number of allowed attribute values or sub-attribute instances within a complex attribute, S-100 uses the concept of multiplicity. This defines lower and upper limits for the number of values, whether the order of the instances is significant, and if an attribute is mandatory. Common examples are shown in Table 2-4:

Format: MinOccurs, MaxOccurs (a * indicates that infinite instances are possible, the term (ordered) indicates that the order of the provided instances is significant)

Table 2-4 — Multiplicity of attributes

Multiplicity	Explanation
0,1	An instance is not required; if provided there must only be one instance.
1,1	An instance is required and there must only be one instance.
0,*	An instance is not required and there can be an infinite number of instances.
1,*	An instance is required and there can be an infinite number of instances.
1,* (ordered)	An instance is required and there can be an infinite number of instances, the order of which is significant.
2,2	Two instances are required and there must be no more than two.

2.4.6 Spatial attribute types

Spatial attribute types must contain a referenced geometry and may be associated with spatial quality attributes. Each spatial attribute instance must be referenced by a feature instance or another spatial attribute instance.

2.4.6.1 Quality of spatial attributes

The quality of spatial attributes in S-122 is described in a **QualityOfNonBathymetricData** meta-feature (clause 4.4). This meta-feature defines areas within which uniform assessment exists for the quality. It is described in detail later in this document.

If the spatial quality attributes for an individual instance of a spatial primitive differ from the quality indicated in the overlying **QualityOfNonBathymetricData** meta-feature, the quality attributes for that instance are carried in an information class called **SpatialQuality**. Only points and curves can be associated with **SpatialQuality**. S-122 does not use multi-points. Currently, no use case for associating surfaces with spatial quality attributes is known, therefore this is prohibited. Vertical uncertainty is prohibited for curves as this dimension is not supported by curves.

NOTE: S-122 does not make use of the S-101 Quality of Bathymetric Data meta-feature since depth range uncertainties are not needed. The **QualityOfNonBathymetricData** meta-feature has all the quality attributes needed by S-122.

The spatial quality of individual spatial primitives can be indicated using the **SpatialQuality** information type (clause 12.2) associated to the individual spatial primitive. Figure 2-1 depicts the conceptual model. This should only be used when it is necessary to override the quality indicated in a covering **QualityOfNonBathymetricData** feature.

A **SpatialQuality** instance should not be created if the quality information to be encoded in it is identical to the quality information in an underlying meta-feature.

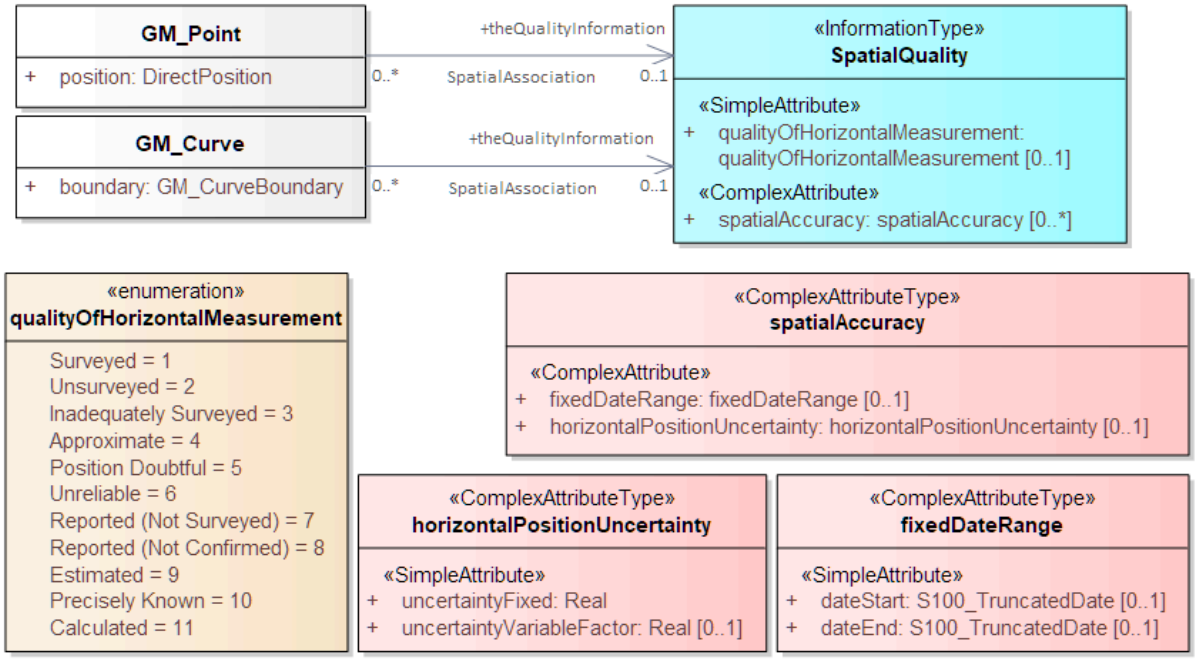

Figure 2-1 — Spatial quality for spatial primitives

2.4.7 Portrayal feature attributes

Marine Protected Areas data products will be used within ECDIS where ENC data is displayed based on the rules defined within the S-101 Portrayal Catalogue. While most ECDIS portrayal is based on attributes describing the instance of a particular feature in the real world, certain feature attributes are used in portrayal rules to provide additional functionality in the ECDIS. Table 2-5 provides a list of attributes which have specific influence on portrayal.

Table 2-5 — Attributes which have effects on portrayal

Attribute	Effects on portrayal
fixedDateRange; periodicDateRange	Population of these complex attributes determines when the feature will be added (sub-attribute dateStart) and/or removed (sub-attribute *dateEnd*) from the display in some ECDIS display settings.
information	Population of this complex attribute will result in the display of the magenta information symbol to highlight additional information to the user.
nameUsage	This sub-attribute determines the priority and level of display (full display or Pick Report only) where multiple instances of the complex attribute feature name are encoded for a single feature instance, based on Mariner's selected ECDIS display settings.
pictorialRepresentation	The population of this Text attribute will result in the display of the magenta information symbol to highlight additional information to the user.
scaleMinimum	value at which the feature will be removed from the display if application of scale minimum is enabled in the ECDIS.
textContent	The population of this complex attribute will result in the display of the magenta information symbol to highlight additional information to the user.

2.4.8 Textual information

Textual information may provide additional information essential to understand the presence of the Marine Protected Areas and other features of an S-122 product. This information may also provide legal information pertaining to the S-122 product features.

The methods to provide textual information vary from the simple provision of short text, to the more structured provision of extensive text. The length of the text determines the method and the attribute selection, see clause 2.4.8.2.

2.4.8.1 Specialized information types for common kinds of textual information

The information types **Restrictions**, **Recommendations**, **Regulations**, or **NauticalInformation** must be used to encode text information when the DCEG allows them to be associated to the feature or information type and the information is of the appropriate kind (a restriction, regulation, etc.).

In exceptional circumstances and only if the use of the information types **Restrictions**, **Recommendations**, or **Regulations** is not sufficient, **NauticalInformation** can be used to encode additional textual information associated to a feature or a group of features.

In some cases, there may be a specialized attribute that is specifically intended for the data in question. If an appropriate specialized attribute is available, it must be used in preference to *information* or *textContent*. For example, feature names will generally be encoded in the *name* sub-attribute of complex attribute *featureName*, instead of *information→text*.

2.4.8.2 Textual information attributes

Textual information which is not appropriate for any of the Text-type attribute (or sub-attribute) allowed for the feature/information type should be encoded using either *information* or *textContent* complex attributes. Generally, either *information* or *textContent* is allowed, but not both.

2.4.8.3 Languages

Both *information* and *textContent* define a language sub-attribute for specifying the language in which the text is encoded.

The default exchange language for textual information should be English; therefore it is not required to populate the sub-attribute *language* for an English version of textual information (unless the *language* attribute is indicated as mandatory).

Languages other than English may be used as a supplementary option, for which language must be populated with an appropriate value to indicate the language.

When a national language is used in the textual attributes, the English translation must also exist.

The specification of the language attribute in the IHO GI registry states “The language is encoded by a 3 character code following ISO 639-2/T.” These codes and the corresponding language names may be obtained from the codelist S100_MD_LanguageCode in the S-100 codelists file, which is part of the S-100 schemas distribution, at the URLs below:

- XML file: https://schemas.s100dev.net/schemas/S100/5.2.0/resources/Codelists/cat/codelists.xml
- Web list: https://schemas.s100dev.net/schemas/S100/5.2.0/resources/Codelists/cat/codelists.html

2.4.8.4 Minimal use of generalized text attributes

The complex attributes *information* and *textContent* must not be used when it is possible to encode the information by means of any other attribute. The population of these attributes provides symbols on an ECDIS screen. Therefore producers should carefully consider use of these attributes as the symbol may contribute significantly to ECDIS screen clutter and text attributes should be populated only when the content conveys useful information.

2.4.8.5 Short textual information

The *text* sub-attribute of complex attribute *information* should generally be used for short notes or to transfer information which cannot be encoded by other attributes, or to give brief information about a feature. The use of the complex attribute *information* as a stand-alone complex attribute is intentionally limited to the information types **ContactDetails**, **Applicability**, **NonStandardWorkingDay**, and **ServiceHours**, which do not need the additional attributes defined in *textContent*. The reason for the limited use of *information* as a stand-alone complex attribute is to provide a structured and harmonised approach to textual information within the S-122 product data sets.

The text populated in *text* attributes must not exceed 300 characters. Character strings contained in *text* attributes must be UTF-8 character encoding.

If the *text* sub-attribute of *information* is populated, the *fileReference* and *fileLocator* sub-attributes must not be populated.

2.4.8.6 Complex or lengthy textual information

More complex encodings of text may use either *information* or *textContent*. The feature catalogue and the feature/information type definitions in this DCEG indicate whether *information* or *textContent* is allowed.

The complex attribute *textContent* also has *information* as a complex sub-attribute. If a short note must be encoded in a feature or information type which has only *textContent* as an attribute, it should be encoded as *textContent→information→text*.

Complex text information, such as text longer than 300 characters, formatted text, or HTML extracts from shipping regulations, must be encoded in a file named in either information→fileReference or *textContent→information→fileReference*. The construction *textContent→information→fileReference* should be used if the feature/information type provides *textContent* as complex attribute.

The complex attribute *information* defines an optional sub-attribute *headline* which may be used for a short title not exceeding 60 characters. The content should be short but informative – if the textual information is divided into sections, the most relevant section header from the referenced content may be a good choice for *headline*.

Multiple levels of headings are permitted when the upper bound of *headline* multiplicity is > 1. Multi-level headings must be encoded according to the heading level structure, that is, the highest level heading must be first, then the second level, then the third, and so on.

The complex attribute *textContent* defines an optional sub-attribute *categoryOfText* for indicating whether the text is the full text from the source, an extract from the source, or a summary prepared by the encoder. Populating *categoryOfText* is recommended whenever the textual information is taken or summarised from a law or regulation.

If it is considered necessary to include a description of the source of the textual information, the sub-attribute *sourceIndication* of *textContent* must be used. Encoding a description of the source is strongly recommended for textual information whose source is considered as information the end-user must have, e.g., because the date of issue must be conveyed or because it cites official regulations which are frequently updated.

NOTE: Some government documents are frequently updated, e.g., the U.S. Electronic Code of Federal Regulations, which is currently updated every working day even though a particular section may be stable for years.

2.4.9 Attributes referencing external files

2.4.9.1 Predefined derived types

Table 2-6 presents the following predefined derived types which are described in S-100 (clause 1-4.6):

Table 2-6 — Predefined derived types

Name	Description	Derived from
URI	A uniform resource identifier which character encoding shall follow the syntax rules as defined in RFC 3986. EXAMPLE https://registry.iho.int	CharacterString
URL	A uniform resource locator (URL) is a URI that provides a means of locating the resource by describing its primary access mechanism (RFC 3986). EXAMPLE https://registry.iho.int	URI
URN	A persistent, location-independent, resource identifier that follows the syntax and semantics for URNs specified in RFC 2141. EXAMPLE urn:iho:s101:1:0:0:AnchorageArea	URI

2.4.9.2 Reference to textual files

The files referenced by complex attribute *information* and its sub-attribute *fileReference* must be .TXT or .HTM files, and may contain formatted text. It is up to the Producing Authority to determine the most

suitable means of encoding a particular piece of text (as text or HTML). The format of the reference to the file should be a “file URI” (S-100 1-4.6).

Besides being bound to certain types, the complex attribute *information* is also a sub-attribute of the complex attribute *textContent*. This means that any type that binds *textContent* as an attribute can also contain a reference to a textual file via an *information* sub-attribute. In S-122, there are several features, information types, and complex attributes that bind either *textContent* or *information*.

The exchange language for textual information should be English. The sub-attribute *language* must be populated with an appropriate value to indicate the language used. Languages other than English may be used as a supplementary option. Generally this means, when a national language is used in the textual attributes, the English translation must also exist.

Files must only use UTF-8 character encoding even when the sub-attribute *language* is populated with a language other than English.

If it is necessary to indicate a specific section within a large text file, this may be done by encoding the location in the *fileLocator* sub-attribute of *information*, as described in Table 2-7.

Producers and application developers should note that the use of the fileLocator attribute enables a single support file to contain separate chunks of text referenced from different features, information types, or complex attribute. Adopting this practice enables producers to reduce the number of external files needed with a dataset.

Table 2-7 — Locators for external files

Format	File extension	Content of fileLocator
Text	TXT	Locators to text files are not permitted; the file should be split into separate text files or an HTML file used instead.
HTML	HTM	The HTML fragment identifier, i.e., the value of the HTML name or id attribute of the target (as defined in the relevant HTML specification).

2.4.9.3 Reference to external sources

References to Internet sources should be encoded using the *onlineResource* sub-attribute of *textContent*. Encoders should be aware that systems may not be able to access the Internet, so *onlineResource* should be used only for non-essential information. Only sources that can be certified as secure and free from malicious downloads should be provided.

2.4.9.4 Reference to graphics

If it is required to indicate a graphic, the complex attribute *graphic* must be used. The sub-attribute *pictorialRepresentation* must be used to indicate the file name (without the path) of the external graphical file. Graphic files that form part of the data product must be content with the characteristics collected in Table 2-8.

Table 2-8 — Graphics characteristics

Characteristics	Values
Recommended Resolution	96 DPI
Minimum Size x,y	200,200 pixels
Maximum Size x,y	800,800 pixels
Bit Depth	8 Bit Indexed Colour
Compression	LZW
Format	Tiff 6.0

File sizes should consider the maximum permitted sizes of datasets and exchange sets.

Additional information about the graphic file may be encoded in other sub-attributes of attribute graphic, as described in clause 2.4.12.

2.4.10 Dates

Dates may be need to be encoded as complete or truncated values, depending on available information and allowed format for the particular attribute. The definition of the attribute will indicate if it must take a complete value (type Date or DA) or is allowed to take a truncated value (type S100_TruncatedDate or TD). Complete and truncated dates are different value types (see S-100 1-4.5.2 Table 1-2).

For attributes that use the complete date type (type Date or DA), all their components (year, month, and day) must be specified.

For attributes that use the truncated date type (type S100_TruncatedDate or TD), zero, one, or two of the year/month/day components may be omitted. If the year component is included, it must be specified using exactly 4 digits.

2.4.10.1 Complete dates

Dates (except truncated dates, see the following clause) must be encoded in conformance with the Date format as specified in S-100 Clause 1-4.5.2 which is the same as the DA format in Table 2-2 in this document. The data values have to be provided in accordance with the Gregorian Calendar starting with four digits for the year, two digits for the month and two digits for the day.

Example: The date 18 September 2010 is encoded as follows:

In the GML format: <date>2010-09-18</date>

Note that since both discovery metadata and GML datasets are XML files, both will use the “GML format” above.

2.4.10.2 Truncated dates

In Truncated Dates one or more components (year, month, or day) of the date is not specified. Truncated date values must be encoded in conformance with the S100_TruncatedDate format or equivalent as specified in S-100 (clauses 1-4.5.2 and 3-9) which is the same as the TD format in Table 2-2 in this document. If encoding attributes which can take truncated date values (e.g., *fixedDateRange*, *periodicDateRange*, *reportedDate*) and no specific year, month, or day is required, the values must be encoded in conformance with the truncated date format as specified in S-100 (clauses 1-4.5.2 and 3-9), using the format-specific type for XML/GML.

To encode partial dates in the XML/GML data format:

Table 2-9 — Date encoding format in XML and GML

Description	ISO 8211	GML
No specific year, same day each year	`----MMDD`	`<gMonthDay>--MM-DD</gMonthDay>`
No specific year, same month each year	`----MM--`	`<gMonth>--MM</gMonth>`
No specific day	`YYYYMM--`	`<gYearMonth>YYYY-MM</gYearMonth>`
No specific month and no specific day	`YYYY----`	`<gYear>YYYY</gYear>`
NOTE: YYYY = calendar year; MM = month; DD = day.		

The dashes (-) indicating that the year, month, or date which is not specified must be included in the encoding (with no space between the dashes).

2.4.10.3 Start and end of ranges

In accordance with S-100 clause 3-8, the start and end instants of a range or period are included in the range or period.

EXAMPLE 1: If the beginning of a date range is encoded as the complete date 01 January 2016, the period begins at 00:00:00 on 1 January 2016, and the whole of New Year’s Day is included in the period.

If the end of the date range is encoded as 01 January 2016, the period ends at 24:00:00 on 1 January 2016, i.e., again the whole of New Year's Day is included in the period.

EXAMPLE 2: If the beginning of a period is encoded in truncated date format as `----01--` (i.e., year and day not specified), the period begins at 00:00:00 on 1 January each year. If the end of the period is encoded as `----01--`, the period ends at 24:00:00 on 31 January each year.

NOTE (1): Particular care should be taken if the start or end date is 28 or 29 February. S-100 3-8.3 explains the implications for end of February. For example, the truncated date `----02--` will be interpreted as 29 February in leap years and 28 February in non-leap years, while `----0228` will be interpreted as 28 February in every year.

NOTE (2): In accordance with ISO practice at the time S-100 date and time formats were defined, the time `00:00:00` means midnight at the start of a day and `24:00:00` means midnight at the end of a day. This continues to be S-100 usage.

Overlapping date, time, or dateTime ranges, whether periodic or fixed, in the same feature, information type, or complex attribute are generally an error.

Null date, time, or dateTime ranges (where the start and end are the same), whether periodic or fixed, in the same feature, information type, or complex attribute are generally an error.

2.4.10.4 Schedules

Weekly service schedules of a feature can be comprehensively described by using the information types **ServiceHours** and **NonStandardWorkingDay**.

EXAMPLE: A feature service is available under normal operation status 24 hours/day on Monday and Wednesday and from 08:00 to 16:00 LT from Thursday to Saturday. The service is available by pre-arrangement on public holidays and the 5th of August of each year when they fall on days which would otherwise be normal working days.

```
ServiceHours
 scheduleByDayOfWeek
  categoryOfSchedule =1 (normal operation)
  timeIntervalsByDayofWeek
   dayOfWeek = 2(Monday), 4(Wednesday)
   dayOfWeekIsRange = 0 (false)
  timeIntervalsByDayofWeek
   dayOfWeek = 5(Thursday), 7(Saturday)
   dayOfWeekIsRange = 1 (true)
   timeOfDayStart = 08:00:00
   timeOfDayEnd = 16:00:00
NonStandardWorkingDay
 dateFixed = ---08-05 (5 August)
 dateVariable = public holidays
 information.text = "By pre-arrangement"
```

The above example can be encoded as follows:

```
    <S122:ServiceHours gml:id="(GML ID of ServiceHours)">
        <scheduleByDayOfWeek>
            <categoryOfSchedule code="1">Normal Operation</categoryOfSchedule>
            <timeIntervalsByDayOfWeek>
                <dayOfWeek code="2">Monday</dayOfWeek>
                <dayOfWeek code="4">Wednesday</dayOfWeek>
                <dayOfWeekIsRange>0</dayOfWeekIsRange>
                <timeOfDayStart>00:00:00</timeOfDayStart>
                <timeOfDayEnd>24:00:00</timeOfDayEnd>
            </timeIntervalsByDayOfWeek>
            <timeIntervalsByDayOfWeek>
                <dayOfWeek code="5">Thursday</dayOfWeek>
                <dayOfWeek code="7">Saturday</dayOfWeek>
                <dayOfWeekIsRange>1</dayOfWeekIsRange>
                <timeOfDayStart>08:00:00</timeOfDayStart>
                <timeOfDayEnd>16:00:00</timeOfDayEnd>
            </timeIntervalsByDayOfWeek>
        </scheduleByDayOfWeek>
```

```
    <partialWorkingDay xlink:href="(reference to NonStandardWorkingDay)"/>
</S122:ServiceHours>

<S122:NonStandardWorkingDay gml:id="(GML ID of NonStandardWorkingDay)">
    <dateFixed><gMonthDay>--08-05</gMonthDay></dateFixed>
    <dateVariable>public holidays</dateVariable>
    <information><text>By pre-arrangement</text</information>
    <theServiceHours_nsdy xlink:href="(reference to ServiceHours)"/>
</S122:NonStandardWorkingDay>
```

If the days of week are known but the hours of availability are unknown, there is no time attribute. Twenty-four availability is indicated by encoding the availability period as 000000-240000. Special cases such as unknown can be explained in the *textContent* or *information* attribute of **ServiceHours**. To encode two or more periods within the same day, repeat the *timeOfDayStart* and *timeOfDayEnd* attributes. If one of the times is not known, it may be nilled as described in clause 2.4.4.

For example, to encode open hours of 8 a.m. to 12 noon and 1 p.m. to 5 p.m. on Thursdays and Saturdays:

```
timeIntervalsByDayofWeek
 dayOfWeek =5(Thursday), 7(Saturday)
 dayOfWeekIsRange =1 (true)
 timeOfDayStart = 08:00:00
 timeOfDayStart = 13:00:00
 timeOfDayEnd = 12:00:00
 timeOfDayEnd = 17:00:00
```

The order of repeated *timeOfDayStart* and *timeOfDayEnd* attributes is significant, since intervals are specified by matching them pairwise in order.

UTC is indicated by the Z suffix. The absence of the Z suffix indicates local time.

The absence of any additional information other than date (fixed or variable) in **NonStandardWorkingDay** should be interpreted as closure on the specified days. Non-standard working days do not need to be associated with **ServiceHours** instances categorized as “closure” (*categoryOfSchedule*=Closure) because the closure is already indicated in the **ServiceHours** instance.

2.4.10.5 Times

If it is required to provide information of the start time and end time of an active period of a feature, it must be encoded using the attributes *timeOfDayStart* and *timeOfDayEnd*. The order has significance.

2.4.11 Combination of date schedules and times

Schedule information can also include time of day. The complex attribute *timeIntervalsByDayofWeek* also includes *timeOfDayStart* and *timeOfDayEnd* attributes to encode the daily start and end times of service. Complete instructions on how to encode schedules are described in clause 2.4.10.4.

2.4.12 Graphic information

A graphic file should be appropriate for the purpose and should supplement the information in terms of navigational relevance. Preferably, the graphic should provide perspective relevant to the view of the mariner. Graphics should be such that all the information in the graphic is legible in the application display. Graphic information must be encoded using the complex attribute *graphic*. The simple sub-attribute *pictureInformation* should be used to provide credits to the picture creator, copyright owner etc. Assuming that graphic information provides a coastal view, mariners are interested in knowing from which point on sea that graphic has been taken. The complex attribute *bearingInformation* (see clause 2.4.12.1) provides all necessary information.

2.4.12.1 Bearing information

The most accurate information should be provided if it is necessary to indicate a position from where a picture has been taken. *information* is a sub-complex attribute of *bearingInformation* and should be used to specify that no bearing information can be provided whenever such is the case. The sub-attribute *orientation* (which provides an attribute for providing uncertainty) can be used to describe a certain level of inaccuracy in the position determination.

2.5 Associations

2.5.1 Introduction

An association expresses a relationship between two classes — features, information types, or a feature and an information type. Objects in the dataset (instances of feature/information types) are related only if the link between them is encoded in the dataset. An association end may have a multiplicity which describes how many instances the feature or information type instance at the other end is allowed to link to.

EXAMPLE: An **Authority** information type provides the responsible authority information to the **VesselTrafficServiceArea** feature. An association named **ServiceControl** is used to relate the two classes; roles are used to convey the meaning of the relationship. Secondly, **VesselTrafficServiceArea** inherits an **AdditionalInformation** association to **NauticalInformation** and thereby can also provide general information in a note.

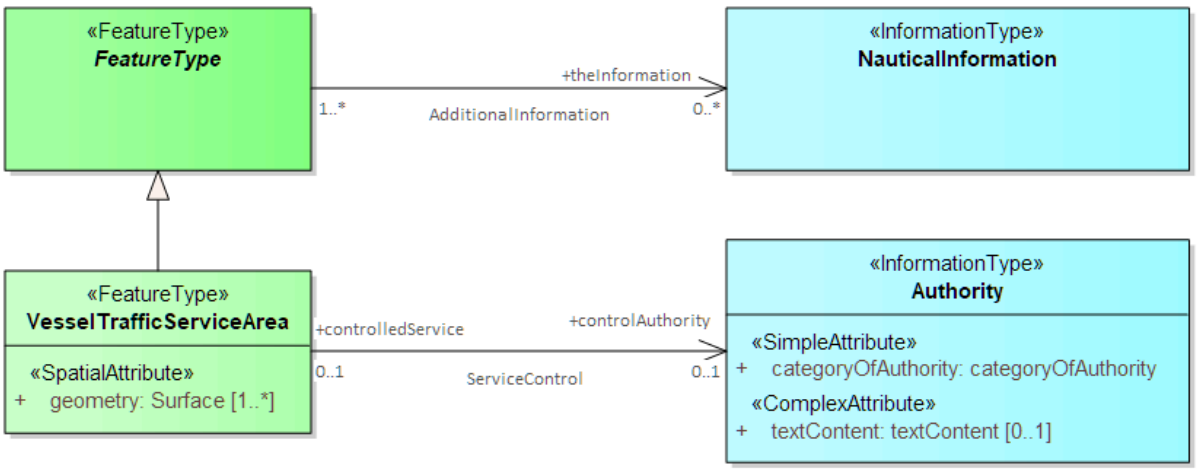

Figure 2-2 — Information association relating a feature to an information type

2.5.2 Association names

The association name is normally provided by the UML diagram at the middle of the connection line/ arrow between the two involved classes and can be obtained from the feature and information type tables provided in this document). Association names may be omitted in the UML diagrams for the following reasons:

1) the association is defined by an association class, see clause 2.5.4 (the name of the association class is used);
2) to avoid cluttering the diagram – however, the name is always documented in the feature/information type tables.

2.5.3 Association roles

Either or both association ends can have a name (role). Roles may be also omitted from the diagram to reduce clutter – again, the role name is documented in the feature/information type tables.

NOTE: Instead of documenting every single role, Product Specifications may describe rules for defining default roles.

2.5.4 Association classes

Association classes allow relationships to be characterized by one or more attributes. The attributes of the association class belong to the association itself, not to any of the features or information types it connects. An association class is both an association and a class. Within an S-122 product the association classes **PermissionType** and **InclusionType** may be used for relating vessel classes to feature and information types.

2.5.4.1 Permission Type

This association class specifies the relationship of the vessel class to a feature, e.g., whether access to a feature (or use of a facility) is prohibited or permitted for a specified class of vessel. The class of vessel

is described by the simple and complex attributes of the information type **Applicability** such as length, cargo, etc. The attributes of the association class describe the nature of the relationship, i.e., whether access to an area is permitted or prohibited, or whether use of a service is required or recommended.

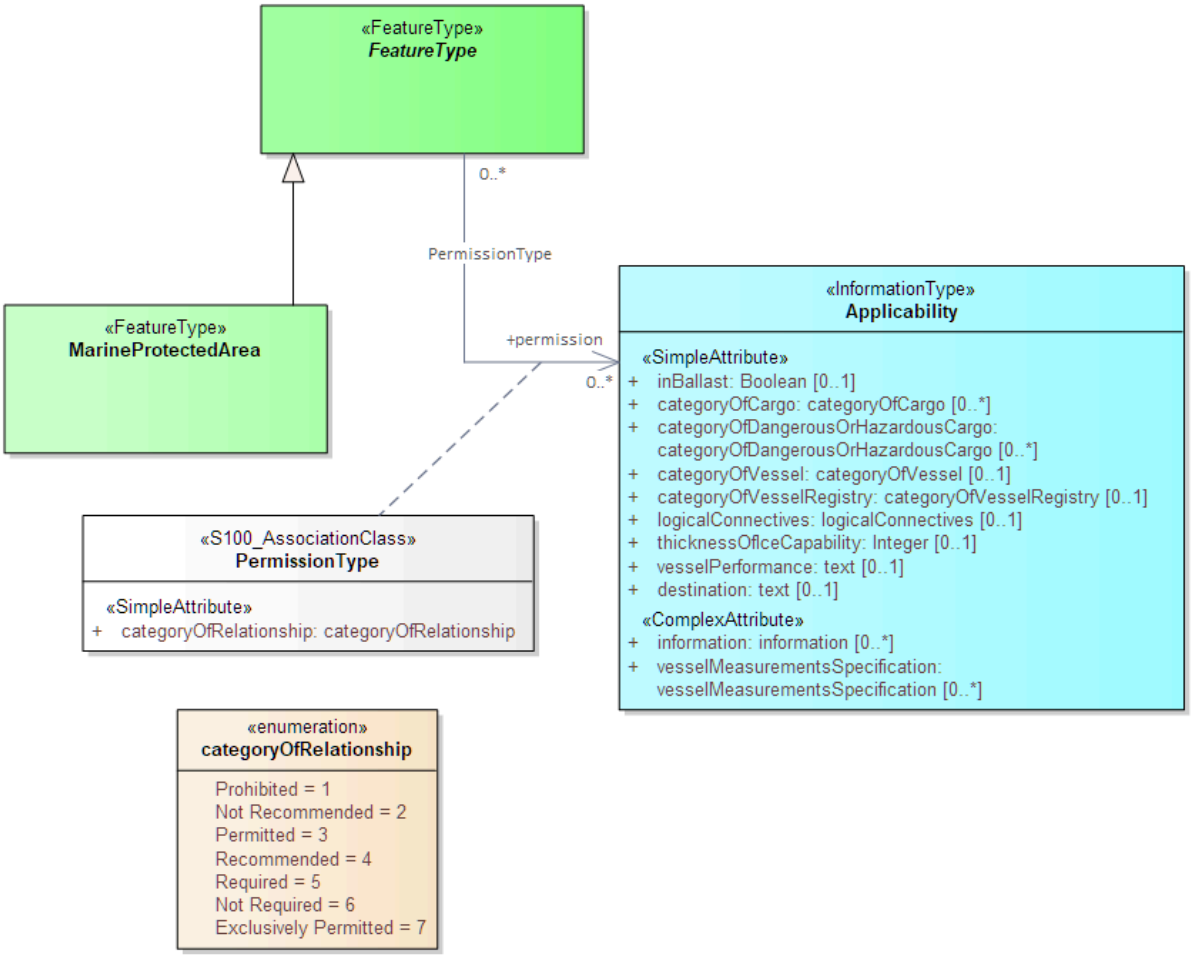

Figure 2-3 — Association class for hypothetical regulations permissions relating to use of a protected area by specified types of vessels

EXAMPLE: An association between an **Applicability** instance with attribute *categoryOfDangerousOrHazardousCargo* = IMDG Code Class 3 and an instance of feature **MarineProtectedArea**, with **PermissionType** attribute *categoryOfRelationship* = *Prohibited*, means that vessels carrying flammable liquids (hazardous cargo type class 3 in the IMDG Code) must not enter the area.

2.5.4.2 Inclusion Type

This association class defines whether a specified customer (class of vessels, as described by **Applicability**) is excluded or included from a particular regulation, recommendation, etc. Again, the attributes of the association class describe the nature of the relationship; in this case whether the vessel is included or excluded from the regulation, etc.

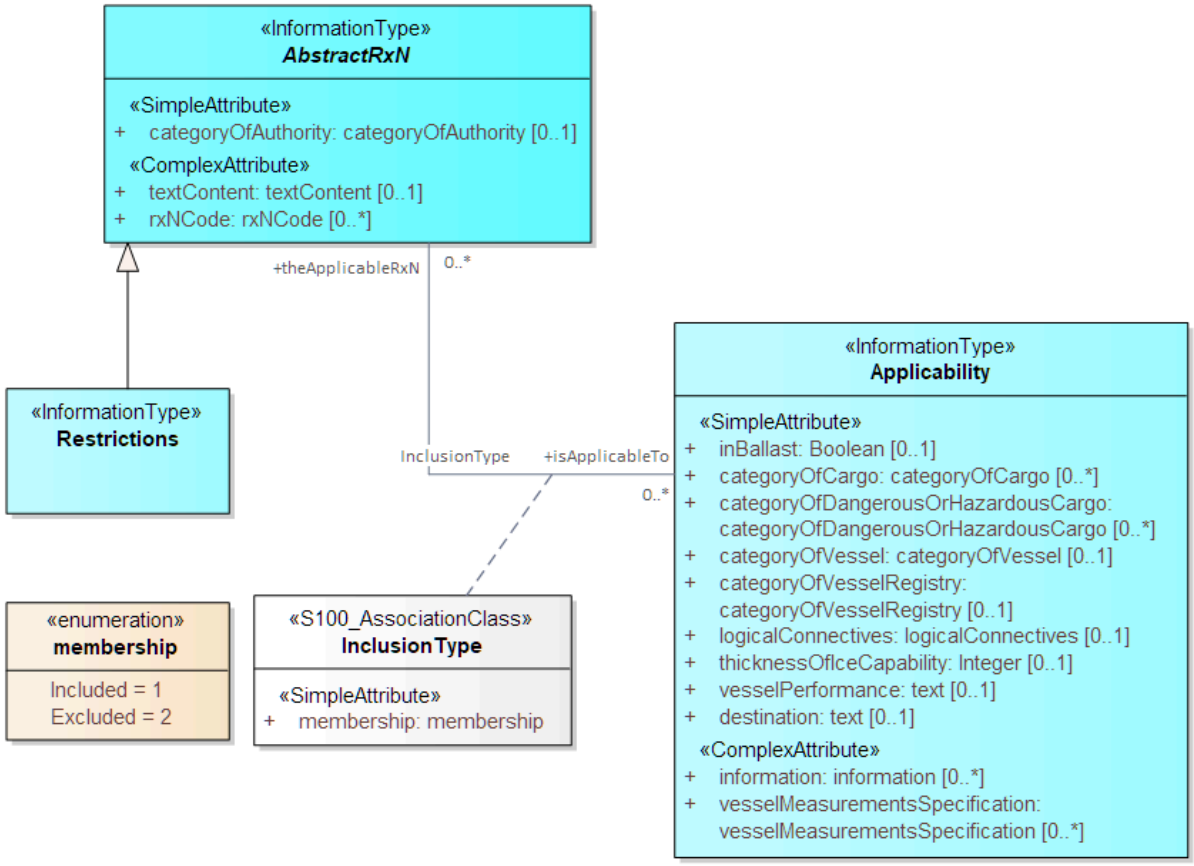

Figure 2-4 — Association class for inclusion of vessel types in regulations

EXAMPLE: An association between an Applicability instance with attribute categoryOfDangerousOrHazardousCargo = IMDG Code Class 3, with Inclusion Type's attribute membership = included, and an association of a Regulation instance to the same Inclusion Type, means that the information provided by the Regulation (a sub-type of AbstractRXN) applies to vessels carrying flammable liquids (hazardous cargo type class 3 in the IMDG Code).

NOTE (1): Since **AbstractRxN** is an abstract type, it cannot have direct instances in the dataset. Only instances of its (non-abstract) sub-types can be used.

NOTE (2): Specific tools may use different presentations in their user interfaces, e.g., as two associations (as described in the text of the example), or one association with an association class also shown (as shown in Figure 2-4).

2.5.5 Use of various associations

2.5.5.1 General

In general, associations must be encoded whenever the relationship is useful for navigation, monitoring, voyage or route planning, or reporting purposes, or any other purpose for which the dataset is intended. The multiplicity lower bound of "0" at an association end means only that the absence of a link to the relevant instance does not invalidate the dataset. The encoding instructions for individual feature and information types describe what associations are allowed and whether they are required or optional.

2.5.5.2 Generic association for uncategorized additional information

Unless other associations are specified, information types are associated to the relevant features using the association name **AdditionalInformation** and the role *theInformation*.

2.5.5.3 Associations to Restrictions, Recommendation, Regulations, and Nautical Information

The Restrictions, Recommendation, Regulations, Nautical Information are associated to the relevant features using the association named **AssociatedRxN** (inherited from their common abstract super-type). The roles at the ends of this association are appliesInLocation and theRxN (the Restriction,

Regulation etc.). If the regulation applies only to a specific class, or if it mentions an exempt class, an additional association to an Applicability object is encoded using the InclusionType association class.

2.5.5.4 Conventional Association

Certain features and information types may be permitted or required to have associations to other feature or information types. The allowed or mandatory associations for a feature/information type are shown in the application schema (clause 4 of the Product Specification) and listed in the documentation for individual types in this Annex (clauses 4 — 12). Definitions of the associations and roles are also given in the DCEG.

2.5.5.5 Where to Encode Associations

The presentation and management of associations will be determined by the user interface of the encoding software tools. Since S-100 permits feature-information associations to be encoded only from the geographic feature to the information type and not vice versa, the information-to-feature link might be unavailable or treated differently from the feature-to-information link.

2.5.5.6 Required Encoding for Associations

Associations must be encoded with the *gml:id* of the target feature or information instance in an *xlink:href* attribute and the alpha code of the association (as given in the Feature Catalogue) of the association in an *xlink:title* attribute.

For example:

```
<S122:controlAuthority xlink:href="#AUTHORITY.0018" xlink:title=
"ServiceControl"/>
```

to encode a **ServiceControl** link from a **MarineProtectedArea** instance to an **Authority** instance where the *gml:id* of the target **AUTHORITY** instance is *AUTHORITY.0018*.

For **InclusionType** and **PermissionType** associations (which are association classes and have association attributes) the attribute must also be included. For example, to link to the **Regulations** instance with *gml:id* = *R01*:

```
<S122:theApplicableRxN xlink:href="#R01" xlink:title="InclusionType">
    <S122:InclusionType>
        <S122:membership code="1">Included</S122:membership>
    </S122:InclusionType>
</S122:theApplicableRxN>
```

2.6 Datasets

2.6.1 Types of Datasets

A dataset is a grouping of features, attributes, geometry and metadata which comprises a specific coverage. Table 2-10 shows the types of datasets which may be produced and contained within an exchange set:

Table 2-10 — Dataset types

Dataset	Explanations
New dataset	Data for an area different (in coverage and/or extent) to existing datasets.
New Edition of a dataset	A re-issue plus new information which has not been previously distributed by Updates. Each New Edition of a dataset must have the same name as the dataset that it replaces and should have the same spatial extents.
Update dataset	Updated or new information. Contains information about objects being added, modified, or deleted.
Re-issue	Includes all the updates applied to the original dataset up to the date of the re-issue.
Cancellation	There is no cancellation dataset. Cancellations are done by means of discovery metadata records in exchange catalogue files.

2.6.2 Overlay data sets

S-122 datasets are intended to be used together with S-101 ENC (or similar data products) which will act as a base layer. The base layer is expected to provide navigational and visual context. Generally, an overlay dataset like S-122 does not provide “skin of the earth” coverage and there will be large areas with no data coverage because the S-122 application schema does not include any feature for designating a region as “other”, or “not an S-122 area” (i.e., there is no S-122 feature equivalent to the S-101 Unsurveyed Area). Further, an overlay dataset does not include features that provide auxiliary information such as bathymetry within a protected area.

2.6.3 Data coverage

A Marine Protected Areas dataset can contain one or more **DataCoverage** features (see clause 4.3). The data boundary is defined by the extent of the **DataCoverage** meta features. Data must only be present within **DataCoverage** meta features. When a feature extends across datasets of overlapping scale ranges, its geometry must be split at the boundaries of the **DataCoverage** features and its complete attribute description must be repeated in each dataset. An Update dataset must not extend the data coverage for the base dataset to which it applies. Where the extent of the data coverage for a base dataset is to be changed, this must be done by issuing a New Edition of the dataset.

2.6.4 Discovery metadata

Discovery metadata is intended to allow applications to find out important information about datasets and accompanying support files to be examined without accessing the data itself (or without reading the support file). Discovery metadata includes, but is not limited to:

- information identifying the product specification and encoding format;
- edition and version numbers, production/release date, and other details of data creation and updating;
- data coverage of the dataset;
- summary descriptions of content, purpose, use, and limitations;
- identification and contact information for the producer and distributor of the dataset.

Discovery metadata is encoded in the exchange catalogue. S-122 uses the same classes and attributes for discovery metadata as S-100, but adds certain product-specific restrictions. The classes and attributes for generic discovery metadata are defined in S-100 Part 17. Constraints and restrictions specific to S-122 are defined in the S-122 Product Specification. The schema for the exchange catalogue file (CATALOG.XML) for S-122 is available from the schema server (https://schemas.s100dev.net).

2.6.5 Dataset header metadata

Dataset header metadata contains structural and discovery metadata that apply to the whole dataset and are encoded in the dataset file. The elements are described in S-100 Part 10b.

2.6.6 Dataset units

The depth, height and positional uncertainty units in a dataset must be metres.

2.6.7 Dataset Coverage

Marine Protected Areas datasets are spatially limited.

In areas which include neighbouring producer nations, producing agencies should co-operate to agree on dataset boundaries and ensure no data overlap. Where possible, adjoining nations should agree on common data boundaries within a technical arrangement based on cartographic convenience and benefit to the mariner.

If an MPA feature extends outside the product coverage and the adjoining object does not exist, e.g. due to delay in the production of the neighbouring HO product, an indication should be placed at the outer edge of the product.

2.6.8 Overlaps

The **DataCoverage** features within a dataset must not overlap, however **DataCoverage** features from different datasets may overlap if they have differing maximum display scales or the datasets are for different ports.

MPA does not envisage multiple datasets for the same port, and does not anticipate overlapping datasets for a single port.

Overlapping datasets are possible in the case where there are two or more ports in close proximity (which may, for example, have overlapping approaches). In the latter case, consideration should be given to creating a single dataset that covers all the ports in the region in question, but overlapping datasets may be created as necessary. In case of overlapping datasets, the ECDIS should display an indicator and allow the user to select one dataset for display.

2.6.9 Feature Object Identifiers

Each feature and information instance within a dataset must have a unique universal Feature Object Identifier [FOID]. This is mapped to the gml:id attribute of the feature in the dataset (FOID and gml:id may not be identical due to XML restrictions on the format of gml:id attributes). Where a real-world feature has multiple geometric elements within a single dataset due to the dataset scheme, the same FOID may be used to identify multiple instances of the same feature. Since gml:id attributes in the same file must be unique, the mapping between FOID and gml:id must allow for a one-to-many mapping if needed. Features within a dataset may carry multiple geometries. Features split across multiple datasets may be identified by the same FOID. Features repeated in different scale ranges may be identified by the same FOID. FOID must not be reused, even when a feature has been deleted. However, the same feature can be deleted and added again later using the same FOID.

NOTE (1) (informative): The current format of FOID is defined in S-101 as a concatenation of subfields Producing Agency, Feature Identification Number and Feature Identification Subdivision. The identifier is currently formatted as a string value. The identifier may eventually be replaced with an identifier adhering to the scheme for Maritime Resource Names (MRN) which is based on the format of URNs.

NOTE (2): S-122 uses gml:id as a proxy representing FOID. S-122 does not define a rule for the structure or generation of gml:id values or their relation to identifiers in S-57, S-101 or other sources. Producers may generate gml:id values according to any desired scheme or schemes

2.6.10 180° Meridian of Longitude

Datasets must not cross the 180° meridian of longitude.

2.7 Geographic names

2.7.1 Feature names

If it is required to encode an international or national geographic name, it must be done using complex attribute *featureName*.

If it is required to encode a geographic name for which there is no existing feature, an appropriate area feature must be created. In order to minimise the data volume, these features should, where possible, use the geometry of existing features.

Geographic names should be encoded with the complex attribute *featureName*. The complex attribute *featureName* consists of the simple sub-attributes *language*, *name* and a usage attribute to indicate the intended display use of that particular name.

National geographic names can be left in their original national language in a non-English iteration of the complex attribute *featureName* (but only if the national language can be expressed using lexical level 0 or 1), or transliterated or transcribed and used in an English iteration of the complex attribute *featureName*, in which case the national name should be populated in an additional iteration of the *featureName* with sub-attribute language populated with the relevant national language value in accordance with ISO 639-2/T.

All area and point features within a Marine Protected Areas product should be encoded using *featureName* if a name is available.

A group of features, associated with a particular geographic name, should have the name encoded using *featureName* on an aggregation feature (of type surface or point, or no geometry, as appropriate). The name should not be encoded on the individual hydrographic features.

A group of service or forecast areas with the same attribute values associated with the same name should be encoded as spatial attributes of the same feature (so there would be only one feature with multiple spatial attributes for location).

Named features listed in Hydrographic Office's Sailing Directions or other documents that may assist in locating service information should be encoded using feature name on the relevant feature (e.g., **VesselTrafficServiceArea**). In all instances, if the exact extent of the feature to be named is known, a feature must be created. If the exact extent is not known, or the area is too small, an existing or specifically encoded point feature should be used to encode the geographic name.

2.7.2 Text placement

The cartographic feature **TextPlacement** (see clause 7.2) is used specifically to place text cartographically. The properties of the **TextPlacement** feature are described as follows:

- Geometry (point) – the spatial point location of the text string.
- text type – the classification of the text being placed based on attribution of the target feature(s) (mandatory).
- text offset bearing and text offset distance – the bearing and distance (in millimetres in the ECDIS display) used to position the text relative to the feature.

The **TextPlacement** feature is associated to the feature which carries the text being placed. The mandatory attribute text type identifies the text string(s) to be placed. The **TextPlacement** feature may provide functionality such that, as an ECDIS screen rotates from its optimum position in "north up" display mode (for example, if display is set to "course up") text can remain readable, or clear other important charted information.

The **TextPlacement** feature is associated to the feature which carries the text being placed. The attribute *textType* determines which text string is to be displayed if more than one is present. The **TextPlacement** feature ensures that as the screen rotates from "north up" (e.g. if display is set to "course up") text can remain readable, or clear other important charted information.

2.8 Scale policy

2.8.1 General policy

Marine Protected Areas data must be compiled in the best applicable scale.

2.8.2 Usage of scale attributes in displays

2.8.2.1 Scale attributes on domain features

The attributes *scaleMinimum* and *scaleMaximum* define the range of display scales within which features will be portrayed on the display if these scale minimum/maximum functions are enabled in the ECDIS or another GIS device. A geo feature with one or more spatial attributes can utilize the *scaleMinimum* and *scaleMaximum* attributes on the link to the spatial object (see the S-100 General Feature Model, S-100 Part 3, Figure 3-1 and 3-5.3.5). There are essentially two ways in which these attributes may be used.

- A producer may decide to use only a *scaleMinimum* value. This option is employed when the data producer wishes to turn off the display of a feature above certain scales. This is particularly useful in areas with high data density, and when it is expected that the data will be used a larger scale where data clutter might become an issue. Features are therefore encoded with an applicable value, which represents the scale at which the producer wishes to turn off the feature.
- A producer may decide to provide several pairs of *scaleMinimum* and *scaleMaximum* values. This decision may be based on the fact that for one particular feature different spatial instances in different scale ranges should be provided to supply this particular feature with more detailed geographic representation at larger scales.

An example can be a building which has two spatial objects associated, first one with only *scaleMinimum* value encoded at 21999, and the second spatial object encoded with *scaleMaximum* at 22000 and *scaleMinimum* encoded with 999999. These values would enable the use of a highly-detailed geometry at larger scales than 22000, and a less detailed geometry at scales of 22000 and less, while the building would be turned off at scales of 999999 and less.

A similar strategy can be followed to enable boundaries to conform to a scale-dependent geometry such as a coastline. Conformance at different scales can achieved by using minimum/maximum scales on spatial attributes to indicate which particular geometry should be used at a given scale.

NOTE: This edition of S-122 does not bind *scaleMaximum* as a feature attribute; it is, however, possible to specify both minimum and maximum scales on links to spatial objects as specified in the S-100 General Feature Model.

2.8.2.2 Scale attributes on DataCoverage features

The meta feature **DataCoverage** (clause 4.3) is used to provide ECDIS with the scale information needed for the determination of dataset loading and unloading in relation to the user-selected viewing scale of the ECDIS. The mandatory attribute *maximumDisplayScale* is used to indicate the largest intended viewing scale for the data. The mandatory attribute *minimumDisplayScale* is used to indicate the smallest intended viewing scale for the data.

All data within a dataset must have the same minimum display scale, but portions of a dataset can have a different maximum display scale, depending on the best scale required in an area for the operational purpose of the data.

S-122 does not prescribe specific values for *maximumDisplayScale* and *minimumDisplayScale*. Instead, producers should refer to the S-101 DCEG for values, and use values appropriate to the S-101 ENCs underlying the S-122 dataset.

2.8.3 Scale minimum values

The guidance in this clause pertains to *scaleMinimum* attributes bound to certain feature types in the S-122 feature catalogue.

Scale minimum values must be chosen from the list in S-101 to ensure visual compatibility between comparable underlying S-101 ENCs and S-122 data products. The scale minimum values used in the actual comparable underlying ENCs should be used, and in case of differences with the list below, the values in the actual ENCs prevail. “Comparable” ENCs for the purpose of this requirement means ENCs containing **RestrictedArea** or **VesselTrafficServiceArea** features, since both S-101 and S-122 have these feature types (though not necessarily with the same attribute bindings).

Data producers should cooperate to determine a scale minimum policy that results in suitable and consistent display of trans-boundary features, especially trans-boundary protected areas and areas beyond national jurisdictions.

Table 2-11 — Scale minimum values

Scale
19999999
9999999
4999999
3499999
1499999
999999
699999
499999
349999
259999
179999
119999

Scale
89999
59999
44999
29999
21999
17999
11999
7999
3999
2999
1999
999

Scale minimum values should be chosen so as to avoid situations where a feature disappears and then reappears as the user zooms out on their ECDIS display. Scale minimum values should also be chosen so to avoid situations where protected areas disappear from part of the display. If additional guidance about the choice of scale minimum values is needed, producers should consult the guidance for scale minimum policy for ENC features in the S-101 DCEG.

Alignment of S-122 scale policies for datasets as well as comparable individual features to the scale policies of the underlying ENC(s) may be a viable strategy to avoid the above problems and achieve consistent display; however this Product Specification does not mandate such alignment. Data producers should use their discretion as to alignment with S-101 ENCs.

2.8.4 Scale policy for feature types

Unlike S-101, S-122 does not define scale minimum values or steps for individual geo feature types.

As in S-101, the *scaleMinimum* value on a **TextPlacement** cartographic feature must be less than or equal to that of the associated geo feature.

2.9 Masking

In order to best determine the appropriate level of masking required for an dataset, it is recommended that the dataset be viewed in an ECDIS. The masking policy in S-122 is the same as for ENC features, described in the S-101 DCEG (clause 2.5.10) and is summarised below.

The following sub-clauses describe scenarios where masking is recommended should be considered by compilers.

2.9.1 Surface features crossing ENC cell boundaries

When a single feature of type surface crosses the boundaries of adjoining datasets, mask the edge where it shares the geometry of the boundary in each dataset:

This allows the features to be displayed as a single feature of type surface rather than being divided at the cell boundary and having the representation of two separate features. Note that some production software will automatically truncate (mask) features at the cell boundary.

NOTE: Occasionally an edge of the boundary of an area actually coincides with the dataset boundary. Where this occurs and the production system applies automatic truncation (masking) of this edge, the compiler must “unmask” that edge so as to avoid the appearance of the area to be “open ended”.

Where features of type surface extend beyond the entire limit of data coverage for the dataset, all edges of these area features should be masked:

Where a cell contains an area of no data coverage and the production software applies automatic truncation (masking) of features extending beyond the limit of data coverage of the dataset, edges of area features extending beyond the internal limit of the area of no data coverage may need to be masked manually.

Table 2-12 lists those features of type surface that should have edges masked where the boundary of the area crosses or extends beyond the dataset limit or the area of data coverage of the dataset.

Table 2-12 — Features of which edges have to be masked when crossing the dataset boundary

Feature Type	Comment
MarineProtectedArea	
RestrictedArea	
VesselTrafficServiceArea	
InformationArea	

2.9.2 Surface features having ECDIS symbol pattern fill

Surfaces symbolised in ECDIS with a patterned fill, and for which the outer edge of the surface has no significance (or is subject to change or intermittent), may have the boundary of the surface masked to reduce screen clutter.

Compilers must take care that the surface is large enough at the optimum display scale of the data (and at smaller optimum display scales at which it is intended that the feature should be displayed) so that at least one pattern symbol is displayed in the area. If this is not the case, the boundary of the surface should not be masked. Alternatively, a point feature may be encoded instead of the surface feature. It may be useful to load and display the data in an ECDIS in order to assist with making decisions as to the best encoding option to adopt in individual circumstances.

2.9.3 Routeing measures – entrance and exit edges

The S-101 DCEG describes the masking policy for routeing measures such as Traffic Separation Schemes (TSS), Two-Way Routes and Deep Water Routes. Since S-122 does not include routeing measures, masking at the “ends” of routeing measures as described in S-101 is not applicable to S-122 data.

2.10 Linear surface features

If it is required to encode a linear feature when the only allowable primitive for the relevant feature type is surface (e.g. a service area along a track, or channel), a very narrow surface should be encoded. The suggested extent is 0.3mm wide at viewing scales (keeping in mind that S-100 permits different spatial attributes at different scales.) An edge of this surface should correspond to the position of the line. All other edges should be masked.

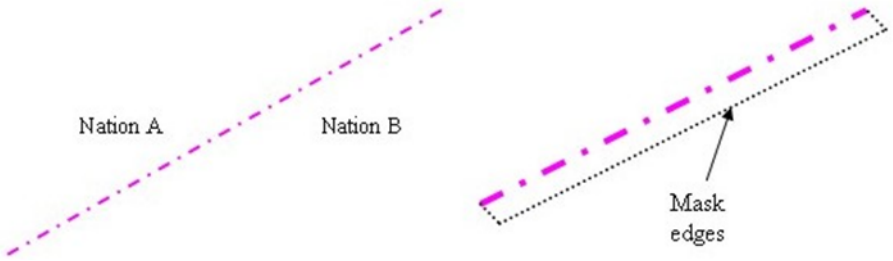

Figure 2-5 — Linear features

Page intentionally left blank

3 Description of table format for feature and information types

The tables describing feature and information types are based on the template below.

X.X Feature Name

IHO Definition: (Definition) (followed by Remarks if any)				
S-122 [Geo Feature/Information Type]: S-1XX Feature or Information Type (followed by (Abstract) if abstract type)				
Super Type: (supertype)				
Primitives: (allowed spatial primitives)				
Real World	*Paper Chart Symbol*	*ECDIS Symbol*		
(Reserved)	*(Reserved)*	*(Reserved)*		
S-1XX Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
(This section lists the allowabe **local** attributes)		(Allowed values for enumeration and codelist attributes)		
Inherited Attributes				
S-1XX Attribute	**Inherited From**		**Type**	**Multiplicity**
(attribute)	(supertype where defined)			

Feature/information associations (permitted associations)				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
Role of target class	Name of the Association if inherited, “(Inherited from …)”	Name of the target (the feature or information type referenced by the link)	aggregation / composition / aggregation	(How many target instances a single source instance can link to.)
If the association is listed in only one of the feature or information types in the asspciation, it means the association is unidirectional, that is, the binding for the association is only in one of the participating features or information types. This is sometimes the case for information associations that link a feature to an information type — the feature type has a binding to the information type, but not vice versa. Associations to or from any type are inherited by all sub-types of the type at any level unless explicitly prohibited in the relevant encoding instructions. Hyphens in roles and association names (camel-case codes) are only for document formatting and should be ignored for production purposes.				

INT 1 Reference (optional): The INT 1 location(s) of the Feature – by INT1 Section and Section Number.

(Encoding instructions are provided in sub-clauses following the table.)

X.X.X General

General guidance for encoding.

Introductory remarks. Includes information regarding the real world entity/situation requiring the encoding of the Feature in the product, and where required nautical cartographic principles relevant to the Feature to aid the compiler in determining encoding requirements.

X.X.X.X Sub-clause heading(s) (if needed) If applicable — S-4 reference

Additional encoding guidance relevant to the feature.

Clauses related to specific encoding scenarios for the Feature (if required).

X.X.X Remarks

Guidance for encoding specific attributes.

Remarks:

S-122 Attribute: Indentation of attributes indicates sub-attributes of complex attributes. Complex attributes may also be sub-attributes of complex attributes. Complex sub-attributes are generally not expanded to show their own sub-attributes, because expanding sub-attributes produces tables of inordinate length, but if expansion is done, further levels of indentation are applied to the sub-attributes. Inherited attributes are shown separately from locally defined attributes. Inherited complex attributes are not expanded to show their sub-attributes.

Allowable Encoding Value: For (EN) type attributes, the enumerates listed are only those allowable for the particular occurrence of the attribute relevant to the feature. Allowable values may vary for the attribute depending on the feature to which the attribute is bound. Such bindings are defined in the S-122 Feature Catalogue. The full list of enumerates that may be assigned to an attribute in S-122 can be found in the Simple Attributes section of the printed feature catalogue document.

Type: The prefix **C** indicates that the attribute is a complex attribute. Complex attributes are aggregates of other attributes that can be simple type or complex type. The prefix **S** indicates that the attribute is a sub-attribute of a complex attribute. Complex attributes that are sub-attributes of a complex attribute, and their sub-attributes, are indicated by indentation of the attribute name in the S-122 Attribute column.

Introductory clauses may depict associations using a UML diagram showing the relationships that apply to the class and its super-classes (generalizations). Relationships which are inherited from super-classes are shown by including the super-classes and their associations in the diagram.

The usual UML conventions apply. For explanations of standard UML notations, see S-100 Part 1.

Association ends and multiplicities: A lower bound of 0 in the multiplicity at any end of an association indicates only that the association is not mandatory for any particular instance of the feature at the other end (i.e., it is not mandatory for an instance of “that” feature type to have an association to a feature of “this” type). A lower bound of “1” means that if an instance of “that” type exists, it must be associated to an instance of “this” type. If the association is actually encoded then it amounts to saying that “this relationship exists between these two instances” and there must be an appropriate feature instance at both ends. Associations that are not mandatory should be encoded if and only if they convey useful information.

4 Meta-Features

4.1 Introduction

Meta-features are used to reduce the need to code quality and datum attributes in individual features, as well as to delimit the extent of data in the dataset. In a base dataset, some meta-features are mandatory (clause 4.2).

Horizontal and vertical uncertainties that apply to the majority of features are encoded as attributes of one or more **QualityOfNonBathymetricData** features together covering the same extent as the spatial union of the **DataCoverage** features in the dataset. (Typically, there would be one **DataCoverage** feature and one **QualityOfNonBathymetricData** feature, having the same spatial extent.) Exceptional horizontal and vertical uncertainties are encoded in a **SpatialQuality** information type associated to particular spatial primitives.

4.2 Mandatory meta features

The mandatory meta features are:

- DataCoverage
- QualityOfNonBathymetricData

4.3 Data Coverage

IHO Definition: A geographical area that describes the coverage and extent of spatial objects.				
S-122 Geo Feature: DataCoverage				
Super Type:				
Primitives: surface				
Real World	*Paper Chart Symbol*	*ECDIS Symbol*		
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Maximum Display Scale			IN	1,1
Minimum Display Scale			IN	1,1
Optimum Display Scale			IN	0,1
Interoperability Identifier			URN	0,*

Inherited Attributes			
S-122Attribute	**Inherited From**	**Type**	**Multiplicity**
No inherited attributes			

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**

Feature associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**

4.3.1 General

The meta feature **DataCoverage** encodes the area covered by the dataset. This feature is also used to provide the ECDIS with the scale information necessary for the determination of dataset loading and unloading in relation to the user selected viewing scale in the ECDIS.

DataCoverage features must cover at least the total extent of all geographic features in the dataset, and must not overlap.

4.3.2 Scale attributes

The use of S-122 data is scale-independent and minimum display scale will normally be (null) and maximum display scale 1000 (the extreme values in the table of scales in the S-101 ENC, see Table 2-11). Should a producer need to encode different maximum and minimum display scales from the extreme (i.e., create scale-dependent datasets), the values of maximum and minimum display scales should be harmonized with base layer S-101 datasets (see the S-101 DCEG).

The attribute *optimumDisplayScale* is used to indicate the intended viewing scale for the data. The value populated for *optimumDisplayScale*, therefore, provides a reference for the user selected viewing scale in the ECDIS at which the overscale warning will be displayed as the Mariner continues to zoom in if there is no larger optimum display scale MPA dataset available.

The mandatory attribute *maximumDisplayScale* is used to indicate the largest intended viewing scale for the data.

The mandatory attribute *minimumDisplayScale* is used to indicate the smallest intended viewing scale for the data. Where an empty (null) value is populated for *minimumDisplayScale*, the ECDIS will continue to display the data regardless of how small the user selected viewing scale becomes. The value populated for *minimumDisplayScale*, therefore, is intended to be used in a series of datasets covering a geographic area to determine the dataset rendering (display) priority as the user selected viewing scale becomes larger.

The values of maximum and minimum display scales should be harmonized with comparable base layer S-101 datasets (Table 4-1[1]). This serves to harmonize the loading strategy of MPA information with that for the underlying ENCs. However, use of the same values as S-101 datasets is not mandatory in S-122.

Table 4-1 — Maximum, optimum and minimum display scale values (from S-101 Ed. 2.0.0 DCEG)

maximum display scale	optimum display scale	minimum display scale
Any value	10,000,000	empty (null)
	3,500,000	10,000,000

[1] The current S-101 DCEG should be consulted to take into account any revisions to S-101 since the preparation of this Product Specification.

	1,500,000	3,500,000
	700,000	1,500,000
	350,000	700,000
	180,000	350,000
	90,000	180,000
	45,000	90,000
	22,000	45,000
	12,000	22,000
	8,000	12,000
	4,000	8,000
	3,000	4,000
	2,000	3,000
	1,000	2,000

NOTE: The selection of values for *maximumDisplayScale* and *minimumDisplayScale* for any selected *optimumDisplayScale* are at the discretion of the Data Producer. That is, any value listed for *maximumDisplayScale* and *minimumDisplayScale* above may be selected from any of the listed values, with the only restriction being that *maximumDisplayScale* must be a smaller value than/equal to *optimumDisplayScale* which must be a smaller value than *minimumDisplayScale* (or any value if *minimumDisplayScale* is populated with an empty (null) value).

Given that S-122 data will overlay ENC and possibly other datasets, the conditions described in the S-101 DCEG “Data Coverage” clause for displaying overscale warnings and setting the viewing scale may be overridden by interoperability constraints or the presence of higher-priority datasets. The specification of such behaviour is out of scope for this document (the S-100 interoperability specification should address it for ECDIS).

4.3.3 Number of feature instances

There must be a minimum of one **DataCoverage** feature in a dataset.

Typically, only a single **DataCoverage** feature should be used in a dataset. However, if the *optimumDisplayScale* is different for discrete areas within a single dataset, this must be indicated by encoding separate, non-overlapping **DataCoverage** features, each having a different value populated for *optimumDisplayScale*. Producing Authorities are to note, however, that excessive use of multiple **DataCoverage** features having different values of *optimumDisplayScale* within a single dataset should be avoided. Where different values of *optimumDisplayScale* are used, this should be restricted only to data compiled in order to achieve the intended navigational usage for the entire dataset.

4.3.4 Compatibility of scale values

Datasets must have the same value for *minimumDisplayScale* for all **DataCoverage** features in the dataset. Datasets may have different values populated for *maximumDisplayScale* for the **DataCoverage** features in the dataset; these values are typically populated as the value corresponding to 2 x the scale (or half the denominator) value populated for *optimumDisplayScale*, but are at the discretion of the data producer. For example, the value for *maximumDisplayScale* may be set to the same value as *optimumDisplayScale* to have the “grossly overscaled” warning appear at any larger user selected viewing scale than *optimumDisplayScale*; or populated as the value corresponding to the *minimumDisplayScale* value for the next largest scale dataset(s) in the ENC portfolio.

Where a series of differing *optimumDisplayScale* datasets are compiled covering the same geographic area, the smallest scale value populated for *optimumDisplayScale* for **DataCoverage** feature(s) in the dataset should correspond to the *minimumDisplayScale*, where populated, for the next largest *optimumDisplayScale* dataset. The largest scale value populated for *optimumDisplayScale* for

DataCoverage feature(s) in the dataset must not be a larger scale value than the *optimumDisplayScale* for the next largest *optimumDisplayScale* dataset, where such a dataset exists.

4.3.5 Remarks

- This meta feature is intended to support an indication of coverage and facilitate the loading and rendering (display) of datasets in the end-user system.
- S-122 being intended as an overlay for S-101 data, an S-122 dataset, when initially loaded, will be displayed in the ECDIS at the current setting for display scale.
- If the dataset is not loaded an an overlay, and more than one **DataCoverage** feature exists for a dataset, the dataset, when initially loaded, will be displayed in the ECDIS at a display scale corresponding to the largest scale value populated for *optimumDisplayScale*.
- Where a dataset consists of only one **DataCoverage** feature, the value for the *maximumDisplayScale* populated in the dataset discovery metadata must be the same as the value populated for *maximumDisplayScale* on the **DataCoverage**.
- For any **DataCoverage** feature, *maximumDisplayScale* < *minimumDisplayScale*.
- Except for the largest scale dataset coverage, datasets with multiple **DataCoverage** features must not have excessive differences in the values populated for *optimumDisplayScale* between the **DataCoverage** features. Typically, this should be interpreted as there being no more than one scale step value as defined in Table 4-1 above between the optimum display scale values in a single dataset.

4.4 Quality of Non-Bathymetric Data

IHO Definition: An area within which a uniform assessment of the quality of the non-bathymetric data exists.				
S-122 Geo Feature: QualityOfNonBathymetricData				
Super Type:				
Primitives: surface				
Real World	*Paper Chart Symbol*	*ECDIS Symbol*		
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Temporal Variation		1: Extreme Event 4: Likely to Change 5: Unlikely to Change 6: Unassessed	EN	0,1
Horizontal Distance Uncertainty			RE	0,1
Horizontal Position Uncertainty			C	0,1
Uncertainty Fixed			(S) RE	1,1
Uncertainty Variable Factor			(S) RE	0,1
Orientation Uncertainty			RE	0,1
Interoperability Identifier			URN	0,*
Source Indication			C	0,1
Category of Authority		2: Border Control 3: Police 4: Port 5: Immigration 6: Health 7: Coast Guard 8: Agricultural	(S) EN	0,1

		9: Military 10: Private Company 11: Maritime Police 12: Environmental 13: Fishery 14: Finance 15: Maritime 16: Customs		
Country Name			(S) TE	0,1
Source			(S) TE	0,1
Source Type		1: Law or Regulation 2: Official Publication 7: Mariner Report, Confirmed 8: Mariner Report, Not Confirmed 9: Industry Publications and Reports 10: Remotely Sensed Images 11: Photographs 12: Products Issued by HO Services 13: News Media 14: Traffic Data	(S) EN	0,1
Reported Date			(S) TD	0,1
Feature Name			(S) C	0,*
Survey Date Range			C	0,1
Date Start			(S) TD	0,1
Date End			(S) TD	1,1
Information			C	0,*
File Locator			(S) TE	0,1
File Reference			(S) TE	0,1
Headline			(S) TE	0,* (ordered)
Language			(S) TE	0,1
Text			(S) TE	0,1

Inherited Attributes			
S-122Attribute	**Inherited From**	**Type**	**Multiplicity**
No inherited attributes			

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**

Feature associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**

4.4.1 General

The meta feature **QualityOfNonBathymetricData** may be used to provide an indication of the overall uncertainty of position for all non-bathymetric features. It must not be used to provide the uncertainty of bathymetric information.

The attribute *horizontalPositionUncertainty* may be applied to any spatial type, in order to qualify the location of a feature.

HorizontalDistanceUncertainty and horizontal position uncertainty must not be applied to the spatial type of any geo feature if they are identical to the horizontal distance uncertainty and position uncertainty values of the underlying meta feature.

Position uncertainty on the **QualityOfNonBathymetricData** applies to non-bathymetric data situated within the area, while position uncertainty on the associated spatial types qualifies the location of the **QualityOfNonBathymetricData** feature itself.

4.4.2 Remarks

- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.
- If the complex attribute *information* is present at least one of its sub-attributes must be populated.

5 Abstract Geo Features

5.1 Introduction

This clause describes the single abstract feature type in S-122. The abstract type cannot be used directly, but define attributes and associations inherited by their sub-types. The encoding remarks in the description of each abstract feature apply to its sub-types but may be overridden by remarks in the sub-type.

The abstract feature types are depicted in Figure 5-1. In S-122 there is a single abstract feature type named **FeatureType**, from which all feature types except cartographic and meta-features inherit several attributes. This means that any Geo feature in S-122 can have any of the several attributes in the **FeatureType** box. This type also has information associations to three information types, and a feature association to **TextPlacement** which, as for attributes, allows any S-122 Geo feature to have the same associations.

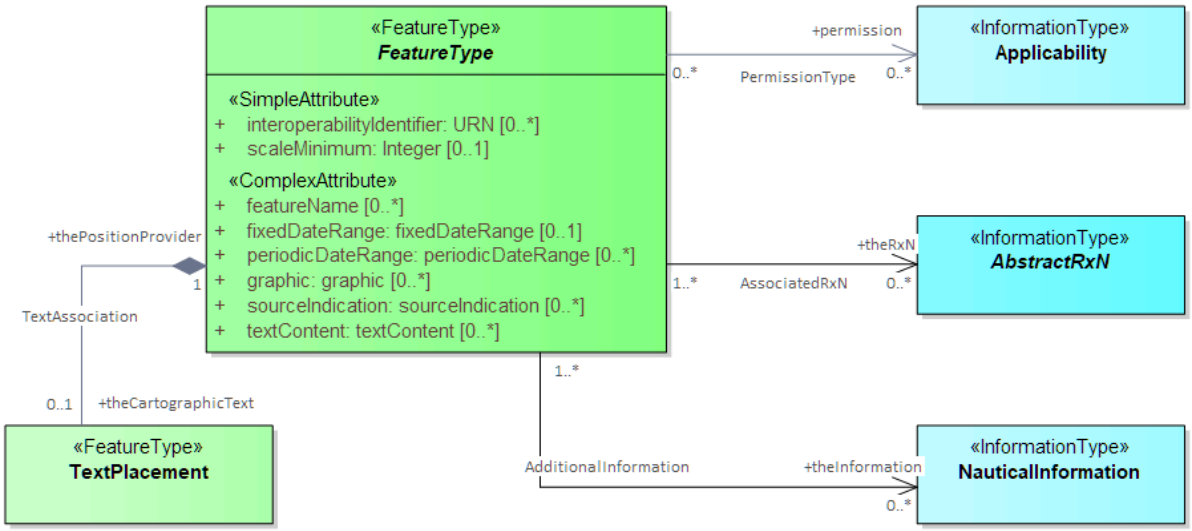

Figure 5-1 — Abstract feature hierarchy

Cartographic and meta-features are not derived from this abstract type and do not inherit its attributes and associations.

5.2 Feature Type

IHO Definition: Generalized feature type which carries all the common attributes.				
S-122 Geo Feature: FeatureType (Abstract)				
Super Type:				
Primitives: noGeometry				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Interoperability Identifier			URN	0,*
Feature Name			C	0,*
Language			(S) TE	1,1

Name			(S) TE	1,1
Name Usage		1: Default Name Display 2: Alternate Name Display 3: No Chart Display	(S) EN	0,1
Fixed Date Range			C	0,1
Date Start			(S) TD	0,1
Date End			(S) TD	0,1
Periodic Date Range			C	0,*
Date Start			(S) TD	1,1
Date End			(S) TD	1,1
Scale Minimum			IN	0,1
Graphic			C	0,*
Pictorial Representation			(S) TE	1,*
Picture Caption			(S) TE	0,1
Source Date			(S) DA	0,1
Picture Information			(S) TE	0,1
Bearing Information			(S) C	0,1
Source Indication			C	0,*
Category of Authority		2: Border Control 3: Police 4: Port 5: Immigration 6: Health 7: Coast Guard 8: Agricultural 9: Military 10: Private Company 11: Maritime Police 12: Environmental 13: Fishery 14: Finance 15: Maritime 16: Customs	(S) EN	0,1
Country Name			(S) TE	0,1
Source			(S) TE	0,1
Source Type		1: Law or Regulation 2: Official Publication 7: Mariner Report, Confirmed 8: Mariner Report, Not Confirmed 9: Industry Publications and Reports 10: Remotely Sensed Images 11: Photographs 12: Products Issued by HO Services 13: News Media 14: Traffic Data	(S) EN	0,1
Reported Date			(S) TD	0,1

Feature Name			(S) C	0,*
Text Content			C	0,*
Category of Text		1: Abstract or Summary 2: Extract 3: Full Text	(S) EN	0,1
Information			(S) C	0,*
Online Resource			(S) C	0,1
Source Indication			(S) C	0,*
Inherited Attributes				
S-122Attribute	**Inherited From**		**Type**	**Multiplicity**
No inherited attributes				

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
permission	PermissionType	Applicability	association	0,*
theRxN	AssociatedRxN	AbstractRxN	association	0,*
theInformation	AdditionalInformation	NauticalInformation	association	0,*

Feature associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
theCartographicText	TextAssociation	TextPlacement	association	0,1

5.2.1 General

Where a complex attribute has all its sub-attributes optional (e.g., multiplicity 0..1 or 0..*), at least one of the sub-attributes must be populated if the complex attribute is present.

The *featureName* attribute in complex attribute *sourceIndication* is intended for the name of the source.

The **AdditionalInformation** association to a **NauticalInformation** object can be used to attach an additional chunk of information to an information type, and there is no applicable specific information type or association. This should be used sparingly if at all.

The **PermissionType** association is used to encode permission information (e.g., whether use or entry is prohibited, etc) for vessels with different characteristics, if such permissions or requirements exist for a feature.

The **AssociatedRxN** association allows (mostly) textual information pertaining to regulations, etc., to be associated to features.

As an abstract type, instances of **FeatureType** cannot be directly encoded in datasets. However, the encoding instructions for this type apply to all its sub-types unless explicitly overriden in the encoding instructions for any particular sub-type.

5.2.2 Remarks

- Regulations, recommendations, restrictions, or general nautical information must be encoded in the appropriate associated information type (see clauses 8.2 and 9). The ability to encode *textContent* as an attribute of features must not be used to avoid encoding instances of **Regulations**, **Restrictions**, **Recommendations**, or **NauticalInformation**, because encoding the same type of information using

different methods or different structures in the same dataset or data product makes it more difficult for the mariner to find information.

- When encoding text information in the complex attribute *textContent*, it is not necessary to encode the entire content in a single instance of the information sub-attribute. Instead, the information should be organized so that each instance of information deals with a distinct topic or sub-topic, each with an appropriate heading in the *headline* attribute. This will make it easier for readers to find a topic. Part, chapter, section and sub-section headings in the source material may be used in either verbatim or condensed form, ordered according to the hierarchy in the source.
- Multiple instances of *textContent* should be used when the encoded material bears different relationships to the source (abstract/extract vs. summary vs. full text).
- Multiple instances of *textContent* may be used to distinguish information available purely as an external reference (in the *onlineResource* sub-attribute) from information encoded within the dataset (in the *information.text* sub-attribute or in a support file).
- In general, encoders may use the multiplicities of *textContent* and its sub-attributes to organize textual information so as to facilitate structuring text by topic, avoid flooding end-user screens with large blocks of unorganized text, and improve its accessibility to the mariner.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.

6 Domain Features

6.1 General

6.1.1 Spatial primitives

Encoding MPA features as surfaces (polygonal areas) in preference to points or curves is preferred, as being more useful for route planning and route monitoring applications. Restrictions apply over a defined spatial extent which requires a polygon. On the other hand, a small-scale dataset covering large regions and not intended for navigation may find it more efficient to encode the nominal locations of protected areas as curves or simple bounding box polygons (in the absence of point primitives as a permitted spatial primitive).

6.1.2 Purpose-guided encoding of MPA data

Data producers are reminded that S-122 is intended to act as an overlay to S-101 data and should supplement the information provided in underlying S-101 datasets. Consideration may need to be given to the scale ranges of underlying S-101 datasets and the effects on the visibility or otherwise of protected areas on-screen in determining whether an S-122 dataset should be developed.

Decisions about what information should be included in the dataset should be guided by the intended purpose of the dataset.

- For the IHO use case, when the dataset is intended to be used in route monitoring or route planning, encoding restrictions that bear directly on navigation(entry, speed, anchoring) and on-water activities (types of fishing, discharge) is critical.
  - Restrictions should be mapped to the listed values in available enumeration and codelist attributes.
  - A reference to the original regulation (where an online source is available) may be provided but mariners cannot be realistically expected to read the regulations in full.
  - Text-heavy encodings should be minimised. This means making a judicious use of support files and limiting the length of text provided in support files.
- For broader application of S-122 where encoding more detailed management would be required, extra detail can be provided and more liberal use can be made of support files.

6.2 Information Area

IHO Definition: An area for which general information regarding navigation, but not directly related to safety of navigation, is available.				
S-122 Geo Feature: InformationArea				
Super Type: FeatureType				
Primitives: surface				
Real World	*Paper Chart Symbol*	*ECDIS Symbol*		
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Relationship		1: Prohibited 3: Permitted	EN	1,1
Action or Activity		17: Ballast Water Exchange	CL	1,1

Inherited Attributes			
S-122Attribute	**Inherited From**	**Type**	**Multiplicity**
Interoperability Identifier	FeatureType	URN	0,*
Feature Name	FeatureType	C	0,*
Fixed Date Range	FeatureType	C	0,1
Periodic Date Range	FeatureType	C	0,*
Scale Minimum	FeatureType	IN	0,1
Graphic	FeatureType	C	0,*
Source Indication	FeatureType	C	0,*
Text Content	FeatureType	C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
permission	PermissionType (inherited from FeatureType)	Applicability	association	0,*
theRxN	AssociatedRxN (inherited from FeatureType)	AbstractRxN	association	0,*
theInformation	AdditionalInformation (inherited from FeatureType)	NauticalInformation	association	0,*

Feature associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
theCartographicText	TextAssociation (inherited from FeatureType)	TextPlacement	association	0,1

6.2.1 General

Ballast water management areas defined under the Ballast Water Management Convention can be encoded using **InformationArea** features. Areas where ballast water exchange is permitted can be encoded as features with *categoryOfRelationship* = *Permitted* and areas where ballast water exhange is prohibited can be encoded as features with *categoryOfRelationship* = *Prohibited*.

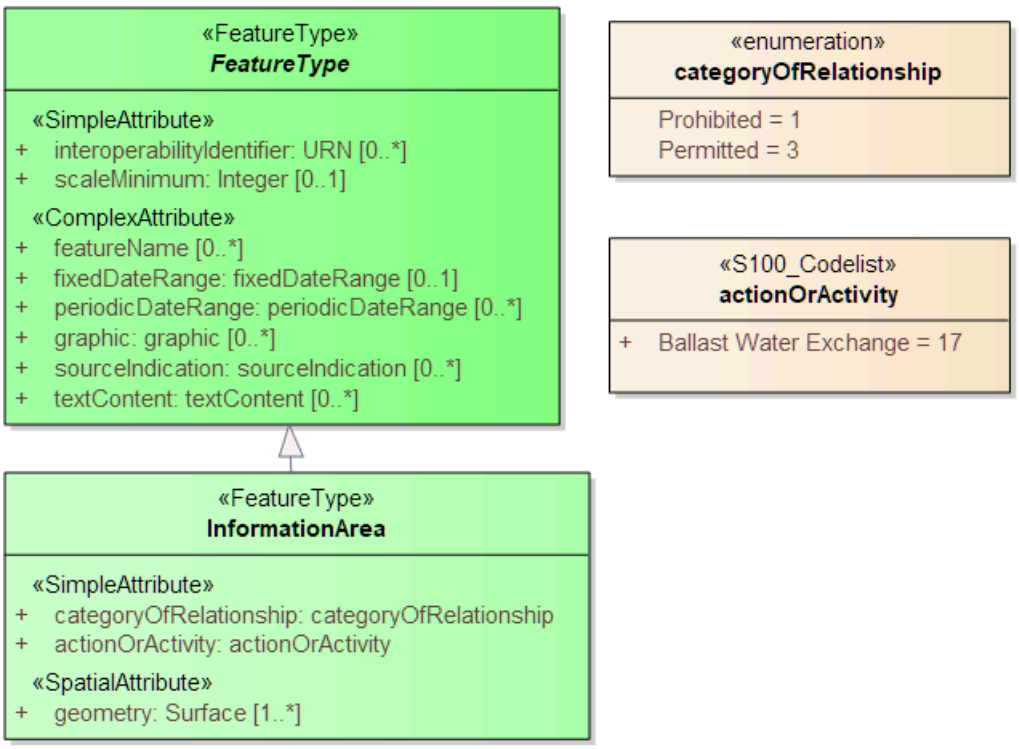

Figure 6-1 — Ballast water management

As a sub-type of **FeatureType**, the attributes and associations of **FeatureType** are inherited by **InformationArea** and available to the cartographer. For example, regulations may be encoded in a **Regulations** type referenced by the **InformationArea** feature instance.

Seasonal periodicity or fixed temporal limits may be encoded using *periodicDateRange* and *fixedDateRange*, which are inherited from **FeatureType**.

6.2.2 Remarks

- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.

6.3 Marine Protected Area

IHO Definition: Any area of the intertidal or sub-tidal terrain, together with its overlying water and associated flora, fauna, historical and cultural features, which has been reserved by law or other effective means to protect part or all of the enclosed environment.		
S-122 Geo Feature: MarineProtectedArea		
Super Type: FeatureType		
Primitives: curve surface		
Real World	*Paper Chart Symbol*	*ECDIS Symbol*

S-122 Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Marine Protected Area		1: IUCN Category Ia 2: IUCN Category Ib 3: IUCN Category II 4: IUCN Category III 5: IUCN Category IV 6: IUCN Category V 7: IUCN Category VI	CL	0,*
Category of Restricted Area		1: Offshore Safety Zone 4: Nature Reserve 5: Bird Sanctuary 6: Game Reserve 7: Seal Sanctuary 10: Historic Wreck Area 20: Research Area 22: Fish Sanctuary 23: Ecological Reserve 27: Environmentally Sensitive Sea Area 28: Particularly Sensitive Sea Area 31: Coral Sanctuary 32: Recreation Area 33: Ship Pollution Emission Control	EN	0,*
Jurisdiction		1: International 2: National 3: National Sub-Division	EN	0,1
Restriction		1: Anchoring Prohibited 2: Anchoring Restricted 3: Fishing Prohibited 4: Fishing Restricted 5: Trawling Prohibited 6: Trawling Restricted 7: Entry Prohibited 8: Entry Restricted 9: Dredging Prohibited 10: Dredging Restricted 11: Diving Prohibited 12: Diving Restricted 13: No Wake 14: Area To Be Avoided 15: Construction Prohibited 16: Discharging Prohibited 17: Discharging Restricted 18: Industrial or Mineral Exploration/Development Prohibited 19: Industrial or Mineral Exploration/Development Restricted 20: Drilling Prohibited 21: Drilling Restricted 22: Removal of Historical Artefacts Prohibited 23: Cargo Transhipment (Lightening) Prohibited 24: Dragging Prohibited 25: Stopping Prohibited	EN	0,*

		26: Landing Prohibited 27: Speed Restricted 38: Use of Spuds Prohibited 39: Swimming Prohibited 40: SOx Emission Restricted 41: NOx Emission Restricted 42: Power-Driven Vessels Prohibited		
Status		1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 9: Mandatory 18: Existence Doubtful 28: Buoyed 13: Historic 14: Public 18: Existence Doubtful 28: Buoyed	EN	0,*
Designation			C	0,*
Designation Scheme			(S) TE	0,1
Designation Identifier			(S) TE	0,1
Jurisdiction		1: International 2: National 3: National Sub-Division	(S) EN	0,1
Text			(S) TE	0,1
Inherited Attributes				
S-122Attribute	**Inherited From**		**Type**	**Multiplicity**
Interoperability Identifier	FeatureType		URN	0,*
Feature Name	FeatureType		C	0,*
Fixed Date Range	FeatureType		C	0,1
Periodic Date Range	FeatureType		C	0,*
Scale Minimum	FeatureType		IN	0,1
Graphic	FeatureType		C	0,*
Source Indication	FeatureType		C	0,*
Text Content	FeatureType		C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
responsibleAuthority	ProtectedAreaAuthority	Authority	association	0,*
permission	PermissionType (inherited from FeatureType)	Applicability	association	0,*

Information associations				
theRxN	AssociatedRxN (inherited from FeatureType)	AbstractRxN	association	0,*
theInformation	AdditionalInformation (inherited from FeatureType)	NauticalInformation	association	0,*

Feature associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
theCartographicText	TextAssociation (inherited from FeatureType)	TextPlacement	association	0,1

6.3.1 General

The Marine Protected Area *categoryOfMarineProtectedArea* attribute is an open enumeration codelist whose 'standard' values are specified by the IUCN categorization.

National or other categorisations may be encoded using the *designation* complex attribute.

If there is a formal categorisation according to a defined scheme under which unique identifiers are assigned to protected areas, the name or other official identifer of the scheme must be populated in the sub-attribute *designationScheme* and the identifier of the area under the scheme must be populated in the sub-attribute *designationIdentifier*.

If there is no defined scheme, the *designation.text* attribute must be populated with an identifier for the protected area.

Encoding the applicable restrictions in a *restriction* attribute is strongly recommended. If the allowed values of the *restriction* attribute are considered insufficient to provide necessary details, a **Restrictions** information type should be created and linked to the **MarineProtectedArea** feature.

Duplicative encoding in **MarineProtectedArea** and **RestrictedArea** features should be avoided. If the information to be encoded in the two is exactly the same, the boundaries are coincident, and each feature type can encode all the available information, encoding as a **MarineProtectedArea** should be preferred.

Constraints on navigation and human activities as restricted, allowed, prohibited, or unspecified should be mapped to the S-122 enumerated restrictions and action categories. Short text summaries of the key regulations should be provided if the content cannot be directly mapped to the enumerated categories.

6.3.2 Marine management areas

Marine management areas should be encoded using *designation* with:

- *designation.designationScheme* = (type of management area)
- *designation.designationIdentifier* = (Site code if assigned by designating authority) or *designation.text* = (geographic identifier)
- *designation.jurisdiction* = (according to designating government authority or agency)

If the area has been given a name by the designating authority but no other identifier, the name may be encoded in *designation.text* with "unspecified".

This method can be used to encode marine mammal management areas and similar areas designated for protection of animals or environment.

If the protection or designation is seasonal or otherwise time-dependent, the dates between which the protection is operative should be encoded using *fixedDateRange* or *periodicDateRange* complex attribute. NOAA slow zones or dynamic management areas may be thus indicated, since the designation is for a specified period.

For example:

- *designation.designationScheme* = Seasonal Management Area
- *designation.text* = Migratory Route and Calving Grounds — Ports of New York/New Jersey

- *designation.jurisdiction* = National

6.3.3 Natura 2000 sites

Natura 2000 is a network of protected areas on land and at sea designated under the European Union Birds and Habitats Directives. Descriptive lists are maintained by national authorities and a European database are based on the information that national authorities have submitted to the European Environment Agency, for each of the Natura 2000 sites. The list is supposed to be updated annually and each annual list may be identified by a citation identifier. Natura 2000 sites should be encoded with:

- *designation.designationScheme* = Natura2000 <optional citation identifier>
- *designation.designationIdentifier* = (Natura 2000 site code)
- *designation.jurisdiction* = (of designating authority — *National* if the designation orignates from a national government)

For example:

- *designation.designationScheme* = Natura2000 eea_t_natura2000_p_2023_v01_r00
- *designation.designationIdentifier* = NL2008001
- *designation.jurisdiction* = National

6.3.4 Other Effective Area-Based Conservation Measures

Other Effective Area-Based Conservation Measures (OECMs) should use

- *designation.designationScheme* = Other Effective Area-Based Conservation Measure
- *designation.designationIdentifier* = (Site code if assigned) or *designation.text* = (name oe geographic identifier)
- *designation.jurisdiction* = (according to designating government authority or agency)

6.3.5 Ecosystem Protection Zones

Ecosystem Protection Zones (EPZ) may be populated using complex attribute *designation* with:

- *designation.designationScheme* = Ecosystem Protection Zone
- *designation.designationIdentifier* = (Site code if assigned by designating authority) or *designation.text* = (name oe geographic identifier)
- *designation.jurisdiction* = (according to designating government authority or agency)

6.3.6 Restrictions and regulations for protected areas

Navigation within marine protected areas can be limited by regulations/restrictions and recommendations. That information is usually provided by relevant authorities. If the *restriction* attribute suffices to encode such information, it must be used; otherwise an appropriate **Restrictions**, **Regulations**, **Recommendations**, or **NauticalInformation** information type can be associated to the **MarineProtectedArea**.

Information that is conditional on vessel characteristics may be encoded using the **PermissionType** association to an information type that defines the set of vessels to which the conditions apply. (See the S-122 Product Specification and clauses 11 and 9 of this DCEG for more information about coding such conditions.)

6.3.7 Jurisdiction information

The *jurisdiction* attribute describes the type of jurisdiction over the area (national, sub-national, or international). It should be used to describe the type of authority or type of protection scheme under which the location was designated as a protected area.

The attribute *jurisdiction* is an optional attribute which is available in two places (as a direct local attribute of **MarineProtectedArea** and as a sub-attribute of *designation*). It should be populated either directly in **MarineProtectedArea** if the complex attribute *designation* is not present, or as a sub-attribute of *designation* when that complex attribute is present.

6.3.8 Remarks

- Multiple *categoryOfMarineProtectedArea* and/or *designation* attributes may be encoded in the same **MarineProtectedArea** feature if a site is protected under different protection schemes. The geometry must be the same under all schemes or authorities; if there is a difference under different schemes or

in the precise areas designated by different authorities, overlapping **MarineProtectedArea** features may be encoded.

- If complex attribute *designation* is present, one of sub-attributes *designationIdentifier* or *text* must be populated.
- If the complex attribute *designation* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.

6.4 Restricted Area

IHO Definition: A specified area designated by an appropriate authority within which navigation is restricted in accordance with certain specified conditions.				
S-122 Geo Feature: RestrictedArea				
Super Type: FeatureType				
Primitives: surface				
Real World	*Paper Chart Symbol*	*ECDIS Symbol*		
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Restricted Area		1: Offshore Safety Zone 4: Nature Reserve 5: Bird Sanctuary 6: Game Reserve 7: Seal Sanctuary 10: Historic Wreck Area 20: Research Area 22: Fish Sanctuary 23: Ecological Reserve 27: Environmentally Sensitive Sea Area 28: Particularly Sensitive Sea Area 31: Coral Sanctuary 32: Recreation Area 33: Ship Pollution Emission Control	EN	0,*
Restriction		1: Anchoring Prohibited 2: Anchoring Restricted 3: Fishing Prohibited 4: Fishing Restricted 5: Trawling Prohibited 6: Trawling Restricted 7: Entry Prohibited 8: Entry Restricted 9: Dredging Prohibited 10: Dredging Restricted 11: Diving Prohibited 12: Diving Restricted 13: No Wake 14: Area To Be Avoided 15: Construction Prohibited 16: Discharging Prohibited	EN	1,*

		17: Discharging Restricted 18: Industrial or Mineral Exploration/Development Prohibited 19: Industrial or Mineral Exploration/Development Restricted 20: Drilling Prohibited 21: Drilling Restricted 22: Removal of Historical Artefacts Prohibited 23: Cargo Transhipment (Lightening) Prohibited 24: Dragging Prohibited 25: Stopping Prohibited 26: Landing Prohibited 27: Speed Restricted 38: Use of Spuds Prohibited 39: Swimming Prohibited 40: SOx Emission Restricted 41: NOx Emission Restricted 42: Power-Driven Vessels Prohibited		
Status		1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 9: Mandatory 18: Existence Doubtful 28: Buoyed 13: Historic 14: Public 18: Existence Doubtful 28: Buoyed	EN	0,*

Inherited Attributes

S-122Attribute	Inherited From	Type	Multiplicity
Interoperability Identifier	FeatureType	URN	0,*
Feature Name	FeatureType	C	0,*
Fixed Date Range	FeatureType	C	0,1
Periodic Date Range	FeatureType	C	0,*
Scale Minimum	FeatureType	IN	0,1
Graphic	FeatureType	C	0,*
Source Indication	FeatureType	C	0,*
Text Content	FeatureType	C	0,*

Information associations

S-122 Role	S-122 Association Name	Associated to	Type	Mult.
permission	PermissionType (inherited from FeatureType)	Applicability	association	0,*
theRxN	AssociatedRxN	AbstractRxN	association	0,*

Information associations				
	(inherited from FeatureType)			
theInformation	AdditionalInformation (inherited from FeatureType)	NauticalInformation	association	0,*

Feature associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
theCartographicText	TextAssociation (inherited from FeatureType)	TextPlacement	association	0,1

6.4.1 General

(see S-4 – B-431.4; B-435.7; B-435.11; B-437.1-7; B-439.2-4; B-445.9; B-448; B-448.1 and B-449.5)

There are many types of areas within which certain activities are discouraged or prohibited, or from which certain classes of vessels are excluded. The general term for all areas in which certain aspects of navigation may be restricted or prohibited by regulations is “Restricted Area”, or equivalent. The word “prohibited”, or its equivalent, may appear in terms relating to activities which are contrary to the regulations, e.g. “Anchoring Prohibited”, “Entry Prohibited”.

If it is required to encode a restricted area, it must be done using the feature **RestrictedArea**, or other features having the attribute *restriction* (specifically, **MarineProtectedArea**) if such other features are more representative of conditions in the area (that is, the reason for the restriction is because it is a marine protected area).

RestrictedAreas should be encoded in S-122 datasets only when they are one of the listed categories or otherwise related to marine protected areas, environmental, or wildlife protection.

Duplicative encoding in **MarineProtectedArea** and **RestrictedArea** features should be avoided. If all the available information can be encoded in a **MarineProtectedArea**, it is not necessary to encode a coincident **RestrictedArea**. A **RestrictedArea** feature overlaying a **MarineProtectedArea** feature should be encoded only if the boundaries and/or thematic attributes differ.

6.4.1.1 Nature reserves (see S-4 – B-437.3)

If it is required to encode a marine nature reserve area with navigational restrictions, it must be done using a **RestrictedArea** feature, with attribute *categoryOfRestrictedArea* = 4 (nature reserve).

6.4.1.2 Speed limits (see S-4 – B-430.2)

Speed is often limited inside MPAs in order to protect the species that inhabit the area. If it is required to encode this restriction, it must be done using a **RestrictedArea** feature, with the attribute *restriction* = 27 (speed restricted), with the speed limit and its unit of measurement encoded using an associated instance of the information type **Regulations**(see clause 9.4).

6.4.1.3 Areas to be avoided (see S-4 – B-435.7)

If it is required to encode an IMO designated Area to be Avoided, it must be done using a **RestrictedArea** feature, with attribute *restriction* = 14 (area to be avoided).

6.4.1.4 Environmentally Sensitive Sea Areas (see S-4 – B-437)

If it is required to encode an Environmentally Sensitive Sea Area with navigational restrictions, it must be done using a **Restricted Area** feature, with attribute *categoryOfRestrictedArea* = 27 (ESSA) or 28 (PSSA).

An Environmentally Sensitive Sea Area that is shown on the source as a point symbol should be encoded using a small surface **RestrictedArea** feature.

6.4.2 Remarks

- The attribute *categoryOfRestrictedArea* is used to describe the type of area, while the attribute *restriction* describes the restrictions.

- An associated instance of the information types **Restrictions**, **Regulations**, **Recommendations** and **NauticalInformation**, complex attributes text content sub-attribute *information* or solely attribute *information* may be used to provide an additional explanation about the restriction, where required.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.

6.5 Vessel Traffic Service Area

IHO Definition: The area of any service implemented by a relevant authority primarily designed to improve safety and efficiency of traffic flow and the protection of the environment. It may range from simple information messages, to extensive organisation of the traffic involving national or regional schemes.				
S-122 Geo Feature: VesselTrafficServiceArea				
Super Type: FeatureType				
Primitives: surface				
Real World	*Paper Chart Symbol*	*ECDIS Symbol*		
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Inherited Attributes				
S-122Attribute	**Inherited From**		**Type**	**Multiplicity**
Interoperability Identifier	FeatureType		URN	0,*
Feature Name	FeatureType		C	0,*
Fixed Date Range	FeatureType		C	0,1
Periodic Date Range	FeatureType		C	0,*
Scale Minimum	FeatureType		IN	0,1
Graphic	FeatureType		C	0,*
Source Indication	FeatureType		C	0,*
Text Content	FeatureType		C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
controlAuthority	ServiceControl	Authority	association	0,1
permission	PermissionType (inherited from FeatureType)	Applicability	association	0,*
theRxN	AssociatedRxN (inherited from FeatureType)	AbstractRxN	association	0,*
theInformation	AdditionalInformation (inherited from FeatureType)	NauticalInformation	association	0,*

Feature associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
theCartographicText	TextAssociation (inherited from FeatureType)	TextPlacement	association	0,1

6.5.1 General

Traffic control and other vessel traffic service areas are encoded in S-122 datasets only when they are related to protected area information in some way, for example if it is necessary to provide information about reporting entry into and departure from a protected area.

6.5.2 Remarks

- The *serviceAccessProcedure* and *requirementsForMaintenanceOfListeningWatch* attributes used in other product specifications are not available in S-122.
- Any similar requirements pertaining specifically to protected areas should be encoded in the *textContent* attribute. (There may be a general requirement in the applicable national shipping regulations instead of or in addition to a specific requirement.)
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.

7 Cartographic Features

7.1 Introduction

This product specification uses the **TextPlacement** cartographic features derived from S-101. The structure of the feature and its usage are the same as in S-101 but the feature specification in S-122 omits elements which are not relevant to marine harbour infrastructure, for example, 'light characteristic' is omitted as a listed value for the attribute *textType*.

7.2 Text Placement

IHO Definition: The Text Placement feature is used in association with the Feature Name attribute or a light description to optimize text positioning in ECDIS.				
S-122 Geo Feature: TextPlacement				
Super Type:				
Primitives: point				
Real World	*Paper Chart Symbol*	*ECDIS Symbol*		
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Text Offset Bearing			IN	1,1
Text Offset Distance			IN	1,1
Text Rotation			BO	0,1
Text Type		1: Name	EN	1,1
Scale Minimum			IN	0,1
Inherited Attributes				
S-122Attribute	**Inherited From**		**Type**	**Multiplicity**
No inherited attributes				

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**

Feature associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
thePositionProvider	TextAssociation	FeatureType	composition	1,1

7.2.1 General

If it is required to place text to improve clarity and readability of display, it must be done using the cartographic feature **TextPlacement**. In navigationally relevant areas such as shipping channels and dredged areas, where default ECDIS text positioning may cover other features, Data Producers should consider using **TextPlacement**. The **TextPlacement** feature must be associated with the relevant geo feature using the composition **TextAssociation** (see clause 13.1).

While the feature associations table above indicates the abstract type **FeatureType** as the target of feature associations from **TextPlacement**, the actual association in any **TextPlacement** instance should be a reverse link to the non-abstract feature type (sub-type of **FeatureType** at any level) which links to the particular **TextPlacement** instance.

NOTE: Where an associated instance of **TextPlacement** has not been related to a feature having the attribute name and/or the attributes associated with the characteristics of a feature populated, the text will be positioned in the ECDIS display in accordance with the default position for text strings defined in the Portrayal Catalogue.

7.2.2 Remarks

- The **TextPlacement** cartographic feature is used by the ECDIS to optionally position text in ECDIS, which has been populated using an attribute(s) for the associated feature. The attribute(s) is identified by populating the mandatory attribute *textType*.
- Where two instances of text type are populated for a **TextPlacement** instance[2], the feature name and characteristics as derived from the target feature attribution will be vertically aligned in the ECDIS display in accordance with the defined text offset bearing and distance. If it is required to position the feature name and the feature characteristics independently, this must be done by associating two instances of **TextPlacement**, one having *textType* = 1 (name) and the other having *textType* = 2 (feature characteristic), to the target feature. Note, however, that independent vertical or horizontal alignment of both the name and the characteristic of a feature is not recommended, as the text will overlap as the Mariner zooms to smaller scales than the optimum display scale for the data.
- The attributes *textOffsetBearing* and *textOffsetDistance* define the bearing (related to true north) and distance of the anchor point of the text, in millimetres in the ECDIS display, to be displayed from the associated feature. The values populated for these attributes must be determined based on the desired position of the text at the optimum display scale of the ENC data. Note that the attribute *textOffsetBearing* does not rotate the text itself, but determines the alignment of the anchor point (or justification) for the text location (horizontal (left, centred or right) and vertical (bottom, centre or top)) based on the encoded bearing. Displayed text will always appear horizontal regardless of the display mode set by the mariner (north-up or course-up), unless the Boolean attribute *textRotation* is set to True.
- The Boolean attribute *textRotation*, when populated as True, will rotate the text on the ECDIS display to align along the bearing populated for the attribute *textOffsetBearing*.
- Data Producers are advised to determine the best positioning for text at the optimum display scale for the data; and based on “north-up” ECDIS display.
  - While *textOffsetBearing*, *textOffsetDistance* and *textRotation* will position the text at the same location relative to the associated feature at all Mariner’s Selected Viewing Scales (MSVS), Data Producers are advised that, as the Mariner zooms out to smaller viewing scales, text may unintentionally cover other charted detail. Therefore, as an alternative, Data Producers may experiment with positioning the text so that it clears the majority of other charted features at the smallest scale at which the text is intended to be displayed, and populating the attribute *scaleMinimum* accordingly (see bullet below).
  - Data Producers are also advised that optimum results may not be achieved when the Mariner has set the display setting for the ECDIS to screen rotations other than “north-up”. Encoders should also consider the positioning of the name of a feature where the name is encoded in multiple languages, as the name displayed may be of varying character length based on the Mariner’s language settings (see clause 2.7.1).
- The attribute *scaleMinimum* (if permitted by the data format) may be used to determine a scale at which the text string is no longer visible in the ECDIS when *scaleMinimum* functionality is enabled. Where populated, the value for *scaleMinimum* on **TextPlacement** must not be set to a smaller scale value than the value populated for the associated feature.
- **TextPlacement** should normally be associated with features of type point, but may be used for features of type curve and surface.

[2] Retained for compatibility and potential future use, but not a practical possibility in this edition of the Product Specification because only the only re-positioned text in this edition of the Product Specification is the name.

8 Abstract Information Types

The abstract information types are depicted in Figure 8-1 below. At the root is the type named **InformationType**, from which all information types except **SpatialQuality** and **QualityOfNonbathymetricData** inherit several attributes. This means that any information type in S-122 except **SpatialQuality** can have any of the several attributes in the **InformationType** box. The information types **AbstractRxN** adds attributes and associations inherited by the four types **Regulations**, **Restrictions**, **Recommendations**, and **NauticalInformation**.

The abstract information type hierarchy in S-122 is harmonised with the abstract hierarchy in other nautical publications Product Specifications.

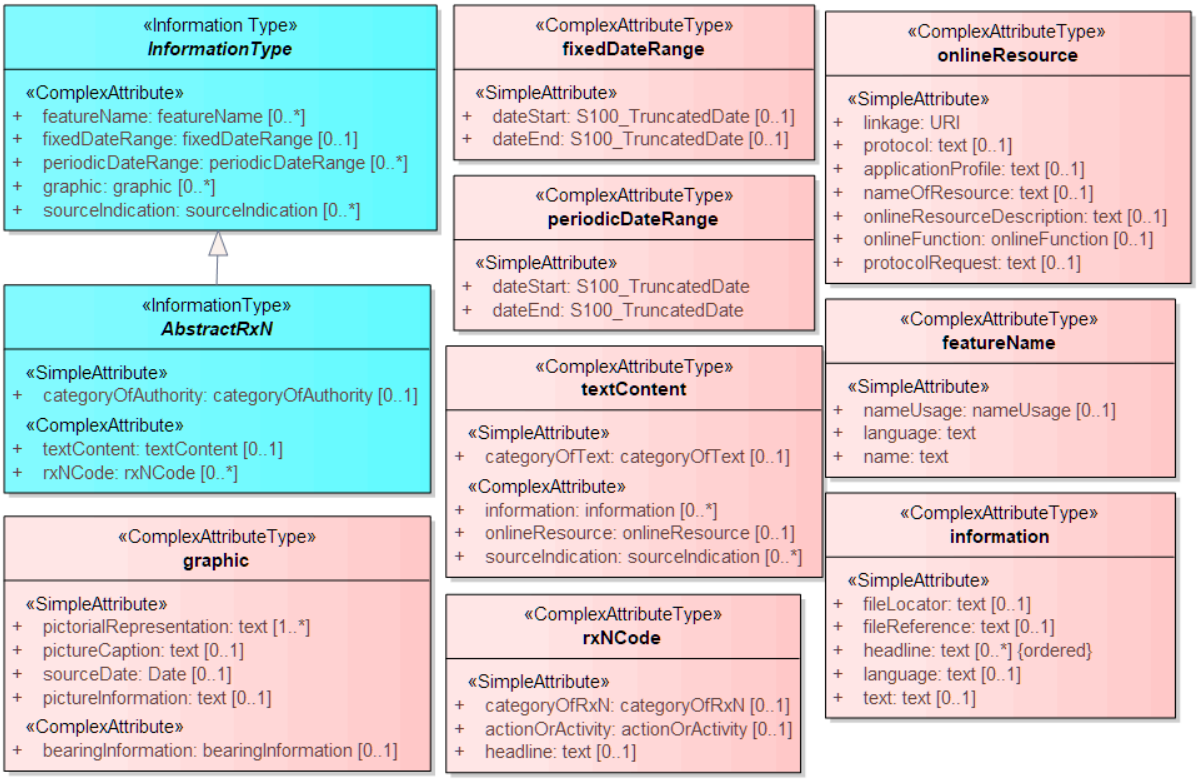

Figure 8-1 — Abstract Information Types

8.1 Information Type

IHO Definition: Generalized information type which carries all the common attributes.				
S-122 Information Type: InformationType (Abstract type)				
Super Type:				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Feature Name			C	0,*
Language			(S) TE	1,1
Name			(S) TE	1,1
Name Usage		1: Default Name Display 2: Alternate Name Display	(S) EN	0,1

		3: No Chart Display		
Fixed Date Range			C	0,1
Date Start			(S) TD	0,1
Date End			(S) TD	0,1
Periodic Date Range			C	0,*
Date Start			(S) TD	1,1
Date End			(S) TD	1,1
Graphic			C	0,*
Pictorial Representation			(S) TE	1,*
Picture Caption			(S) TE	0,1
Source Date			(S) DA	0,1
Picture Information			(S) TE	0,1
Bearing Information			(S) C	0,1
Source Indication			C	0,*
Category of Authority		2: Border Control 3: Police 4: Port 5: Immigration 6: Health 7: Coast Guard 8: Agricultural 9: Military 10: Private Company 11: Maritime Police 12: Environmental 13: Fishery 14: Finance 15: Maritime 16: Customs	(S) EN	0,1
Country Name			(S) TE	0,1
Source			(S) TE	0,1
Source Type		1: Law or Regulation 2: Official Publication 7: Mariner Report, Confirmed 8: Mariner Report, Not Confirmed 9: Industry Publications and Reports 10: Remotely Sensed Images 11: Photographs 12: Products Issued by HO Services 13: News Media 14: Traffic Data	(S) EN	0,1
Reported Date			(S) TD	0,1
Feature Name			(S) C	0,*

Inherited Attributes			
S-122 Attribute	**Inherited From**	**Type**	**Multiplicity**
No inherited attributes			

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**

8.1.1 General

Where a complex attribute has all its sub-attributes optional (e.g., multiplicity 0..1 or 0..*), at least one of the sub-attributes must be populated.

The *featureName* attribute of an instance of an information type can be used for a short title that is either a proper name (if such is relevant) or which describes the instance. For example, the *featureName* attribute of an **Authority** information type can be the name of a government agency.

The *featureName* attributes of information types should not duplicate the geographic feature name of an associated feature, but should pertain to the information instance itself.

The *featureName* attribute should be populated only if the value conveys useful information to the end user. Some examples of such situations are:

- providing the name of an organisation, such as the name of an **Authority**.
- distinguishing between instances – if multiple instances of the same information type are associated to the same feature type (or another information type), the different instances may be given descriptive names to make it easier for the mariner to distinguish their content.

Some information instances are associated to multiple features, in which case its name should be general enough to be relevant to all the features.

For example, if naming **Regulations** instances describing regulations, consider whether (for example) there is a general regulation applicable to all areas in a jurisdiction and an exceptional regulations object associated to a single area or a subset of areas in the jurisdiction. In this situation, the general regulations may be encoded with the name “General regulations for (feature type) Areas” and associated to several features, while a specific feature can also have a specific regulation whose name is “Special regulations for (named area)”.

8.1.2 Remarks

- If the complex attribute *bearingInformation* is present, at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

8.2 AbstractRxN

IHO Definition: An abstract superclass for information types that encode rules, recommendations, and general information in text or graphic form.				
S-122 Information Type: AbstractRxN (Abstract type)				
Super Type: InformationType				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Authority		2: Border Control 3: Police	EN	0,1

		4: Port 5: Immigration 6: Health 7: Coast Guard 8: Agricultural 9: Military 10: Private Company 11: Maritime Police 12: Environmental 13: Fishery 14: Finance 15: Maritime 16: Customs		
RxN Code			C	0,*
Category of RxN		1: Navigation 2: Communication 3: Environmental Protection 4: Wildlife Protection 5: Security 6: Customs 7: Cargo Operation 8: Refuge 9: Health 10: Natural Resources or Exploitation 11: Port 12: Finance 13: Agriculture	(S) CL	0,1
Action or Activity		1: Navigating With a Pilot 2: Entering Port 3: Leaving Port 4: Berthing 5: Slipping 6: Anchoring 7: Weighing Anchor 8: Transiting 9: Overtaking 10: Reporting 11: Working Cargo 12: Landing 13: Diving 14: Fishing 15: Discharging Overboard 16: Passing 17: Ballast Water Exchange 18: Hull Cleaning 19: Scientific Research 20: Tourism 21: Education 22: Infrastructure Maintenance	(S) CL	0,1
Headline			(S) TE	0,1
Text Content			C	0,*
Category of Text		1: Abstract or Summary 2: Extract 3: Full Text	(S) EN	0,1
Information			(S) C	0,*

Online Resource			(S) C	0,1
Source Indication			(S) C	0,*
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Feature Name	InformationType		C	0,*
Fixed Date Range	InformationType		C	0,1
Periodic Date Range	InformationType		C	0,*
Graphic	InformationType		C	0,*
Source Indication	InformationType		C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
isApplicableTo	InclusionType	Applicability	association	0,*
theOrganisation	RelatedOrganisation	Authority	association	0,*

8.2.1 General

AbstractRxN is the supertype of the four types intended primarily for encoding information from regulatory or other text sources. The attributes *categoryOfRxN* and *actionOrActivity* should be encoded wherever possible in order to allow software to classify the content according to the type of regulation (*categoryOfRxN*) and its effects on common maritime activities by both commercial and recreational vessels.

The complex attribute *rxNCode* can be used to classify regulations (or recommendations, etc.) according to their principal subject (sub-attribute *categoryOfRxN*) and the type of vessel activity affected (sub-attribute *actionOrActivity*), as well as provide a sequence of brief topic headings (sub-attribute *headline*). The *rxNCode* attribute is intended to be used to allow mariners to obtain information relevant to particular subjects or to particular kinds of vessel operations.

- As an abstract type, instances of **AbstractRxN** cannot be directly encoded in datasets. However, the encoding instructions for this type apply to all its sub-types unless explicitly overriden in the encoding instructions for any particular sub-type.

8.2.2 Remarks

- At least one of the attributes *textContent* and *graphic* must be populated. Populating both is permitted.
- If the complex attribute *rxNCode* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

Page intentionally left blank

9 Textual Regulations

9.1 Introduction

The information types **Regulations**, **Restrictions**, **Recommendations**, and **NauticalInformation** all inherit the attributes of their immediate abstract superclass **AbstractRxN**, which provides attributes *textContent* and *graphic* for textual and pictorial material respectively. The sub-attributes of its complex attribute *rxnCode* allow optional classification of the material encoded in *textContent* or *graphic* according to the type of material and the kind of nautical activity affected by it. The classifications in *rxNCode* sub-attributes *categoryOfRxN* and *actionOrActivity* are intended to facilitate software queries for information, while the sub-attribute *headline* provides additional topic headings for subject matter.

These four information types also inherit the attributes of abstract superclass InformationType, which allows encoding of the effective and expiry dates, if any, and the source of information , if it is necessary to encode that data.

The content of the regulation (recommendation, etc.) should be encoded in the *textContent* attribute, which is also inherited from the abstract superclass **InformationType**. It may be encoded inline (*textContent.information.text*) or in an external file (*textContent.information.fileReference*) depending on its length, on whether it is unique to the feature instance, and on whether the producer decides to include a support file containing multiple sections referenced from different places in the dataset. (See also clauses 2.4.8 and 2.4.9 for general guidance on encoding textual information.)

These four information types are intended primarily for encoding textual information, such as that which derives from textual source material such as port handbooks, national or local laws or official publications.

The four types for textual information are depicted in Figure 9-1

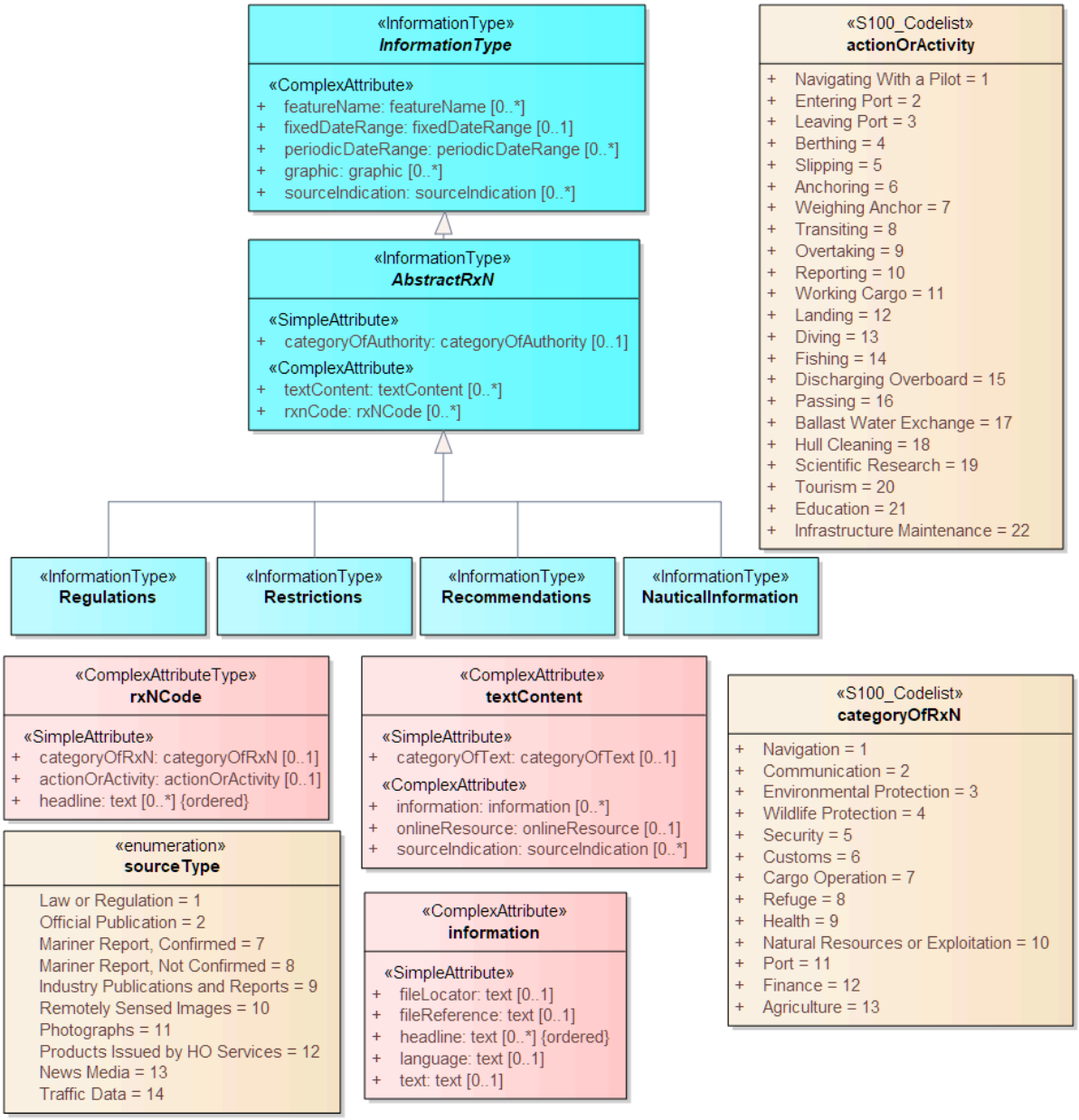

Figure 9-1 — Types for textual information concerning regulations, etc.

Where possible, these types should be classified using the *categoryOfRxN* and *actionOrActivity* codelists in the rxnCode complex attribute. Being open enumeration codelists, they allow for additional categories not listed among their standard values. For example, an "under repair" activity might be encoded in the *actionOrActivity* attribute (as "other: underRepair", following the syntax rule for encoding "extra" values in open enumerations[3]).

Producers should note that such extra values will merely be displayed and not processed (for example, the user interface will not use extra values to choose symbols or filter instances of Regulations[4], whereas it may apply filters to the standard values and/or them in portrayal).

[3] S-100 3-6.7 specifies the format as "The word 'other' followed by a colon and a single space character (that is 'other: ' without quotes), followed by one or more alphanumeric strings separated by single spaces."

[4] In the interest of brevity, "Regulations" in this sub-clause stands for any one of the four types described by this section.

9.2 Regulations, etc., for specific locations

All geo features may have an association to any of **Regulations** or its sibling information types. This association is **AssociatedRxN** and it is inherited from the root feature type **FeatureType**.

If it is necessary to identify an authority or organization related to a particular regulation (restriction, etc.) object, this may be done using the **RelatedOrganisation** association between **Regulations**, etc., and an **Authority** object.

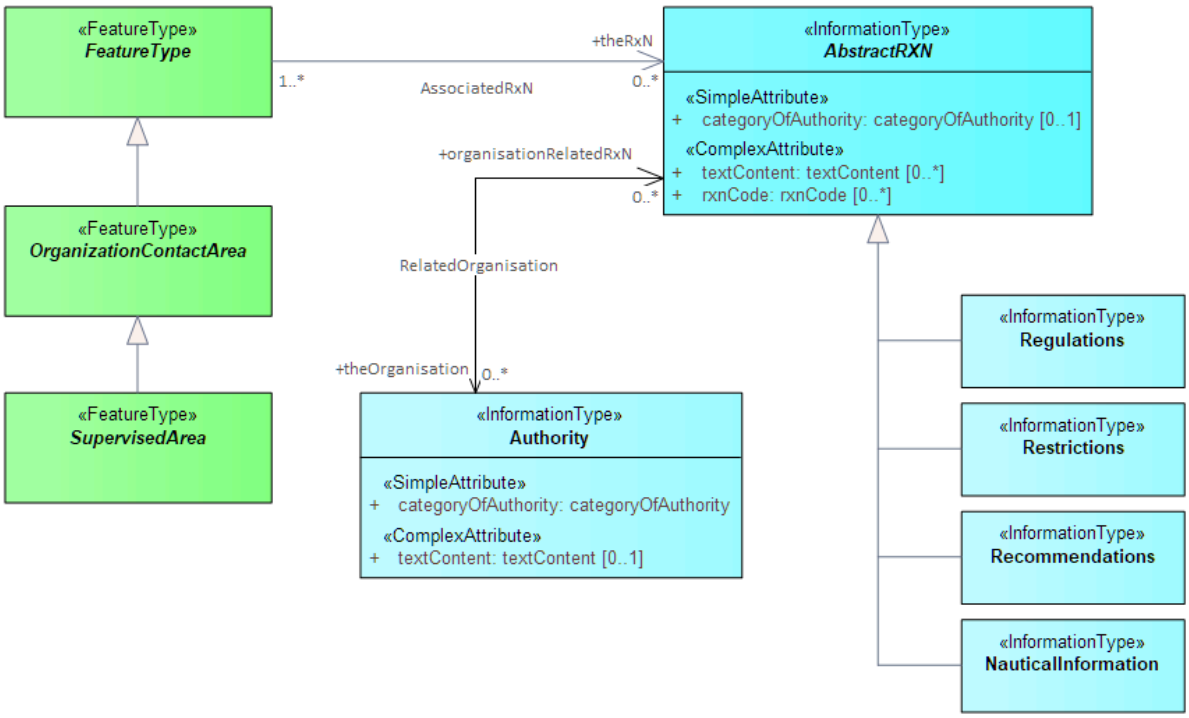

Figure 9-2 — Regulations, etc., for geo features

9.3 Regulations applying only to vessels with specific characteristics or cargoes

Regulations applying only to vessels of specified types, exceeding specified dimensions, or carrying specified cargoes (or other limitations which apply only to subsets of vessels) are encoded by defining the subset of vessels using an **Applicability** instance and associating the **Regulations** object to that **Applicability**.

For information on the use of **Applicability** to define subsets of vessels, see clause 11 in this DCEG and clause 4 in the main PS.

9.4 Regulations

IHO Definition: Regulations for a related area or facility.				
S-122 Information Type: Regulations (Abstract type)				
Super Type: AbstractRxN				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Category of Authority	AbstractRxN		EN	0,1

RxN Code	AbstractRxN	C	0,*
Text Content	AbstractRxN	C	0,*
Feature Name	InformationType	C	0,*
Fixed Date Range	InformationType	C	0,1
Periodic Date Range	InformationType	C	0,*
Graphic	InformationType	C	0,*
Source Indication	InformationType	C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
isApplicableTo	InclusionType (inherited from AbstractRxN)	Applicability	association	0,*
theOrganisation	RelatedOrganisation (inherited from AbstractRxN)	Authority	association	0,*

9.4.1 General

Regulations is intended to be used for official rules, laws, and similar source material, i.e., sources that have the force of law or are mandated by a controlling authority. They will generally originate from some kind of administration or authority, including port authorities.

A single instance of **Regulations** can be referenced by multiple feature instances.

9.4.2 Remarks

- Instances of **Regulations** do not encode a reverse link to the geographic features which reference them. While an association from geographic feature to information type can be encoded in the geographic feature instance, the reverse association from the information type to the geographic feature is omitted from the information type instance.
- If the complex attribute *rxNCode* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

9.5 Restrictions

IHO Definition: Restrictions for a related area or facility.				
S-122 Information Type: Restrictions (Abstract type)				
Super Type: AbstractRxN				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Category of Authority	AbstractRxN		EN	0,1

RxN Code	AbstractRxN	C	0,*
Text Content	AbstractRxN	C	0,*
Feature Name	InformationType	C	0,*
Fixed Date Range	InformationType	C	0,1
Periodic Date Range	InformationType	C	0,*
Graphic	InformationType	C	0,*
Source Indication	InformationType	C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
isApplicableTo	InclusionType (inherited from AbstractRxN)	Applicability	association	0,*
theOrganisation	RelatedOrganisation (inherited from AbstractRxN)	Authority	association	0,*

9.5.1 General

Restrictions is intended for restrictions that constrain the activities of vessels temporarily with or without the legal force, or for longer terms without the force of law; they may be issued by a local authority such as a port captain or US Coast Guard district.

A single instance of **Restrictions** can be referenced by multiple feature instances.

Modelling of restrictions where the default is all activities are prohibited and only exceptions are listed should be done as follows:

- Create **Restrictions**.*textContent.information.text* = “All activities except those specifically permitted” (**Regulations** instead of **Restrictions** also works)
- Assign **Restrictions**.*textContent.information.headline* = “Prohibited Activities”
- Associate it to the feature via an **AssociatedRxN**
- If it applies only to some kinds of vessels, also create an **Applicability** and associate it to the **Restrictions**.
- Create a **Regulations** object with *rxnCode.actionOrActivity* listing the permitted activities and *rxnCode.headline* = “Permitted Activities” and link it to the feature with an **AssociatedRxN** (and link it to the same **Applicability** if defined)

9.5.2 Remarks

- Instances of Restrictions do not encode a reverse link to the geographic features which reference them. While an association from geographic feature to information type can be encoded in the geographic feature instance, the reverse association from the information type to the geographic feature is omitted from the information type instance.
- If the complex attribute *rxNCode* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

9.6 Recommendations

IHO Definition: Recommendations for a related area or facility.				
S-122 Information Type: Recommendations (Abstract type)				
Super Type: AbstractRxN				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Category of Authority	AbstractRxN		EN	0,1
RxN Code	AbstractRxN		C	0,*
Text Content	AbstractRxN		C	0,*
Feature Name	InformationType		C	0,*
Fixed Date Range	InformationType		C	0,1
Periodic Date Range	InformationType		C	0,*
Graphic	InformationType		C	0,*
Source Indication	InformationType		C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
isApplicableTo	InclusionType (inherited from AbstractRxN)	Applicability	association	0,*
theOrganisation	RelatedOrganisation (inherited from AbstractRxN)	Authority	association	0,*

9.6.1 General

Recommendations is intended for encoding suggestions, limitations, or preferred procedures that are not mandatory.

For example, a recommendation for approaching a particular berth at a given orientation may be encoded in a Recommendations object associated to the Berth feature with an **AssociatedRxN** association from the Berth to the **Recommendations** object. If it is a port rule rather than a recommendation, it should be encoded as a **Restrictions** or **Regulations** object instead, with the same association from the Berth feature.

A single instance of **Recommendations** can be referenced by multiple feature instances.

9.6.2 Remarks

- Instances of **Recommendations** do not encode a reverse link to the geographic features which reference them. While an association from geographic feature to information type can be encoded in the geographic feature instance, the reverse association from the information type to the geographic feature is omitted from the information type instance.
- If the complex attribute *rxNCode* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.

- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

9.7 Nautical Information

IHO Definition: Nautical information about a related area or facility.				
S-122 Information Type: NauticalInformation (Abstract type)				
Super Type: AbstractRxN				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Category of Authority	AbstractRxN		EN	0,1
RxN Code	AbstractRxN		C	0,*
Text Content	AbstractRxN		C	0,*
Feature Name	InformationType		C	0,*
Fixed Date Range	InformationType		C	0,1
Periodic Date Range	InformationType		C	0,*
Graphic	InformationType		C	0,*
Source Indication	InformationType		C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
isApplicableTo	InclusionType (inherited from AbstractRxN)	Applicability	association	0,*
theOrganisation	RelatedOrganisation (inherited from AbstractRxN)	Authority	association	0,*

9.7.1 General

NauticalInformation is intended for material that is largely informative in nature, of which does not fit into the category of regulation, recommendation, or restriction.

A single instance of **NauticalInformation** can be referenced by multiple feature instances.

9.7.2 Remarks

- Instances of **NauticalInformation** do not encode a reverse link to the geographic features which reference them. While an association from geographic feature to information type can be encoded in the geographic feature instance, the reverse association from the information type to the geographic feature is omitted from the information type instance.
- In theory, **NauticalInformation** can be associated with any geographic feature through either an **AdditionalInformation** or **AssociatedRxN** association. **AdditionalInformation** should be used only when the information encoded in **NauticalInformation** is general in nature and does not supplement information encoded in a **Regulations**, Restrictions*, or **Recommendations** object associated to the same feature.

- If the complex attribute *rxNCode* is present at least one of its sub-attributes must be populated.
- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

10 Services, Organisations and Schedules

10.1 Work schedules and holidays

Operating schedules and business hours of organizations are encoded by associating a **ServiceHours** instance to an **Authority**. Partial work schedules on holidays or other special days are encoded by associating a **NonstandardWorkingDay** instance to the **ServiceHours** instance.

Similarly, operating schedules for a facility are encoded by associating a **ServiceHours** to the geo feature representing the facility, and associating a **NonstandardWorkingDay** to the **ServiceHours** to encode partial working days. The types and associations are depicted in Figure 10-1 (note that this figure does not show inherited attributes, which are also available to the encoder).

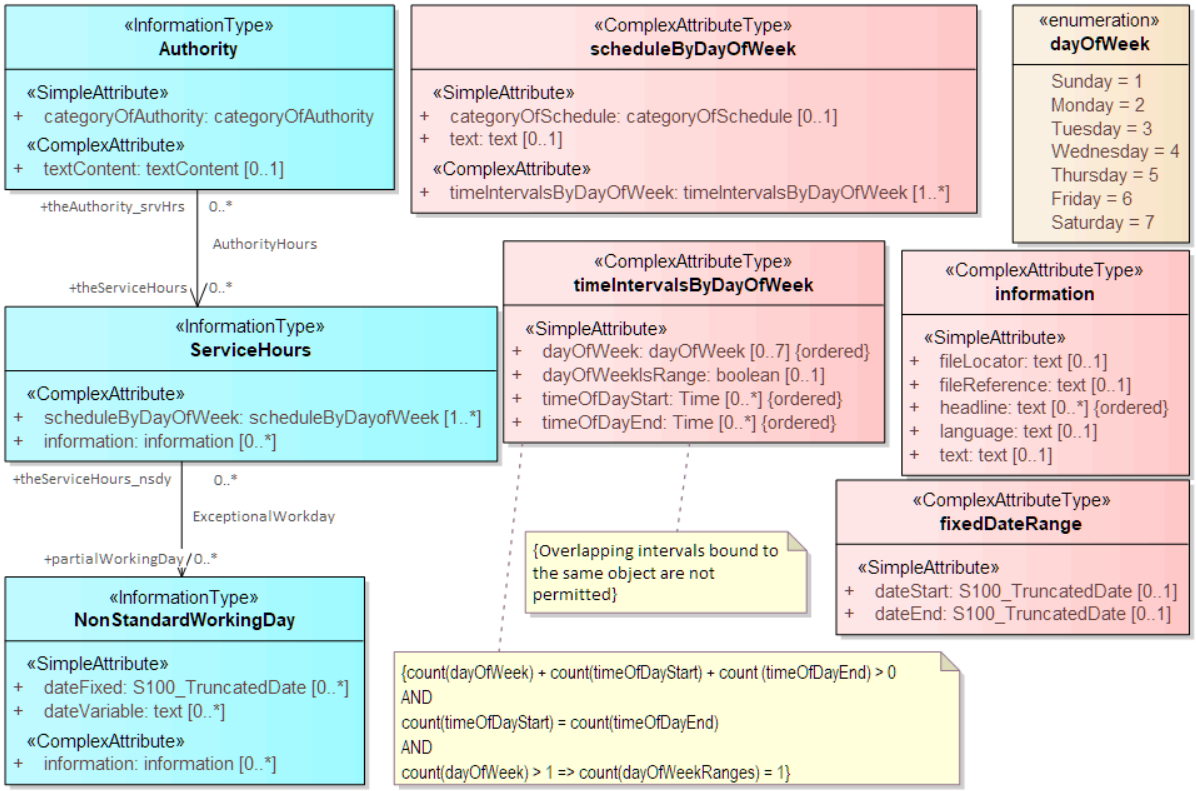

Figure 10-1 — Operating schedules

For further guidance and examples, see clause 2.4.10.4.

10.2 Contact information

Contact information for service operators, controllers or facilities should be encoded in instances of the **ContactDetails** information type, which may be linked from multiple instances of geographic features or information types. Any S-122 geographic feature except meta and cartographic features can be associated to an instance of ContactDetails. S-122 geographic features inherit the association to **ContactDetails** from the abstract feature type **OrganizationContactArea**, as shown in Figure 10-2 (note that this figure does not show inherited attributes, which are also available to the encoder).

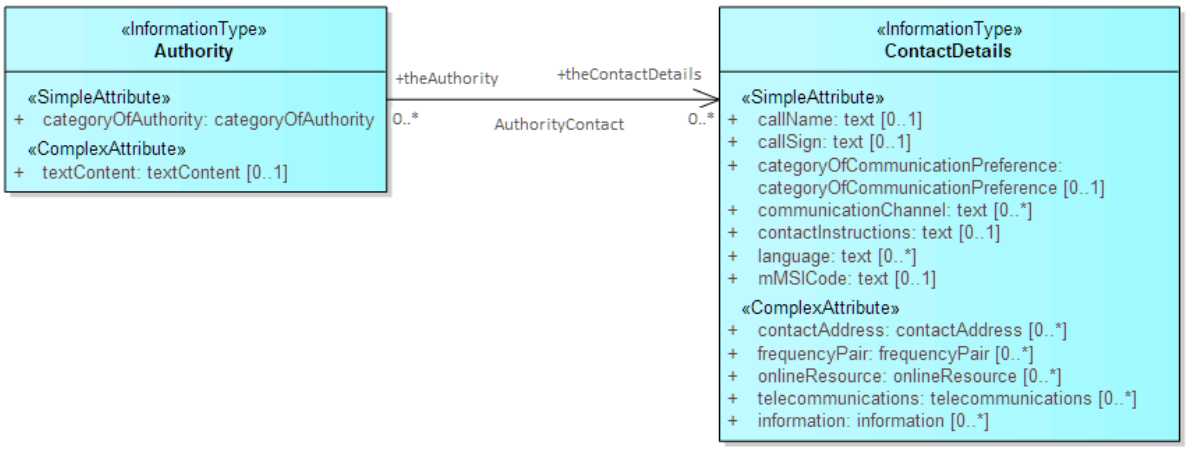

Figure 10-2 — Associations to contact information

Contact information must not be encoded directly in the feature or information type instance using a *textContent* or *information* complex attribute bound directly to the feature or information type. An instance of **ContactDetails** must be created instead. The exception to this rule is when contact-related attributes such as *communicationChannel* are bound to the feature or information type, in which case a **ContactDetails** instance should be created only if it is necessary to provide contact information which cannot be coded in the contact-specific attributes bound to the feature.

10.3 Authority

IHO Definition: A person or organisation having political or administrative power and control.				
S-122 Information Type: Authority (Abstract type)				
Super Type: InformationType				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Authority		2: Border Control 3: Police 4: Port 5: Immigration 6: Health 7: Coast Guard 8: Agricultural 9: Military 10: Private Company 11: Maritime Police 12: Environmental 13: Fishery 14: Finance 15: Maritime 16: Customs	EN	1,1
Text Content			C	0,1
Category of Text		1: Abstract or Summary 2: Extract 3: Full Text	(S) EN	0,1
Information			(S) C	0,*
Online Resource			(S) C	0,1
Source Indication			(S) C	0,*

Inherited Attributes			
S-122 Attribute	**Inherited From**	**Type**	**Multiplicity**
Feature Name	InformationType	C	0,*
Fixed Date Range	InformationType	C	0,1
Periodic Date Range	InformationType	C	0,*
Graphic	InformationType	C	0,*
Source Indication	InformationType	C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
theContactDetails	AuthorityContact	ContactDetails	association	0,*
organisationRelatedRxN	RelatedOrganisation	AbstractRxN	association	0,*
theServiceHours	AuthorityHours	ServiceHours	association	0,*

10.3.1 General

The **Authority** information type is used for encoding information about organizations, including official authorities (port and other) as well as private organizations which control or operate areas or facilities.

For encoding the contact details for an organization, use an associated **ContactDetails** information type (see the information associations table below).

For encoding the general operating hours of an organization, use an associated **ServiceHours** information type (see clause 4 in the main Product Specification).

For encoding the supervisory or operating organization for a facility or area, such as a **MarineProtectedArea**, use an information association from the geo feature to **Authority**.

10.3.2 Remarks

- If the complex attribute *textContent* is present at least one of its sub-attributes must be populated.
  - If the *textContent* sub-attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

10.4 Contact Details

IHO Definition: Information on how to reach a person or organisation by postal, internet, telephone, telex and radio systems.				
S-122 Information Type: ContactDetails (Abstract type)				
Super Type: InformationType				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Call Name			TE	0,1
Call Sign			TE	0,1

Category of Communication Preference		1: Preferred Calling 2: Alternate Calling 3: Preferred Working 4: Alternate Working	EN	0,1
Communication Channel			TE	0,*
Contact Instructions			TE	0,1
Language			TE	0,*
MMSI Code			TE	0,1
Contact Address			C	0,*
Delivery Point			(S) TE	0,* (ordered)
City Name			(S) TE	0,1
Administrative Division			(S) TE	0,1
Country Name			(S) TE	0,1
Postal Code			(S) TE	0,1
Frequency Pair			C	0,*
Frequency Shore Station Receives			(S) IN	0,1
Frequency Shore Station Transmits			(S) IN	1,1
Information			C	0,*
File Locator			(S) TE	0,1
File Reference			(S) TE	0,1
Headline			(S) TE	0,* (ordered)
Language			(S) TE	0,1
Text			(S) TE	0,1
Online Resource			C	0,*
Linkage			(S) URI	1,1
Protocol			(S) TE	0,1
Application Profile			(S) TE	0,1
Name of Resource			(S) TE	0,1
Online Resource Description			(S) TE	0,1
Protocol Request			(S) TE	0,1
Online Function		1: Download 3: Offline Access 4: Order 5: Search 6: Complete Metadata 7: Browse Graphic 8: Upload 9: Email Service 10: Browsing 11: File Access	(S) EN	0,1

Telecommunications			C	0,*
Category of Communication Preference		1: Preferred Calling 2: Alternate Calling 3: Preferred Working 4: Alternate Working	(S) EN	0,1
Telecommunication Identifier			(S) TE	1,1
Telecommunication Carrier			(S) TE	0,1
Contact Instructions			(S) TE	0,1
Telecommunication Service		1: Voice 2: Facsimile 3: SMS 4: Data 5: Streamed Data 6: Telex 7: Telegraph 8: Email	(S) EN	0,*
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Feature Name	InformationType		C	0,*
Fixed Date Range	InformationType		C	0,1
Periodic Date Range	InformationType		C	0,*
Graphic	InformationType		C	0,*
Source Indication	InformationType		C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
theAuthority	AuthorityContact	Authority	association	0,*

10.4.1 General

The **ContactDetails** information type provides several attributes for encoding different types of contact details.

ContactDetails may be associated to:

- An **Authority** information type via an information association (**AuthorityContact**), in which case it encodes the contact information for the organization in general.
- A geo feature via a feature association **ServiceContact**, in which case it encodes contact information particular to the specific feature, either because further information about the controlling authority is not available or because the contact is specific to the feature.

A single instance of **ContactDetails** may be referenced from multiple feature or information type instances.

At least one of the attributes of **ContactDetails** must be populated.

10.4.2 Remarks

- If it is required to encode different call name, call sign, communication preference or contact instructions in different languages, this must be done by creating and associating a different instance of ContactDetails per language. The *language* attribute must be used to designate the language(s)

of each instance. If there is no difference in these attributes for different languages, a singe instance of **ContactDetails** should be created and all the languages indicated using *language* attributes.

- For attributes which allow multiplicity > 1 (*contactAddress*, *frequencyPair*, *information*, *onlineResource*, and *telecommunications*), information that is different for different languages may be encoded using different attribute instances taking care to indicate the language in each attribute instance. Where there is no language sub-attribute, use another appropriate sub-attribute (for example, *contactInstructions* or *onlineResourceDescription*) to indicate the language.
- The name of the contact (for example, the name of the agency, pilot service, office, etc.) should be encoded in the *featureName* attribute, which is inherited from **InformationType**.
- Reverse links from **ContactDetails** to a geo feature referencing it are not encoded, since S-100 feature catalogues does not provide information-to-feature bindings.
- If the complex attribute *contactAddress* is present at least one of its sub-attributes must be populated.
- If the complex attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

10.5 Service Hours

IHO Definition: The time when a service is available and known exceptions.				
S-122 Information Type: ServiceHours (Abstract type)				
Super Type: InformationType				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Schedule by Day of Week			C	1,*
Category of Schedule		1: Normal Operation 2: Closure 3: Unmanned Operation	(S) EN	0,1
Text			(S) TE	0,1
Time Intervals by Day of Week			(S) C	1,*
Information			C	0,*
File Locator			(S) TE	0,1
File Reference			(S) TE	0,1
Headline			(S) TE	0,* (ordered)
Language			(S) TE	0,1
Text			(S) TE	0,1
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Feature Name	InformationType		C	0,*
Fixed Date Range	InformationType		C	0,1
Periodic Date Range	InformationType		C	0,*
Graphic	InformationType		C	0,*
Source Indication	InformationType		C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
partialWorkingDay	ExceptionalWorkday	NonStandardWorkingDay	association	0,*
theAuthority_srvHrs	AuthorityHours	Authority	association	0,*

10.5.1 General

See clause 2.4.10.4 for more information on how to use **ServiceHours** to encode schedules.

Seasonal variations in service hours can be encoded using multiple **ServiceHours** instances with appropriate *periodicDateRange* values.

Ovelapping or null work periods in the same schedule are generally an error and should be avoided.

10.5.2 Remarks

- If none of the listed values of *categoryOfSchedule* applies, *categoryOfSchedule* must be omitted and its co-sub-attribute *text* used to describe the nature of the schedule.
- If the complex attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

10.6 Non-Standard Working Day

IHO Definition: Days when many services are not available. Often days of festivity or recreation or public holidays when normal working hours are limited, especially a national or religious festival, etc.				
S-122 Information Type: NonStandardWorkingDay (Abstract type)				
Super Type: InformationType				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Date Fixed			TD	0,*
Date Variable			TE	0,*
Information			C	0,*
File Locator			(S) TE	0,1
File Reference			(S) TE	0,1
Headline			(S) TE	0,* (ordered)
Language			(S) TE	0,1
Text			(S) TE	0,1
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Feature Name	InformationType		C	0,*
Fixed Date Range	InformationType		C	0,1
Periodic Date Range	InformationType		C	0,*
Graphic	InformationType		C	0,*

Source Indication	InformationType	C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**

10.6.1 General

An instance of **NonStandardWorkingDay** is used in conjunction with **ServiceHours** to indicate exceptions to normal work or operating schedules.

10.6.2 Remarks

- Non-standard workdays which cannot be represented using fixed or variable dates should be encoded using the information complex attribute, preferably as a short description in the text sub-attribute of information. The information attribute can also be used for encoding any additional explanatory information if the explanation is essential knowledge for specifying the day.
- The two date range attributes (fixed and periodic date range) should be used if the non-standard day applies only in specific years or periods (e.g., seasonally).
- At least one of the attributes *dateFixed*, *dateVariable*, or *information* should be populated.
- If the complex attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

11 Limitations

11.1 Introduction

Certain regulations, recommendations, etc., apply only to vessels of specified dimensions, types, or carrying specified cargo, etc. Similarly, certain features have specific significance for vessels of specified dimensions (e.g., different speed limits for vessels carrying specified cargoes or exceeding specified dimensions, or entry prohibitions for certain vessel types).

11.2 Defining subsets of vessels by dimensions, cargo, and other characteristics

This is modelled by first defining the relevant subset of vessels according to the dimension, type, cargo, etc., and then associating that subset to the appropriate feature or information type. The subset of vessels is modelled using the **Applicability** class, which contains attributes for the most common vessel characteristics used in nautical publications. These include measurements (length, beam, draught), type of cargo, displacement, etc. Constraints which cannot be modelled using the attributes of **Applicability** can be described in plain text in its information attribute.

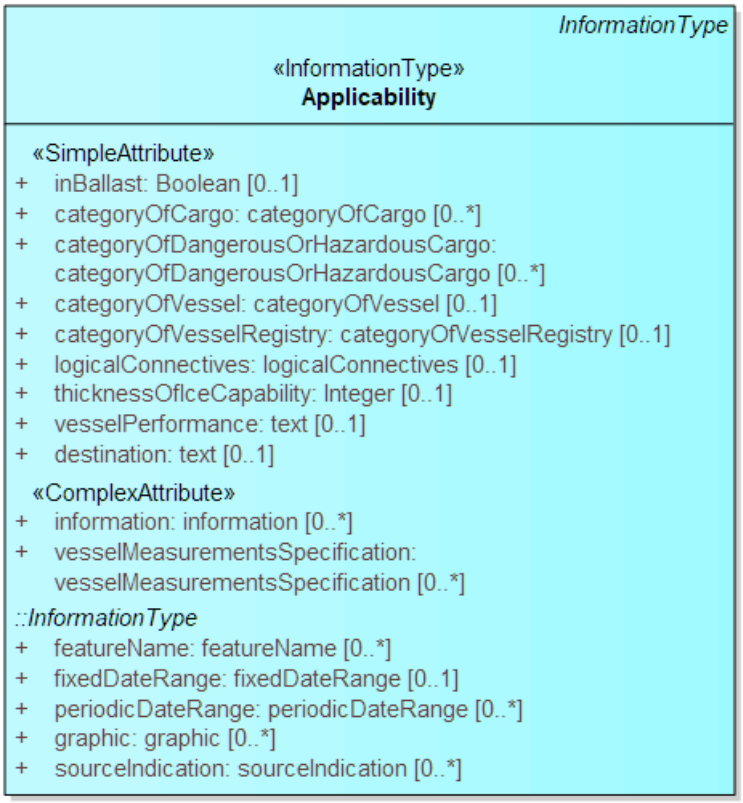

Figure 11-1 — Characteristics and dimensions defining sets of vessels

Conditions relating to vessel dimensions are modelled by the complex attribute *vesselMeasurementsSpecification*, which has sub-attributes for naming the dimension and indicating the limit (whether the condition applies to a vessel which exceeds or falls below the limit).

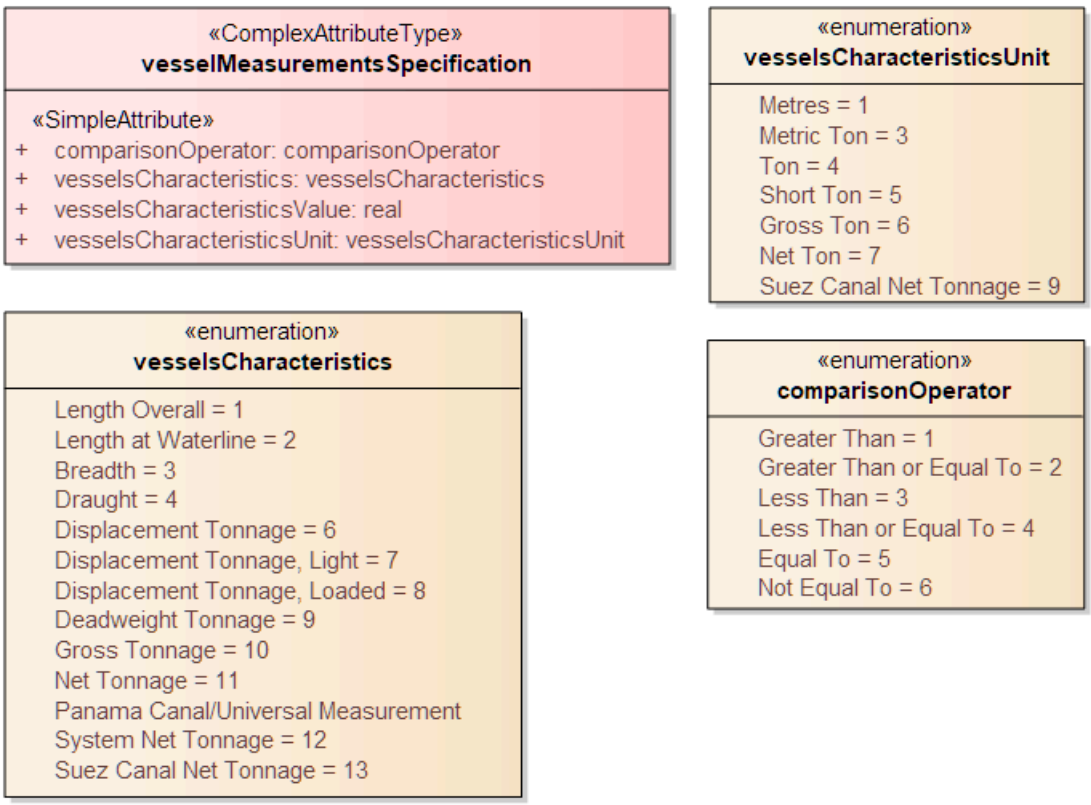

Figure 11-2 — Attributes for specifying vessel dimensions

For example, the combinations in Table 11-1 below describe the conditions “length overall > 50 m” (Condition 1); “length overall < 90 m” (Condition 2); and “breadth > 20 m” (Condition 3).

Table 11-1 — Examples of conditions based on vessel dimensions

Attribute	Condition 1	Condition 2	Condition 3
vesselsCharacteristics	length overall	length overall	breadth
comparisonOperator	greater than	less than	greater than
vesselsCharacteristicsValue	50	90	20
vesselsCharacteristicsUnit	metre	metre	metre

The *logicalConnectives* attribute of **Applicability** is used to indicate how multiple conditions are combined. Combinations may be cumulative (conjunctive, AND) or alternatives (disjunctive, OR).

EXAMPLE 1: Encoding *logicalConnectives=AND* combined with Conditions 1 and 2 above describes a vessel of length between 50 and 90 metres.

EXAMPLE 2: Encoding *logicalConnectives=OR* combined with Conditions 1 and 3 describes a vessel of length greater than 50 metres or beam greater than 20 metres.

This modelling cannot represent subsets defined by both AND and OR combinations, but it is always possible to convert such complex conditions into multiple combinations each using only AND (‘conjunctive normal form’) or OR (‘disjunctive normal form’), and model the subset using more than one **Applicability** object.

11.3 Characterizing the relationship between the vessel set and the feature or regulation

The relationship between a set of vessels and a geographic feature may be one of several different mandate levels ranging from prohibition on use of entry into a geographic location to mandatory use of a feature (such as vessels exceeding certain dimensions being required to board pilots at an outer boarding place).

The relationship between a set of vessels and a regulation information type (or recommendation, restriction, or special note) may be one of inclusion or specific exclusion — either the regulation (recommendation, etc.) specifically applies to the specified set of vessels, or the specified set of vessels is explicitly excluded from the regulation. (If a regulation does not apply to a set of vessels but there is no explicit exemption stated in the source material, there is no relationship that needs to be encoded.)

The association classes **PermissionType** and **InclusionType** (Figures 11-3 and 11-4) characterize these relationships using values of their attributes *categoryOfRelationship* and *membership* respectively.

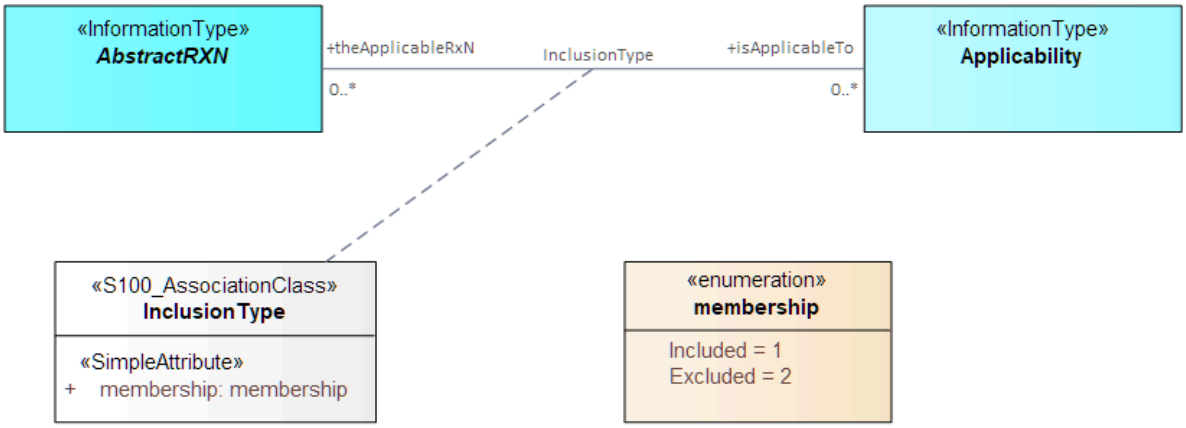

Figure 11-3 — Permission relationship

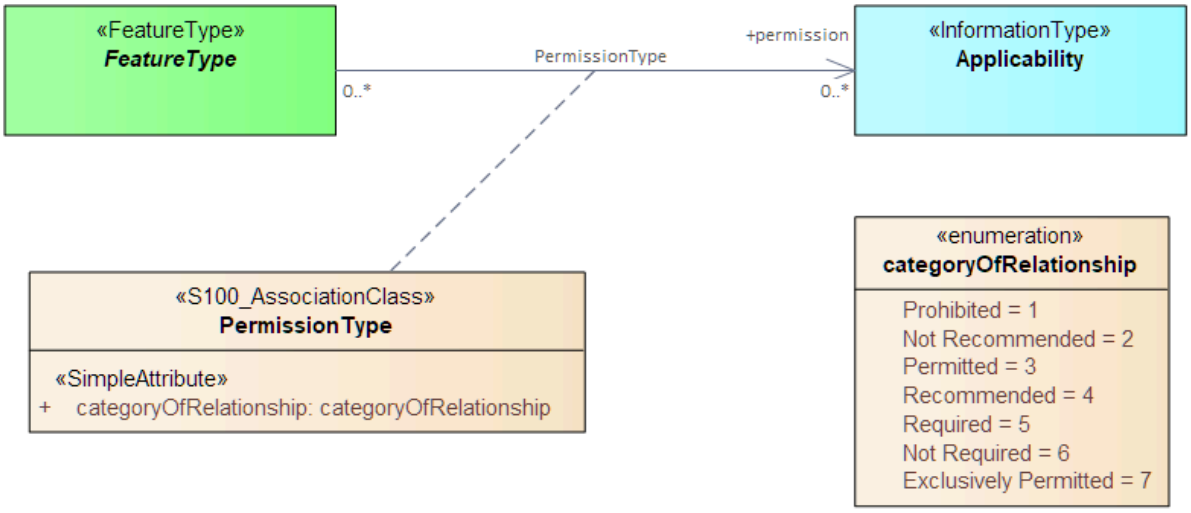

Figure 11-4 — Inclusion/exclusion relationship

EXAMPLE 1: A specified set of vessels is COVERED by a regulation and another set of vessels is EXEMPT from the regulation — described by the membership attribute values “included” and “excluded” respectively.

EXAMPLE 2: Vessels with specified cargo and dimensions MUST use a specified berth, vessels of smaller dimensions are RECOMMENDED to use the berth, and naval transports are EXEMPT from using the berth — described by the categoryOfRelationship attribute values “required”, “recommended” and “recommended” respectively.

11.4 Production hints and recommended practices (informative)

11.4.1 Capturing the application of a regulation, recommendation, etc. to specified kinds of vessels

Encoders may find it easiest to capture the application of a regulation (recommendation , etc.) to a class or set of vessels in three phases:

1) Encode the operative part of the regulation (the part that describes what the vessels subject to the regulation must or must not do), creating an instance of **Regulations** (or **Recommendations**, etc., as appropriate). Descriptions of what kinds of vessels are subject to the regulation must be excluded from the content of the **Regulations** instance.
2) Create an **Applicability** information type and encode the description of what kinds of vessels are subject to (or exempted from) the regulation.
3) Link the two using an **InclusionType** with *membership=included* if the vessels described by **Applicability** are subject to the regulation, or *membership=excluded* if they are explicitly exempted from the regulation.

It is not necessary to create separate instances of the regulation for inclusion and exclusion.

11.4.2 Capturing the permissibility or otherwise of a geographic feature for specified kinds of vessels

Encoders may find it easiest to capture the permissibility of a feature to specified kinds of vessels in three phases.

1) Create the geographic feature if it does not already exist.
2) Create an **Applicability** information type and encode the description of what kinds of vessels are required to use the geographic feature.
3) Link the two using a **PemissionType** with *categoryOfRelationship = required*.

For the other relationships (prohibited, not recommended, etc.) steps 2 and 3 should be modified accordingly (i.e., if use by certain kinds of vessels is “not recommended” encode the description of that kind of vessels in an **Applicability** and create a linking **PermissionType** with *categoryOfRelationship = not recommended*).

It is not necessary to create a separate instance of the geographic feature for each type of relationship.

11.4.3 Constructing the Applicability information type

Where the source material describes complex conditions, encoders may find it useful to write out the conditions in structured language with grouping parentheses, for example, as “(condition A) AND (condition B) AND (condition C)”, or draw diagrams, before encoding **Applicability** and its associations.

Note that the model limitation on mixing logical connectives means some forms of conditions which use “nesting” cannot be encoded in a single **Applicability** instance and multiple instances must be created.

EXAMPLE: The complex condition “(condition A) AND condition B) OR (condition C” must be encoded as two **Applicability** instances, one with “(condition A) AND (condition B)” and the other with “(condition A) AND (condition C)”.

Table 11-2 — Example of conversion of complex condition to multiple simple conditions

Complex condition	Encode as
(condition A) AND condition B) OR (condition C	Applicability 1: (condition A) AND (condition B) Applicability 2: (condition A) AND (condition C)

Data producers may contact NIPWG with questions about encoding complex conditions.

As a last resort, conditions may be written as phrases in natural language and encoded in the information attribute. It is acceptable for an **Applicability** to have only the *information* attribute populated.

11.5 Applicability

IHO Definition: Describes the relationship between vessel characteristics and: (i) the applicability of an associated information object or feature to the vessel; or, (ii) the use of a facility, place, or service by the vessel; or, (iii) passage of the vessel through an area.				
S-122 Information Type: Applicability (Abstract type)				
Super Type: InformationType				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
In Ballast			BO	0,1
Category of Cargo		1: Bulk 2: Container 3: General 4: Liquid 5: Passenger 6: Livestock 7: Dangerous or Hazardous 8: Heavy Lift 10: Dry Bulk Cargo 11: Liquid Bulk Cargo 12: Reefer Container Cargo 13: Ro-Ro Cargo 14: Project Cargo 15: Break Bulk Cargo	EN	0,*
Category Of Dangerous Or Hazardous Cargo		1: IMDG Code Class 1 Div. 1.1 2: IMDG Code Class 1 Div. 1.2 3: IMDG Code Class 1 Div. 1.3 4: IMDG Code Class 1 Div. 1.4 5: IMDG Code Class 1 Div. 1.5 6: IMDG Code Class 1 Div. 1.6 7: IMDG Code Class 2 Div. 2.1 8: IMDG Code Class 2 Div. 2.2 9: IMDG Code Class 2 Div. 2.3 10: IMDG Code Class 3 11: IMDG Code Class 4 Div. 4.1 12: IMDG Code Class 4 Div. 4.2 13: IMDG Code Class 4 Div. 4.3 14: IMDG Code Class 5 Div. 5.1 15: IMDG Code Class 5 Div. 5.2 16: IMDG Code Class 6 Div. 6.1 17: IMDG Code Class 6 Div. 6.2 18: IMDG Code Class 7 19: IMDG Code Class 8	EN	0,*

		20: IMDG Code Class 9 21: Harmful Substances in Packaged Form		
Category of Vessel		1: General Cargo Vessel 2: Container Carrier 3: Tanker 4: Bulk Carrier 5: Passenger Vessel 6: Roll-On Roll-Off 7: Refrigerated Cargo Vessel 8: Fishing Vessel 9: Service 10: Warship 11: Towed or Pushed Composite Unit 12: Tug and Tow 13: Light Recreational 14: Semi-Submersible Offshore Installation 15: Jack-Up Exploration or Project Installation 16: Livestock Carrier 17: Sport Fishing	CL	0,1
Category of Vessel Registry		1: Domestic 2: Foreign	EN	0,1
Logical Connectives		1: Logical Conjunction 2: Logical Disjunction	EN	0,1
Thickness of Ice Capability			IN	0,1
Vessel Performance			TE	0,1
Destination			TE	0,1
Information			C	0,*
File Locator			(S) TE	0,1
File Reference			(S) TE	0,1
Headline			(S) TE	0,* (ordered)
Language			(S) TE	0,1
Text			(S) TE	0,1
Vessel Measurements Specification			C	0,*
Comparison Operator		1: Greater Than 2: Greater Than or Equal To 3: Less Than 4: Less Than or Equal To 5: Equal To 6: Not Equal To	(S) EN	1,1
Vessels Characteristics		1: Length Overall 2: Length at Waterline 3: Breadth 4: Draught 6: Displacement Tonnage	(S) EN	1,1

		7: Displacement Tonnage, Light 8: Displacement Tonnage, Loaded 9: Deadweight Tonnage 10: Gross Tonnage 11: Net Tonnage 12: Panama Canal/ Universal Measurement System Net Tonnage 13: Suez Canal Net Tonnage		
Vessels Characteristics Value			(S) RE	1,1
Vessels Characteristics Unit		1: Metres 3: Metric Ton 4: Ton 5: Short Ton 6: Gross Ton 7: Net Ton 9: Suez Canal Net Tonnage	(S) EN	1,1
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
Feature Name	InformationType		C	0,*
Fixed Date Range	InformationType		C	0,1
Periodic Date Range	InformationType		C	0,*
Graphic	InformationType		C	0,*
Source Indication	InformationType		C	0,*

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**
theApplicableRxN	InclusionType	AbstractRxN	association	0,*

11.5.1 General

The **Applicability** information type is intended for defining sets of vessels according to their dimensions, capabilities, and cargo. Its attributes are intended for defining different limitation conditions, as described by their definitions in clauses 16 and 17.

Multiple instances of **Applicability** associated to the same feature or regulation are treated as “inclusive OR”, that is, each **Applicability** defines an independent set of vessels to which the regulation, permission or requirement applies (or which is specifically exempted, depending on the attribute encoded in the association class).

Clauses 11.1 to 11.3 contain a comprehensive discussion of the use of **Applicability** to describe subsets of vessels according to dimensions, types, cargo, and other characteristics. The remarks below provide additional guidance.

At least one of the attributes of **Applicability** must be populated.

11.5.2 Remarks

- Multiple values of *categoryOfCargo* and of *categoryOfDangerousOrHazardousCargo* should be treated as “inclusive OR” (i.e., if *categoryOfCargo* = *1* and *2*, then it means vessels with either bulk or container cargo or both).

- Limitations which cannot be expressed using more specific attributes should be encoded in text form in the *information* attribute.
- It is acceptable for an **Applicability** to have only the *information* attribute populated.
- Vessel types which do not conform to any of the listed categoryOfVessel values should be encoded as “other: <text>” where <text> is a producer-supplied type name.
- The attribute *logicalConnectives* has multiplicity lower bound 0 for the case where there is only a single limiting condition (for example, if the only condition is “length overall > 100m”) and must be omitted in such a situation. If there is more than one condition, *logicalConnectives* must be encoded. If *logicalConnectives* is omitted and there is more than one condition, the default value assumed is logical conjunction.
- Mutually inconsistent measurements (e.g., draught > 10m and draught < 5m) are an error.
- The inherited attributes *featureName* and *graphic* may be used to provide supplementary information in the form of a title for the defined set of vessels and sketch or other graphic pertaining to the set, but there being no widely acknowledged use cases for them, their use in **Applicability** is discouraged.
- Encoding the inherited *fixedDateRange* and *periodicDateRange* attributes for **Applicability** is discouraged. The *fixedDateRange* and *periodicDateRange* attributes may theoretically be used to qualify the set defined by the **Applicability** instance, but must not be used to define the commencement, termination, season, etc., of the regulation or feature to which **Applicability** is associated (fixed and periodic date ranges should be encoded in the regulation or feature instance instead).
- If the complex attribute *information* is present at least one of its sub-attributes must be populated.
- If the complex attribute *fixedDateRange* is present at least one of its sub-attributes must be populated.
- If the complex attribute *sourceIndication* is present at least one of its sub-attributes must be populated.

12 Spatial Quality

12.1 Introduction

The spatial quality for individual spatial primitives may be reported using the SpatialQuality information type. The conceptual model is depicted in Figure 2-1.

12.2 Spatial Quality

IHO Definition: The indication of the quality of the locational information for features in a dataset.				
S-122 Information Type: SpatialQuality (Abstract type)				
Super Type:				
S-122 Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Quality of Horizontal Measurement		1: Surveyed 2: Unsurveyed 3: Inadequately Surveyed 4: Approximate 5: Position Doubtful 6: Unreliable 7: Reported (Not Surveyed) 8: Reported (Not Confirmed) 9: Estimated 10: Precisely Known 11: Calculated	EN	0,1
Spatial Accuracy			C	0,*
Fixed Date Range			(S) C	0,1
Horizontal Position Uncertainty			(S) C	0,1
Inherited Attributes				
S-122 Attribute	**Inherited From**		**Type**	**Multiplicity**
No inherited attributes				

Information associations				
S-122 Role	**S-122 Association Name**	**Associated to**	**Type**	**Mult.**

12.2.1 General

The **SpatialQuality** information type allows indication of the spatial quality for individual spatial primitives. Quality information in **SpatialQuality** overrides quality information in covering quality meta-feature(s).

The association to **SpatialQuality** is from the spatial primitive. It is not encoded directly as an information association in the feature instance, but in the spatial primitive .

Information associations to *SpatialQuality from spatial primitives*					
Source	**Role**	**Association Name**	**Associated to**	**Type**	**Mult.**
(point or curve spatial primitive)	theQualityInformation	SpatialAssociation	SpatialQuality	association	0,1

The GML structures for point and curve primiitives are depicted in Figure 12-1 and Figure 12-2.

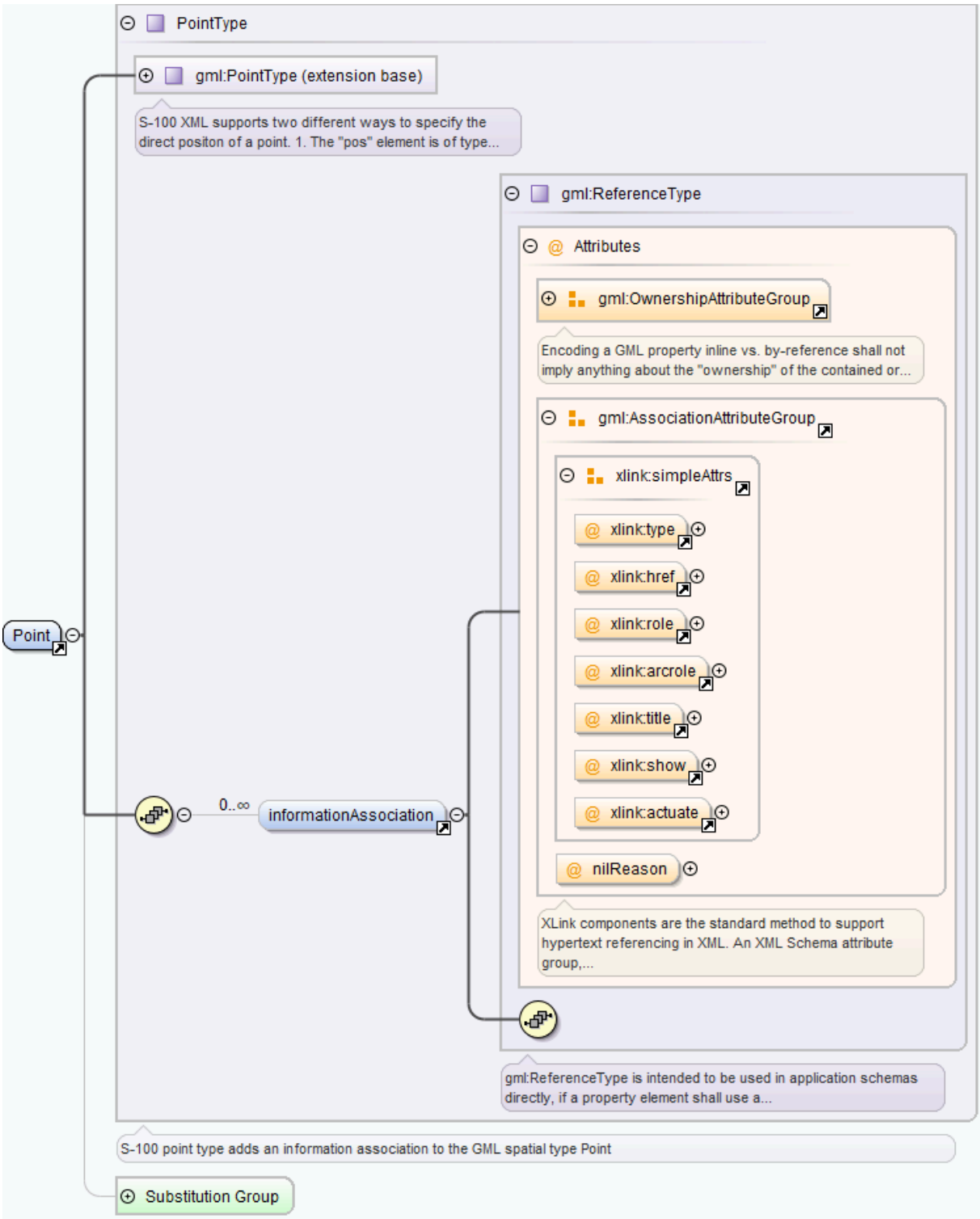

Figure 12-1 — Spatial quality for point spatial primitives

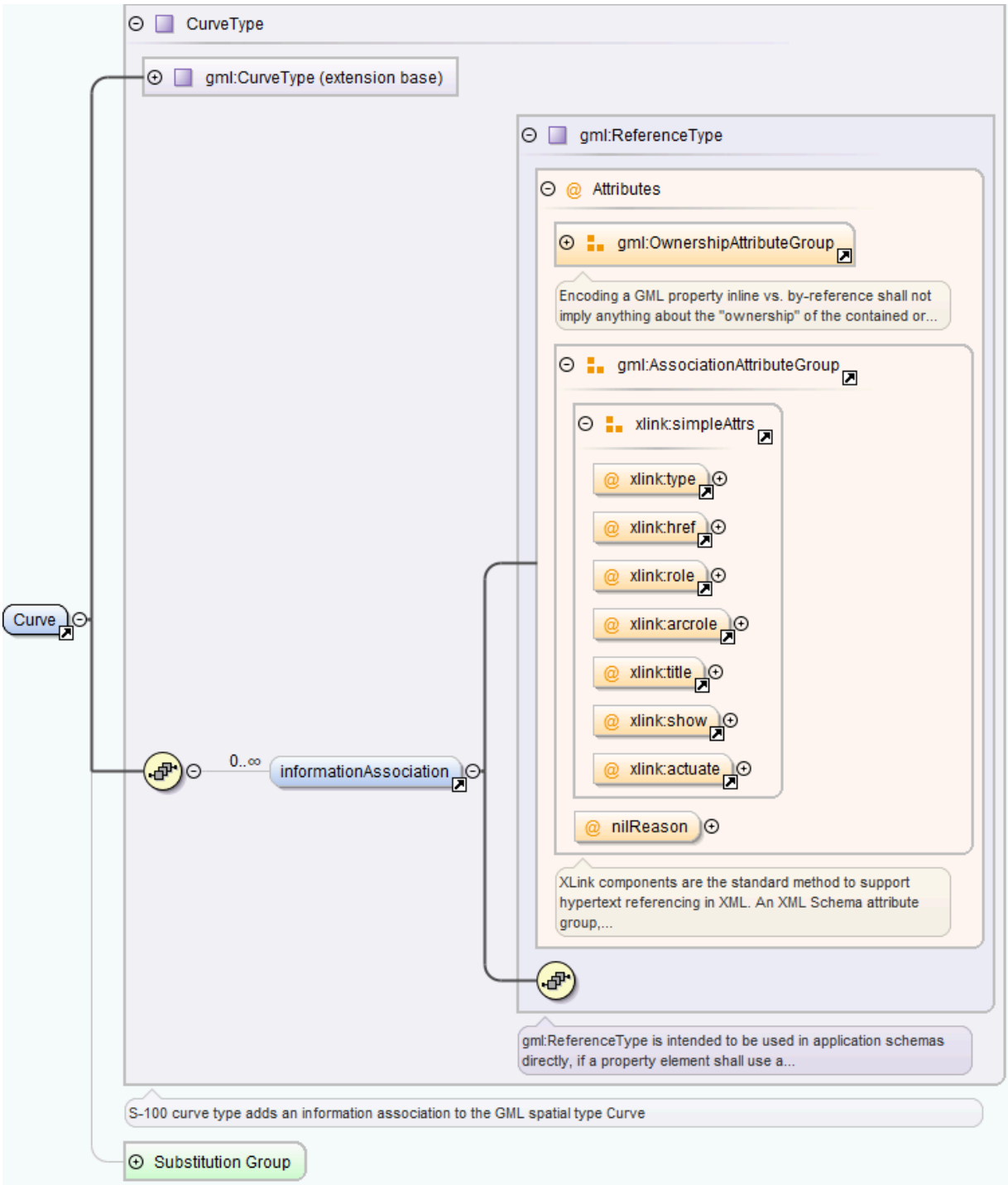

Figure 12-2 — Spatial quality for curve spatial primitives

The association must be encoded using the *informationAssociation* tag with:

- xlink:title = SpatialAssociation
- xlink:href = gml:id of the SpatialQuality instance (using the same prefix convention as for other information associations, for example #SQ00001)
- xlink:arcrole = data:theQualityInformation

The *xlink:show*, *xlink:type* and *xlink:actuate* attributes are not populated. S-100 permits the *xlink:role* attribute to be populated with the "[o]ptional description of the nature of the target resource, given as a URI". However, since the rules for URIs describing target resources are still to be formulated at the time of writing, population of this optional attribute is not recommended.

12.2.2 Remarks

- **SpatialQuality** can only be associated to point and curve types. To indicate the quality of an area boundary, associate **SpatialQuality** to the curve feature for the area boundary.
- **SpatialQuality** associated to Curve or Composite Curve spatial objects cannot have vertical uncertainty attributes.
- If the complex attribute *spatialAccuracy* is present at least one of its sub-attributes must be populated.

Page intentionally left blank

13 Feature Associations

13.1 Text Association

Definition : A feature association for the binding between a geo feature and the cartographically positioned location for text.

CamelCase : TextAssociation

Remarks :

Roles : thePositionProvider theCartographicText

Page intentionally left blank

14 Information Associations

14.1 Additional Information

Definition : an information association for the binding between at least one instance of a geo feature and an instance of an information type.

CamelCase : AdditionalInformation

Remarks :

Roles : theInformation

14.2 Authority Contact

Definition : Contact information for an authority

CamelCase : AuthorityContact

Remarks :

Roles : theAuthority theContactDetails

14.3 Authority Hours

Definition : Service hours for an authority

CamelCase : AuthorityHours

Remarks :

Roles : theAuthority_srvHrs theServiceHours

14.4 Associated RxN

Definition : Association between a geographic location and a regulation, restriction, recommendation, or nautical information

CamelCase : AssociatedRxN

Remarks :

Roles : theRxN

14.5 Exceptional Workday

Definition : Exception to the usual working day

CamelCase : ExceptionalWorkday

Remarks :

Roles : theServiceHours_nsdy partialWorkingDay

14.6 Protected Area Authority

Definition : There may be more than one such authority depending on how responsibilities are divided

CamelCase : ProtectedAreaAuthority

Remarks :

Roles : responsibleAuthority

14.7 Related Organisation

Definition : Related organisation

CamelCase : RelatedOrganisation

Remarks :

Roles : organisationRelatedRxN theOrganisation

14.8 Inclusion Type

Definition : Association class specifying the relationship between the subset of vessels described by an APPLIC data object and a regulation (restriction, recommendation, or nautical information).

CamelCase : InclusionType

Attributes :

Attribute	Acronym	Allowable Encoding Value	Type	Multiplicity
Membership		1: Included 2: Excluded	EN	1,1

Remarks :

Roles : theApplicableRxN isApplicableTo

14.9 Permission Type

Definition : Association class for associations describing whether the subsets of vessels determined by the ship characteristics specified in APPLIC may (or must, etc.) transit, enter, or use a feature.

CamelCase : PermissionType

Attributes :

Attribute	Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Relationship		1: Prohibited 2: Not Recommended 3: Permitted 4: Recommended 5: Required 6: Not Required 7: Exclusively Permitted	EN	1,1

Remarks :

Roles : permission

14.10 Service Control

Definition : The controlling authority for a service area

CamelCase : ServiceControl

Remarks : This is an information association linking a location where a service is provided with an information type describing the provider. Contrast to serviceProvisionArea, which is a feature association

linking the area served with another feature describing the provider. Role controlledService encodable only as a generic inverse association in 3.0.0 datasets as it is an information→feature link

Roles : controlAuthority

14.11 Spatial Association

Definition : An association for the binding between a spatial type and its spatial quality information.

CamelCase : SpatialAssociation

Remarks :

Roles : theQualityInformation

15 Association Roles

15.1 Authority

Definition : A pointer to an Authority object

CamelCase : theAuthority

Remarks :

15.2 Authority Service Hours

Definition : The authority for which service hours are given

CamelCase : theAuthority_srvHrs

Remarks :

15.3 Contact Details

Definition : A pointer to an Contact Details object

CamelCase : theContactDetails

Remarks :

15.4 Control Authority

Definition : The controlling organization or authority for a geographically located service

CamelCase : controlAuthority

Remarks :

15.5 Is Applicable To

Definition : The object or class of objects to which the regulation, restriction, recommendation, or nautical information applies

CamelCase : isApplicableTo

Remarks :

15.6 Organisation-Related RxN

Definition : Reference to regulation, recommendation, restriction or general information related to an organisation

CamelCase : organisationRelatedRxN

Remarks :

15.7 Permission

Definition : Association class for associations describing whether the subsets of vessels determined by the ship characteristics specified in APPLIC may (or must, etc.) transit, enter, or use a feature.

CamelCase : permission

Remarks :

15.8 Partial Working Day

Definition : The work hours for a non-standard workday

CamelCase : partialWorkingDay

Remarks :

15.9 Responsible Authority

Definition : The responsible authority

CamelCase : responsibleAuthority

Remarks :

15.10 Service Hours (reference)

Definition : Service hours for an authority or service provider

CamelCase : theServiceHours

Remarks :

15.11 The Information

Definition : A pointer to an object that provides more information about the referencing feature or information type.

CamelCase : theInformation

Remarks : Registry definition “The information” merely repeats the name.

15.12 The Organisation

Definition : The organisation to which information relates

CamelCase : theOrganisation

Remarks :

15.13 The Quality Information

Definition : A pointer to an information type providing spatial quality information.

CamelCase : theQualityInformation

Remarks :

15.14 The RxN

Definition : The regulation, restriction, recommendation, or nautical information

CamelCase : theRxN

Remarks :

15.15 The Applicable RxN

Definition : The applicable regulation, restriction, recommendation or nautical information

CamelCase : theApplicableRxN

Remarks :

15.16 The Cartographic Text

Definition : A pointer to a specific cartographically positioned location for text.

CamelCase : theCartographicText

Remarks :

15.17 The Position Provider

Definition : A pointer to a specific feature(s).

CamelCase : thePositionProvider

Remarks :

15.18 The service hours for a non-standard workday

Definition : The usual service hours to which an exception applies

CamelCase : theServiceHours_nsdy

Remarks :

Page intentionally left blank

16 Simple Attributes

16.1 Administrative Division

Definition : A generic term for an administrative region within a country at a level below that of the sovereign state.

Type : text

CamelCase : administrativeDivision

Alias :

Remarks :

16.2 Application Profile

Definition : Name of an application profile that can be used with the online resource.

Type : text

CamelCase : applicationProfile

Alias : APPPRF

Remarks :

16.3 Call Name

Definition : The designated call name of a station; for example, radio station, radar station, pilot.

Type : text

CamelCase : callName

Alias :

Remarks : This is the name used when calling a radio station by radio; for example, “Singapore Pilots”.

16.4 Call Sign

Definition : The designated call-sign of a station (radio station, radar station, pilot, …).

Type : text

CamelCase : callSign

Alias : CALSGN

Remarks :

16.5 Cardinal Direction

Definition : Principal and intermediate compass points.

Type : enumeration

CamelCase : cardinalDirection

Alias : CARDIR

Remarks :

Code	Label	Definition
1	North	348.75-011.25 degrees (true north).
2	North Northeast	011.25 — 033.75 degrees.
3	Northeast	033.75 — 056.25 degrees.
4	East Northeast	056.25-078.75 degrees.
5	East	078.75-101.25 degrees.
6	East Southeast	101.25-123.75 degrees.
7	Southeast	123.75-146.25 degrees.
8	South Southeast	146.25-168.75 degrees.
9	South	168.75-191.25 degrees.
10	South Southwest	191.25-213.75 degrees.
11	Southwest	213.75-236.25 degrees.
12	West Southwest	236.25-258.75 degrees.
13	West	258.75-281.25 degrees.
14	West Northwest	281.25-303.75 degrees.
15	Northwest	303.75 — 326.25 degrees.
16	North Northwest	326.25 — 348.75 degrees.

16.6 Category of Authority

Definition : The type of person, government agency or organisation granted powers of managing or controlling access to and/or activity in an area.

Type : enumeration

CamelCase : categoryOfAuthority

Alias : CATAUT

Remarks :

Code	Label	Definition
2	Border Control	The administration to prevent or detect and prosecute violations of rules and regulations at international boundaries.
3	Police	The department of government, or civil force, charged with maintaining public order.
4	Port	Person or corporation, owners of, or entrusted with or invested with the power of managing a port. May be called a Harbour Board, Port Trust, Port Commission, Harbour Commission, Marine Department.
5	Immigration	The authority controlling people entering a country.
6	Health	The authority with responsibility for checking the validity of the health declaration of a vessel and for declaring free pratique.
7	Coast Guard	Organization keeping watch on shipping and coastal waters according to governmental law; normally the authority with responsibility for search and rescue.
8	Agricultural	The authority with responsibility for preventing infection of the agriculture of a country and for the protection of the agricultural interests of a country.

Code	Label	Definition
9	Military	A military authority which provides control of access to or approval for transit through designated areas or airspace.
10	Private Company	A private or publicly owned company or commercial enterprise which exercises control of facilities, for example a calibration area.
11	Maritime Police	A governmental or military force with jurisdiction in territorial waters. Examples could include Gendarmerie Maritime, Carabinierie, and Guardia Civil.
12	Environmental	An authority with responsibility for the protection of the environment.
13	Fishery	An authority with responsibility for the control of fisheries.
14	Finance	An authority with responsibility for the control and movement of money.
15	Maritime	A national or regional authority charged with administration of maritime affairs.
16	Customs	The agency or establishment for collecting duties, tolls.

16.7 Category of Cargo

Definition : Classification of the different types of cargo that a ship may be carrying.

Type : enumeration

CamelCase : categoryOfCargo

Alias : CATCGO

Remarks : If item 7 is used, the nature of dangerous or hazardous cargoes can be amplified with category of dangerous or hazardous cargo.

Code	Label	Definition
1	Bulk	Unpacked homogenous cargo poured loose in a certain space of a vessel, for example oil or grain.
2	Container	One of a number of standard sized cargo carrying units, secured using standard corner attachments and bar.
3	General	Break bulk cargo normally loaded by crane.
4	Liquid	Any cargo loaded by pipeline.
5	Passenger	A fee paying traveller.
6	Livestock	Live animals carried in bulk.
7	Dangerous or Hazardous	Dangerous or hazardous cargo as described by the IMO International Maritime Dangerous Goods code.
8	Heavy Lift	Indivisible heavy items of weight generally over 100 tons, and width or height greater than 100 metres.
10	Dry Bulk Cargo	Commodity cargo that is transported unpackaged in large quantities. These types of goods usually need to be kept dry during the whole transportation period.
11	Liquid Bulk Cargo	Liquids or gases that are transported in bulk and carried unpackaged.
12	Reefer Container Cargo	Cargo transported in refrigerated containers, generally perishable commodities which require temperature-controlled transportation, such as fruit, meat, fish, vegetables, dairy products and other foods.

Code	Label	Definition
13	Ro-Ro Cargo	Wheeled cargo, such as cars, busses, trucks, agricultural vehicles and cranes, that are driven on and off the ship on their own wheels or using a platform vehicle, such as a self-propelled modular transporter.
14	Project Cargo	Project cargo is a term used to broadly describe the national or international transportation of large, heavy, high value, or critical (to the project they are intended for) pieces of equipment. Also commonly referred to as heavy lift, this includes shipments made of various components which need disassembly for shipment and reassembly after delivery.
15	Break Bulk Cargo	Goods that are stowed on board ship in individually counted units, and not in intermodal containers nor in bulk as with oil or grain.

16.8 Category of Communication Preference

Definition : Classification of frequencies, VHF channels, telephone numbers, or other means of communication based on preference.

Type : enumeration

CamelCase : categoryOfCommunicationPreference

Alias :

Remarks :

Code	Label	Definition
1	Preferred Calling	The first choice channel or frequency to be used when calling a radio station.
2	Alternate Calling	A channel or frequency to be used for calling a radio station when the preferred channel or frequency is busy or is suffering from interference.
3	Preferred Working	The first choice channel or frequency to be used when working with a radio station.
4	Alternate Working	A channel or frequency to be used for working with a radio station when the preferred working channel or frequency is busy or is suffering from interference.

16.9 Category Of Dangerous Or Hazardous Cargo

Definition : Classification of dangerous goods or hazardous materials based on the International Maritime Dangerous Goods Code (IMDG Code).

Type : enumeration

CamelCase : categoryOfDangerousOrHazardousCargo

Alias : CATDHC

Remarks :

Code	Label	Definition
1	IMDG Code Class 1 Div. 1.1	Explosives, Division 1: Substances and articles which have a mass explosion hazard.
2	IMDG Code Class 1 Div. 1.2	Explosives, Division 2: Substances and articles which have a projection hazard but not a mass explosion hazard.

Code	Label	Definition
3	IMDG Code Class 1 Div. 1.3	Explosives, Division 3: Substances and articles which have a fire hazard and either a minor blast hazard or a minor projection hazard or both, but not a mass explosion hazard.
4	IMDG Code Class 1 Div. 1.4	Explosives, Division 4: Substances and articles which present no significant hazard.
5	IMDG Code Class 1 Div. 1.5	Explosives, Division 5: Very insensitive substances which have a mass explosion hazard.
6	IMDG Code Class 1 Div. 1.6	Explosives, Division 6: Extremely insensitive articles which do not have a mass explosion hazard.
7	IMDG Code Class 2 Div. 2.1	Gases, flammable gases.
8	IMDG Code Class 2 Div. 2.2	Gases, non-flammable, non-toxic gases.
9	IMDG Code Class 2 Div. 2.3	Gases, toxic gases.
10	IMDG Code Class 3	Flammable liquids.
11	IMDG Code Class 4 Div. 4.1	Flammable solids, self-reactive substances and desensitized explosives.
12	IMDG Code Class 4 Div. 4.2	Substances liable to spontaneous combustion.
13	IMDG Code Class 4 Div. 4.3	Substances which, in contact with water, emit flammable gases.
14	IMDG Code Class 5 Div. 5.1	Oxidizing substances.
15	IMDG Code Class 5 Div. 5.2	Organic peroxides.
16	IMDG Code Class 6 Div. 6.1	Toxic substances.
17	IMDG Code Class 6 Div. 6.2	Infectious substances.
18	IMDG Code Class 7	Radioactive material.
19	IMDG Code Class 8	Corrosive substances.
20	IMDG Code Class 9	Miscellaneous dangerous substances and articles.
21	Harmful Substances in Packaged Form	Harmful substances are those substances which are identified as marine pollutants in the International Maritime Dangerous Goods Code (IMDG Code). Packaged form is defined as the forms of containment specified for harmful substances in the IMDG Code.

16.10 Category of Relationship

Definition : Expresses constraints or requirements on vessel actions or activities in relation to a geographic feature, facility, or service.

Type : enumeration

CamelCase : categoryOfRelationship

Alias :

Remarks :

Code	Label	Definition
1	Prohibited	Use of facility, waterway or service is forbidden.
2	Not Recommended	Use of facility, waterway or service is not recommended.
3	Permitted	Use of facility, waterway, or service is permitted but not required.
4	Recommended	Use of facility, waterway, or service is recommended.
5	Required	Use of facility, waterway, or service is required.
6	Not Required	Use of facility, waterway, or service is not required.
7	Exclusively Permitted	Only vessels of the specified characteristics may use the facility, waterway, or service.

16.11 Category of Restricted Area

Definition : The official legal status of each kind of restricted area defines the kind of restriction(s), for example the restriction for a ‘game reserve’ may be ‘entering prohibited’.

Type : enumeration

CamelCase : categoryOfRestrictedArea

Alias : CATREA

Remarks :

Code	Label	Definition
1	Offshore Safety Zone	The area around an offshore installation within which vessels are prohibited from entering without permission. Special regulations protect installations within a safety zone and vessels of all nationalities are required to respect the zone.
4	Nature Reserve	A tract of land or water managed so as to preserve its flora, fauna, physical features, etc.
5	Bird Sanctuary	A place where birds are bred and protected.
6	Game Reserve	A place where wild animals or birds hunted for sport or food are kept undisturbed for private use.
7	Seal Sanctuary	A place where seals are protected.
10	Historic Wreck Area	An area around certain wrecks of historical importance to protect the wrecks from unauthorized interference by diving, salvage or deposition (including anchoring).
20	Research Area	An area where marine research takes place.
22	Fish Sanctuary	A place where fish (including shellfish and crustaceans) are protected.
23	Ecological Reserve	A tract of land managed so as to preserve the relation of plants and living creatures to each other and to their surroundings.
27	Environmentally Sensitive Sea Area	A generic term which may be used to describe a wide range of areas, considered sensitive for a variety of environmental reasons.

Code	Label	Definition
28	Particularly Sensitive Sea Area	An area that needs special protection through action by IMO because of its significance for regional ecological, socio-economic or scientific reasons and because it may be vulnerable to damage by international shipping activities.
31	Coral Sanctuary	A place where coral is protected.
32	Recreation Area	An area within which recreational activities regularly take place and therefore vessel movement may be restricted.
33	Ship Pollution Emission Control	An area within which the ship pollution emission is controlled.

16.12 Category of Schedule

Definition : The type of schedule, for instance opening, closure, etc.

Type : enumeration

CamelCase : categoryOfSchedule

Alias :

Remarks :

Code	Label	Definition
1	Normal Operation	The service, office, is open, fully manned, and operating normally, or the area is accessible as usual.
2	Closure	The service, office, or area is closed.
3	Unmanned Operation	The service is available but not manned.

16.13 Category of Temporal Variation

Definition : An assessment of the likelihood of change over time.

Type : enumeration

CamelCase : categoryOfTemporalVariation

Alias :

Remarks :

Code	Label	Definition
1	Extreme Event	Indication of the possible impact of a significant event (for example hurricane, earthquake, volcanic eruption, landslide, etc), which is considered likely to have changed the seafloor or landscape significantly.
4	Likely to Change	Continuous or frequent change to non-bathymetric features (for example river siltation, glacier creep/recession, sand dunes, buoys, marine farms, etc).
5	Unlikely to Change	Significant change to the seafloor is not expected.
6	Unassessed	Not having been assessed.

16.14 Category of Text

Definition : Classification of completeness of textual information in relation to the source material from which it is derived.

Type : enumeration

CamelCase : categoryOfText

Alias : CATTXT

Remarks :

Code	Label	Definition
1	Abstract or Summary	A statement summarizing the important points of a text.
2	Extract	An excerpt or excerpts from a text.
3	Full Text	The whole text.

16.15 Category of Vessel Registry

Definition : The locality of vessel registration or enrolment relative to the nationality of a port, territorial sea, administrative area, exclusive zone or other location.

Type : enumeration

CamelCase : categoryOfVesselRegistry

Alias :

Remarks :

Code	Label	Definition
1	Domestic	The vessel is registered or enrolled under the same national flag as the port, harbour, territorial sea, exclusive economic zone, or administrative area in which the object that possesses this attribute applies or is located.
2	Foreign	The vessel is registered or enrolled under a national flag different from the port, harbour, territorial sea, exclusive economic zone, or other administrative area in which the object that possesses this attribute applies or is located.

16.16 City Name

Definition : The name of a town or city.

Type : text

CamelCase : cityName

Alias : CITYNM

Remarks :

16.17 Communication Channel

Definition : A channel number assigned to a specific radio frequency, frequencies or frequency band.

Type : text

CamelCase : communicationChannel

Alias : COMCHA

Remarks : The expected input is the specific VHF-Channel. The attribute ‘communication channel’ encodes the various VHF-channels used for communication.

16.18 Comparison Operator

Definition : Numerical comparison.

Type : enumeration

CamelCase : comparisonOperator

Alias : COMPOP

Remarks : Provides the relation between the value given in the model and the real ship’s value.

Code	Label	Definition
1	Greater Than	The value of the left value is greater than that of the right.
2	Greater Than or Equal To	The value of the left expression is greater than or equal to that of the right.
3	Less Than	The value of the left expression is less than that of the right.
4	Less Than or Equal To	The value of the left expression is less than or equal to that of the right.
5	Equal To	The two values are equivalent.
6	Not Equal To	The two values are not equivalent.

16.19 Contact Instructions

Definition : Instructions provided on how to contact a particular person, organisation or service.

Type : text

CamelCase : contactInstructions

Alias :

Remarks :

16.20 Country Name

Definition : The name of a nation.

Type : text

CamelCase : countryName

Alias :

Remarks :

16.21 Date End

Definition : The latest date on which an object (for example a buoy) will be present.

Type : S100_TruncatedDate

CamelCase : dateEnd

Alias : DATEND

Remarks : The Date End should be encoded using 4 digits for the calendar year (YYYY), 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD). When no specific month and/or day

is required/known, indication of the month and/or day is omitted, and replaced with dashes (-). When no specific year is required (that is, the event or date range ends at the same time each year) the following two cases may be considered:- same day each year: ——MMDD- same month each year: ——MM— This conforms to ISO 8601: 2004. Date End indicates the latest date of an event or the end of a date range. It is used to indicate the end of a fixed date range, the end of a periodic date range, or the removal or cancellation of a feature at a specific date in the future.

16.22 Date Fixed

Definition : The date of an event.

Type : S100_TruncatedDate

CamelCase : dateFixed

Alias :

Remarks :

16.23 Date Start

Definition : The earliest date on which an object (for example a buoy) will be present.

Type : S100_TruncatedDate

CamelCase : dateStart

Alias : DATSTA

Remarks : The Date Start should be encoded using 4 digits for the calendar year (YYYY), 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD). When no specific month and/or day is required/known, indication of the month and/or day is omitted, and replaced with dashes (-). When no specific year is required (that is, the event or date range ends at the same time each year) the following two cases may be considered:- same day each year: ——MMDD- same month each year: ——MM— This conforms to ISO 8601: 2004. Date Start indicates the earliest date of an event or the start of a date range. It is used to indicate the start of a fixed date range, the start of a periodic date range, or the deployment or implementation of a feature at a specific date in the future.

16.24 Date Variable

Definition : A day which is not fixed in the Gregorian calendar.

Type : text

CamelCase : dateVariable

Alias :

Remarks : Examples: The fourth Thursday in November; new moon day of Kartika (Diwali); Easter Sunday.

16.25 Day of Week

Definition : Any one of seven days in a week.

Type : enumeration

CamelCase : dayOfWeek

Alias :

Remarks :

Code	Label	Definition
1	Sunday	The day of the week following Saturday and preceding Monday.
2	Monday	The day of the week following Sunday and preceding Tuesday.
3	Tuesday	The day of the week following Monday and preceding Wednesday.
4	Wednesday	The day of the week following Tuesday and preceding Thursday.
5	Thursday	The day of the week following Wednesday and preceding Friday.
6	Friday	The day of the week following Thursday and preceding Saturday.
7	Saturday	The day of the week following Friday and preceding Sunday.

16.26 Day of Week is Range

Definition : A statement expressing if the days of the week identified define a range or not.

Type : boolean

CamelCase : dayOfWeekIsRange

Alias :

Remarks : A True value is an indication that the identified days of the week define a range between and inclusive of those days.

16.27 Delivery Point

Definition : Details of where post can be delivered such as the apartment, name and/or number of a street, building or PO Box.

Type : text

CamelCase : deliveryPoint

Alias : DELPNT

Remarks :

16.28 Designation Identifier

Definition : An identifier which is an instance of a particular, named designation scheme.

Type : text

CamelCase : designationIdentifier

Alias :

Remarks :

16.29 Designation Scheme

Definition : The name of a scheme under which an identifier is defined.

Type : text

CamelCase : designationScheme

Alias :

Remarks :

16.30 Destination

Definition : The place or general direction to which a vessel is going or directed.

Type : text

CamelCase : destination

Alias :

Remarks : In addition to a place name of a port, harbour area or terminal, the place could include generalities such as The north-west, or upriver.

Max. length: 100

16.31 Distance

Definition : A numeric measure of the spatial separation between two locations.

Type : real

CamelCase : distance

Alias :

Remarks :

Units: Nautical mile **Definition**: A unit of length equal to 1,852 metres. This value was approved by the International Hydrographic Conference of 1929 and has been adopted by nearly all maritime states. **Symbol**: M

Range: Lower Bound (Exclusive): 0.0 Upper Bound: (not specified)

16.32 File Locator

Definition : The location of a fragment of text or other information in a support file.

Type : text

CamelCase : fileLocator

Alias :

Remarks : Application schemas must describe how the associated file is identified. The associated file will commonly be named in a file reference co-attribute of the same complex attribute. Each DCEG must specify requirements for the format of the associated file and the semantics of file locator. For example, the value of file locator may be an HTML ID in an HTML file, line number in a text file) or a bookmark in a PDF file.

16.33 File Reference

Definition : The file name of an externally referenced text file.

Type : text

CamelCase : fileReference

Alias : TXTDSC

Remarks :

16.34 Frequency Shore Station Receives

Definition : The shore station receiver frequency.

Type : integer

CamelCase : frequencyShoreStationReceives

Alias : FRQRXV

Remarks : Examples: 4379.1 kHz becomes 043791; 13162.8 kHz becomes 131628.

Units: Hertz **Definition**: The derived unit of frequency in the metric system. **Symbol**: Hz

Range: Lower Bound (Exclusive): 0 Upper Bound: (not specified)

16.35 Frequency Shore Station Transmits

Definition : The shore station transmitter frequency.

Type : integer

CamelCase : frequencyShoreStationTransmits

Alias : FRQTXM

Remarks : Examples: 4379.1 kHz becomes 043791; 13162.8 kHz becomes 131628.

Units: Hertz **Definition**: The derived unit of frequency in the metric system. **Symbol**: Hz

Range: Lower Bound (Exclusive): 0 Upper Bound: (not specified)

16.36 Headline

Definition : Words set at the head of a passage or page to introduce or categorize.

Type : text

CamelCase : headline

Alias :

Remarks :

16.37 Horizontal Distance Uncertainty

Definition : The best estimate of the horizontal accuracy of horizontal clearances and distances.

Type : real

CamelCase : horizontalDistanceUncertainty

Alias : HORACC

Remarks : The error is assumed to be positive and negative. The plus/minus character must not be encoded.

Units: Metre **Definition**: The basic unit of length in the International System of Units (SI) system. **Symbol**: m

Range: Lower Bound (Inclusive): 0 Upper Bound: (not specified)

16.38 In Ballast

Definition : Whether the vessel is in ballast.

Type : boolean

CamelCase : inBallast

Alias :

Remarks :

16.39 Interoperability Identifier

Definition : A common unique identifier for entities which describe a single real-world feature, and which is used to identify instances of the feature in end-user systems where the feature may be included in multiple data product types.

Type : URN

CamelCase : interoperabilityIdentifier

Alias :

Remarks :

Text pattern: urn:mrn:.+

16.40 Jurisdiction

Definition : The jurisdiction applicable to an administrative area.

Type : enumeration

CamelCase : jurisdiction

Alias : JRSDTN

Remarks :

Code	Label	Definition
1	International	Involving more than one country; covering more than one national area.
2	National	An area administered or controlled by a single nation.
3	National Sub-Division	An area smaller than the nation in which it lies.

16.41 Language

Definition : The method of human communication, either spoken or written, consisting of the use of words in a structured and conventional way.

Type : text

CamelCase : language

Alias :

Remarks : The language is encoded by a 3 character code following ISO 639-2/T.

16.42 Linkage

Definition : Location (address) for online access using a URL/URI address or similar addressing scheme.

Type : URI

CamelCase : linkage

Alias :

Remarks :

16.43 Logical Connectives

Definition : Expresses whether all the constraints described by its co-attributes must be satisfied, or only one such constraint need be satisfied.

Type : enumeration

CamelCase : logicalConnectives

Alias : LOGCON

Remarks : Is intended to be used with co-attributes that encode limits on vessel dimensions, type of cargo, and other characteristics. The combination of constraints described by logicalConnectives and its co-attributes defines a subset of vessels to which information described by a feature or information type instance applies (or does not apply, is required, recommended, etc.). The relationship between the vessel subset and the information is indicated by an association — see PermissionType and InclusionType). The two listed values of logicalConnective are two of the basic operations of Boolean logic. The third basic operation (not) is not used.

Code	Label	Definition
1	Logical Conjunction	All the conditions described by the other attributes of the object, or sub-attributes of the same complex attribute, are true.
2	Logical Disjunction	At least one of the conditions described by the other attributes of the object, or sub-attributes of the same complex attributes, is true.

16.44 Maximum Display Scale

Definition : The value considered by the Data Producer to be the maximum (largest) scale at which the data is to be displayed before it can be considered to be “grossly overscaled”.

Type : integer

CamelCase : maximumDisplayScale

Alias :

Remarks :

Range: Lower Bound (Inclusive): 1 Upper Bound: (not specified)

16.45 Membership

Definition : Indicates whether a vessel is included or excluded from the regulation/restriction/recommendation/nautical information.

Type : enumeration

CamelCase : membership

Alias :

Remarks :

Code	Label	Definition
1	Included	Vessels with these characteristics are included in the regulation/restriction/recommendation/nautical information.
2	Excluded	Vessels with these characteristics are excluded from the regulation/restriction/recommendation/nautical information.

16.46 Minimum Display Scale

Definition : The smallest intended viewing scale for the data.

Type : integer

CamelCase : minimumDisplayScale

Alias :

Remarks :

Range: Lower Bound (Inclusive): 1 Upper Bound: (not specified)

16.47 MMSI Code

Definition : The Maritime Mobile Service Identity (MMSI) Code is formed of a series of nine digits which are transmitted over the radio path in order to uniquely identify ship stations, ship earth stations, coast stations, coast earth stations, and group calls. These identities are formed in such a way that the identity or part thereof can be used by telephone and telex subscribers connected to the general telecommunications network principally to call ships automatically.

Type : text

CamelCase : mMSICode

Alias :

Remarks :

16.48 Name

Definition : The individual name of a feature.

Type : text

CamelCase : name

Alias : OBJNAM

Remarks :

16.49 Name of Resource

Definition : Name of the online resource.

Type : text

CamelCase : nameOfResource

Alias :

Remarks :

16.50 Name Usage

Definition : Classification of the type and display level of the name of a feature in an end-user system.

Type : enumeration

CamelCase : nameUsage

Alias :

Remarks :

Code	Label	Definition
1	Default Name Display	The name is intended to be displayed when the end-user system is set to the default name/text display setting.
2	Alternate Name Display	The name is intended to be displayed when the end-user system is set to an alternate name/text display setting, for example an alternate language.
3	No Chart Display	The name or text is not intended to be displayed.

16.51 Online Function

Definition : Code for function performed by the online resource.

Type : enumeration

CamelCase : onlineFunction

Alias :

Remarks :

Code	Label	Definition
1	Download	Online instructions for transferring data from one storage device or system to another.
3	Offline Access	Online instructions for requesting the resource from the provider.
4	Order	Online order process for obtaining the resource.
5	Search	To make painstaking investigation or examination.
6	Complete Metadata	Complete metadata provided.
7	Browse Graphic	Browse graphic provided.
8	Upload	Online resource upload capability provided.
9	Email Service	Online email service provided.
10	Browsing	Online browsing provided.
11	File Access	Online file access provided.

16.52 Online Resource Description

Definition : Detailed text description of what the online resource is/does.

Type : text

CamelCase : onlineResourceDescription

Alias :

Remarks :

16.53 Optimum Display Scale

Definition : The largest intended viewing scale for the data.

Type : integer

CamelCase : optimumDisplayScale

Alias : CSCALE

Remarks :

Range: Lower Bound (Inclusive): 1 Upper Bound: (not specified)

16.54 Orientation Uncertainty

Definition : The best estimate of the accuracy of a bearing.

Type : real

CamelCase : orientationUncertainty

Alias :

Remarks :

Units: Degree **Definition**: degrees of arc **Symbol**: °

Range: Lower Bound (Inclusive): 0.000 Upper Bound (Inclusive): 360.000

16.55 Orientation Value

Definition : The angular distance measured from true north to the major axis of the feature.

Type : real

CamelCase : orientationValue

Alias : ORIENT

Remarks :

Units: Degree **Definition**: degrees of arc **Symbol**: °

Range: Lower Bound (Inclusive): 0.0 Upper Bound (Inclusive): 360.0

16.56 Pictorial Representation

Definition : The file name of an externally referenced picture file.

Type : text

CamelCase : pictorialRepresentation

Alias : PICREP

Remarks : The ‘pictorial representation’ could be a drawing or a photo. The string encodes the file name of an external graphic file (pixel/vector).

16.57 Picture Caption

Definition : Short description of the purpose of the image.

Type : text

CamelCase : pictureCaption

Alias :

Remarks :

16.58 Picture Information

Definition : A set of information to provide credits to picture creator, copyright owner etc.

Type : text

CamelCase : pictureInformation

Alias :

Remarks :

16.59 Postal Code

Definition : Known in various countries as a postcode, or ZIP code, the postal code is a series of letters and/or digits that identifies each postal delivery area.

Type : text

CamelCase : postalCode

Alias : POSCOD Postcode ZIP Code

Remarks :

16.60 Protocol

Definition : Connection protocol to be used. Example: ftp, http get KVP, http POST, etc.

Type : text

CamelCase : protocol

Alias : PROTCL

Remarks :

16.61 Protocol Request

Definition : Request used to access the resource. Structure and content depend on the protocol and standard used by the online resource, such as Web Feature Service standard.

Type : text

CamelCase : protocolRequest

Alias : PROTRQ

Remarks :

16.62 Quality of Horizontal Measurement

Definition : The degree of reliability attributed to a position.

Type : enumeration

CamelCase : qualityOfHorizontalMeasurement

Alias : QUAPOS

Remarks :

Code	Label	Definition
1	Surveyed	The position(s) was(were) determined by the operation of making measurements for determining the relative position of points on, above or beneath the earth's surface. Survey implies a regular, controlled survey of any date.
2	Unsurveyed	Survey data is does not exist or is very poor.
3	Inadequately Surveyed	Not surveyed to modern standards; or due to its age, scale, or positional or vertical uncertainties is not suitable to the type of navigation expected in the area.
4	Approximate	A position that is considered to be less than third-order accuracy, but is generally considered to be within 30.5 metres of its correct geographic location. Also may apply to an object whose position does not remain fixed.
5	Position Doubtful	Of uncertain position. The expression is used principally on charts to indicate that a wreck, shoal, etc., has been reported in various positions and not definitely determined in any.
6	Unreliable	A feature's position has been obtained from questionable or unreliable data.
7	Reported (Not Surveyed)	An object whose position has been reported and its position confirmed by some means other than a formal survey such as an independent report of the same object.
8	Reported (Not Confirmed)	An object whose position has been reported and its position has not been confirmed.
9	Estimated	The most probable position of an object determined from incomplete data or data of questionable accuracy.
10	Precisely Known	A position that is of a known value, such as the position of an anchor berth or other defined object.
11	Calculated	A position that is computed from data.

16.63 Reported Date

Definition : The date that the item was observed, done, or investigated.

Type : S100_TruncatedDate

CamelCase : reportedDate

Alias : SORDAT

Remarks :

16.64 Restriction

Definition : The official legal statute of each kind of restricted area.

Type : enumeration

CamelCase : restriction

Alias : RESTRN

Remarks : Defines the kind of restriction(s), for example, the restriction for 'a game preserve' may be 'entry prohibited', the restriction for an 'anchoring prohibition' is 'anchoring prohibited'. The complete information about the restriction(s), actually held in handbooks or other publications, may be encoded using an Information type.

Code	Label	Definition
1	Anchoring Prohibited	An area within which anchoring is not permitted.

Code	Label	Definition
2	Anchoring Restricted	A specified area designated by appropriate authority, within which anchoring is restricted in accordance with certain specified conditions.
3	Fishing Prohibited	An area within which fishing is not permitted.
4	Fishing Restricted	A specified area designated by appropriate authority, within which fishing is restricted in accordance with certain specified conditions.
5	Trawling Prohibited	An area within which trawling is not permitted.
6	Trawling Restricted	A specified area designated by appropriate authority, within which trawling is restricted in accordance with certain specified conditions.
7	Entry Prohibited	An area within which navigation and/or anchoring is prohibited.
8	Entry Restricted	A specified area designated by appropriate authority, within which navigation is restricted in accordance with certain specified conditions.
9	Dredging Prohibited	An area within which dredging is not permitted.
10	Dredging Restricted	A specified area designated by appropriate authority, within which dredging is restricted in accordance with certain specified conditions.
11	Diving Prohibited	An area within which diving is not permitted.
12	Diving Restricted	A specified area designated by appropriate authority, within which diving is restricted in accordance with certain specified conditions.
13	No Wake	Mariners must adjust the speed of their vessels to reduce the wave or wash which may cause erosion or disturb moored vessels.
14	Area To Be Avoided	An IMO declared routeing measure comprising an area within defined limits in which either navigation is particularly hazardous or it is exceptionally important to avoid casualties and which should be avoided by all ships, or certain classes of ships.
15	Construction Prohibited	The erection of permanent or temporary fixed structures or artificial islands is prohibited.
16	Discharging Prohibited	An area within which discharging or dumping is prohibited.
17	Discharging Restricted	A specified area designated by an appropriate authority, within which discharging or dumping is restricted in accordance with specified conditions.
18	Industrial or Mineral Exploration/ Development Prohibited	An area within which industrial or mineral exploration and development are prohibited.
19	Industrial or Mineral Exploration/ Development Restricted	A specified area designated by an appropriate authority, within which industrial or mineral exploration and development is restricted in accordance with certain specified conditions.
20	Drilling Prohibited	An area within which excavating a hole on the sea-bottom with a drill is prohibited.
21	Drilling Restricted	A specified area designated by an appropriate authority, within which excavating a hole on the sea-bottom with a drill is restricted in accordance with certain specified conditions.
22	Removal of Historical Artefacts Prohibited	An area within which the removal of historical artefacts is prohibited.
23	Cargo Transhipment (Lightening) Prohibited	An area in which cargo transhipment (lightening) is prohibited.
24	Dragging Prohibited	An area in which the dragging of anything along the bottom, e.g. bottom trawling, is prohibited.

Code	Label	Definition
25	Stopping Prohibited	An area in which a vessel is prohibited from stopping.
26	Landing Prohibited	An area in which landing is prohibited.
27	Speed Restricted	An area within which speed is restricted.
38	Use of Spuds Prohibited	The use of anchoring spuds (telescopic piles) is prohibited.
39	Swimming Prohibited	An area in which swimming is prohibited.
40	SOx Emission Restricted	An area within which the emission of SOx is restricted.
41	NOx Emission Restricted	An area within which the emission of NOx is restricted.
42	Power-Driven Vessels Prohibited	An area within which any vessel propelled by machinery is prohibited.

16.65 Scale Minimum

Definition : The minimum scale at which the feature may be used for example for ECDIS presentation.

Type : integer

CamelCase : scaleMinimum

Alias : SCAMIN

Remarks : The modulus of the scale is indicated, that is 1:1 250 000 is encoded as 1250000.

Additional Remarks:

- The *scaleMinimum* value of a feature determines the display scale below which the feature is no longer displayed. Its purpose is to reduce clutter, to prioritise the display of features and to improve display speed. In encoding its value, the producing authority should consider these factors, as well as the scale at which the feature is no longer likely to be required for navigation.
- *scaleMinimum* only affects the display of a feature on an ECDIS, not its presence in the Electronic Navigational Data Service (ENDS).
- If *scaleMinimum* is not encoded, the feature is displayed at all scales.
- Where *scaleMinimum* is used, it must always be set to a scale less (that is, to a smaller scale) than the optimum display scale of the data. Failure to follow this rule will mean that features will not be displayed on the ECDIS until the overscale indication is activated.
- *scaleMinimum* is not encoded on Meta features.
- If the same feature exists in datasets of different optimum display scales, consideration should be given to whether each occurence of the feature should be assigned the same *scaleMinimum* value.
  - This is a weaker requirement than S-101, which requires that each occurence must be assigned the same *scaleMinimum* value.
  - This recommendation does not apply to *scaleMinimum* values encoded on links to spatial primitives, which are intended to allow for more or less detailed geometry according to scale; however, if *scaleMinimum* values are present on links to spatial primitives, the values should be consistent with any *scaleMinimum* values encoded as feature attributes. In case of conflict, the *scaleMinimum* feature attributes prevail.

Range: Lower Bound (Inclusive): 999 Upper Bound (Inclusive): 19999999

16.66 Source

Definition : The publication, document, or reference work from which information comes or is acquired.

Type : text

CamelCase : source

Alias :

Remarks : May be populated with the corresponding paper chart Notice to Mariners numbers, although other references are permitted.

16.67 Source Date

Definition : The production date of the source; for example the date of measurement.

Type : date

CamelCase : sourceDate

Alias : SORDAT

Remarks :

16.68 Source Type

Definition : Type of the source.

Type : enumeration

CamelCase : sourceType

Alias :

Remarks :

Code	Label	Definition
1	Law or Regulation	Treaty, convention, or international agreement; law or regulation issued by a national or other authority.
2	Official Publication	Publication not having the force of law, issued by an international organisation or a national or local administration.
7	Mariner Report, Confirmed	Reported by mariner(s) and confirmed by another source.
8	Mariner Report, Not Confirmed	Reported by mariner(s) but not confirmed.
9	Industry Publications and Reports	Shipping and other industry publications, including graphics, charts and web sites.
10	Remotely Sensed Images	Information obtained from satellite images.
11	Photographs	Information obtained from photographs.
12	Products Issued by HO Services	Information obtained from products issued by Hydrographic Offices.
13	News Media	Information obtained from news media.
14	Traffic Data	Information obtained from the analysis of traffic data.

16.69 Status

Definition : The condition of an object at a given instant in time.

Type : enumeration

CamelCase : status

Alias : STATUS

Remarks :

Code	Label	Definition
1	Permanent	Intended to last or function indefinitely.
2	Occasional	Acting on special occasions; happening irregularly.
3	Recommended	Presented as worthy of confidence, acceptance, use, etc.
4	Not in Use	Use has ceased, but the facility still exists intact; disused.
5	Periodic/ Intermittent	Recurring at intervals.
6	Reserved	Set apart for some specific use.
7	Temporary	Meant to last only for a time.
9	Mandatory	Compulsory; enforced.
13	Historic	Famous in history; of historical interest.
14	Public	Belonging to, available to, used or shared by, the community as a whole and not restricted to private use.
18	Existence Doubtful	A feature that has been reported but has not been definitely determined to exist.
28	Buoyed	Marked by buoys.

16.70 Telecommunication Carrier

Definition : The name of a provider or type of carrier for a telecommunication service. This service may include land line based, shore based or satellite based radio connections.

Type : text

CamelCase : telecommunicationCarrier

Alias :

Remarks :

16.71 Telecommunication Identifier

Definition : An identifier, such as words, numbers, letters, symbols, or any combination of those used to establish a contact to a particular person, organisation or service.

Type : text

CamelCase : telecommunicationIdentifier

Alias :

Remarks :

16.72 Telecommunication Service

Definition : Classification of methods of communication over a distance by electrical, electronic, or electromagnetic means.

Type : enumeration

CamelCase : telecommunicationService

Alias :

Remarks :

Code	Label	Definition
1	Voice	The transfer or exchange of information by using sounds that are being made by mouth and throat when speaking.
2	Facsimile	A system of transmitting and reproducing graphic matter (as printing or still pictures) by means of signals sent over telephone lines.
3	SMS	Short Message Service is a form of text messaging communication on phones and mobile phones.
4	Data	A representation of facts, concepts or instructions in a formalised manner suitable for communication, interpretation or processing.
5	Streamed Data	Data that is constantly received by and presented to an end-user while being delivered by a provider.
6	Telex	A system of communication in which messages are sent over long distances by using a telephone system and are printed by using a special machine (called a teletypewriter).
7	Telegraph	An apparatus, system or process for communication at a distance by electric transmission over wire.
8	Email	Messages and other data exchanged between individuals using computers in a network.

16.73 Text

Definition : A non-formatted digital text string.

Type : text

CamelCase : text

Alias : INFORM NINFOM

Remarks : Should be used, for example, to hold the information that is for short cautionary or explanatory notes. Therefore, text populated in text must not exceed 300 characters. Text may be in English, or in a national language. No formatting of text is possible within text. If formatted text, or text strings exceeding 300 characters, is required, then an alternate concept should be used.

16.74 Text Offset Bearing

Definition : The angular distance measured from true north that text associated with a feature is positioned from the feature in an end-user system.

Type : integer

CamelCase : textOffsetBearing

Alias :

Remarks :

Units: Degree of Arc **Definition**: 1° = (π/180) rad **Symbol**: °

Range: Lower Bound (Exclusive): 0 Upper Bound (Exclusive): 360

16.75 Text Offset Distance

Definition : The distance that text associated with a feature is positioned from the feature in an end-user system.

Type : integer

CamelCase : textOffsetDistance

Alias :

Remarks :

Units: Millimetre **Definition**: 1 metre = 1000 millimetres **Symbol**: mm

Range: Lower Bound (Exclusive): 0 Upper Bound (Inclusive): 50

16.76 Text Rotation

Definition : A statement that expresses if text associated with a feature is to be rotated in the ECDIS display or not.

Type : boolean

CamelCase : textRotation

Alias :

Remarks :

16.77 Text Type

Definition : The attribute from which a text string is derived.

Type : enumeration

CamelCase : textType

Alias :

Remarks :

Code	Label	Definition
1	Name	The individual name of a feature.

16.78 Thickness of Ice Capability

Definition : The thickness of ice that the ship can safely transit.

Type : integer

CamelCase : thicknessOfIceCapability

Alias :

Remarks :

Units: Centimetre **Definition**: 1 metre = 100 centimetres. **Symbol**: cm

Range: Lower Bound (Exclusive): 0 Upper Bound: (not specified)

16.79 Time of Day End

Definition : The time corresponding to the end of an active period.

Type : time

CamelCase : timeOfDayEnd

Alias :

Remarks : The time of day end must be encoded using 2 digits for the hour (hh), 2 digits for the minutes(mm) and 2 digits for the seconds (ss). This conforms to ISO 8601:2004.

16.80 Time of Day Start

Definition : The time corresponding to the start of an active period.

Type : time

CamelCase : timeOfDayStart

Alias :

Remarks : The time of day start must be encoded using 2 digits for the hour (hh), 2 digits for the minutes(mm) and 2 digits for the seconds (ss). This conforms to ISO 8601:2004.

16.81 Uncertainty Fixed

Definition : The best estimate of the fixed horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances.

Type : real

CamelCase : uncertaintyFixed

Alias : POSACC SOUACC VERACC

Remarks :

Units: Metre **Definition**: The basic unit of length in the International System of Units (SI) system. **Symbol**: m

16.82 Uncertainty Variable Factor

Definition : The factor to be applied to the variable component of an uncertainty equation so as to provide the best estimate of the variable horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances.

Type : real

CamelCase : uncertaintyVariableFactor

Alias :

Remarks :

Range: Lower Bound (Exclusive): 0.00 Upper Bound (Rxclusive): 1.00

16.83 Vessel Performance

Definition : A description of the required handling characteristics of a vessel including hull design, main and auxiliary machinery, cargo handling equipment, navigation equipment and manoeuvring behaviour.

Type : text

CamelCase : vesselPerformance

Alias :

Remarks :

16.84 Vessels Characteristics

Definition : Characteristics of vessels.

Type : enumeration

CamelCase : vesselsCharacteristics

Alias :

Remarks :

Code	Label	Definition
1	Length Overall	The maximum length of the ship.
2	Length at Waterline	The ship's length measured at the waterline.
3	Breadth	The width or beam of the vessel.
4	Draught	The depth of water necessary to float a vessel fully loaded.
6	Displacement Tonnage	A measurement of the weight of the vessel, usually used for warships. (Merchant ships are usually measured based on the volume of cargo space; see tonnage). Displacement is expressed either in long tons of 2,240 pounds or metric tonnes of 1,000 kg. Since the two units are very close in size (2,240 pounds = 1,016 kg and 1,000 kg = 2,205 pounds), it is common not to distinguish between them. To preserve secrecy, nations sometimes misstate a warship's displacement.
7	Displacement Tonnage, Light	The weight of the ship excluding cargo, fuel, ballast, stores, passengers, and crew, but with water in the boilers to steaming level.
8	Displacement Tonnage, Loaded	The weight of the ship including cargo, passengers, fuel, water, stores, dunnage and such other items necessary for use on a voyage, which brings the vessel down to her load draft.
9	Deadweight Tonnage	The difference between displacement, light and displacement, loaded. A measure of the ship's total carrying capacity.
10	Gross Tonnage	The entire internal cubic capacity of the ship expressed in tons of 100 cubic feet to the ton, except certain spaces with are exempted such as: peak and other tanks for water ballast, open forecastle bridge and poop, access of hatchways, certain light and air spaces, domes of skylights, condenser, anchor gear, steering gear, wheel house, galley and cabin for passengers.
11	Net Tonnage	Obtained from the gross tonnage by deducting crew and navigating spaces and allowances for propulsion machinery.
12	Panama Canal/ Universal Measurement System Net Tonnage	The Panama Canal/Universal Measurement System (PC/UMS) is based on net tonnage, modified for Panama Canal purposes. PC/UMS is based on a mathematical formula to calculate a vessel's total volume; a PC/UMS net ton is equivalent to 100 cubic feet of capacity.
13	Suez Canal Net Tonnage	The Suez Canal Net Tonnage (SCNT) is derived with a number of modifications from the former net register tonnage of the Moorsom System and was established by the International Commission of Constantinople in its Protocol of 18 December 1873. It is still in use, as amended by the Rules of Navigation of the Suez Canal Authority, and is registered in the Suez Canal Tonnage Certificate.

16.85 Vessels Characteristics Unit

Definition : The unit used for vessel characteristics attribute.

Type : enumeration

CamelCase : vesselsCharacteristicsUnit

Alias : VSLUNT

Remarks :

Code	Label	Definition
1	Metres	The basic unit of length in the International System of Units (SI) system.
3	Metric Ton	The tonne or metric ton (U.S.), often redundantly referred to as a metric tonne, is a unit of mass equal to 1,000 kg (2,205 lb) or approximately the mass of one cubic metre of water at four degrees Celsius. It is sometimes abbreviated as mt in the United States, but this conflicts with other SI symbols. The tonne is not a unit in the International System of Units (SI), but is accepted for use with the SI. In SI units and prefixes, the tonne is a megagram (Mg). The Imperial and US customary units comparable to the tonne are both spelled ton in English, though they differ in mass. Pronunciation of tonne (the word used in the UK) and ton is usually identical, but is not too confusing unless accuracy is important as the tonne and UK long ton differ by only 1.6.
4	Ton	Long ton (weight ton or imperial ton) is the name for the unit called the “ton” in the avoirdupois or Imperial system of measurements, as used in the United Kingdom and several other Commonwealth countries. It has been mostly replaced by the tonne, and in the United States by the short ton. One long ton is equal to 2,240 pounds (1,016 kg) or 35 cubic feet (0.9911 m) of salt water with a density of 64 lb/ft (1.025 g/ml). It has some limited use in the United States, most commonly in measuring the displacement of ships, and was the unit prescribed for warships by the Washington Naval Treaty for example battleships were limited to a mass of 35,000 long tons (36,000 t; 39,000 ST).
5	Short Ton	A unit of weight equal to 2,000 pounds (907.18474 kg). In the United States it is often called simply ton without distinguishing it from the metric ton (tonne, 1,000 kilograms) or the long ton (2,240 pounds / 1,016.0469088 kilograms); rather, the other two are specifically noted. There are, however, some US applications for which unspecified tons normally means long tons (for example, Navy ships) or metric tons (world grain production figures). Both the long and short ton are defined as 20 hundredweights, but a hundredweight is 100 pounds (45.359237 kg) in the US system (short or net hundredweight) and 112 pounds (50.80234544 kg) in the Imperial system (long or gross hundredweight).
6	Gross Ton	Gross tonnage (GT) is a function of the volume of all ship’s enclosed spaces (from keel to funnel) measured to the outside of the hull framing. There is a sliding scale factor. So GT is a kind of capacity-derived index that is used to rank a ship for purposes of determining manning, safety and other statutory requirements and is expressed simply as GT, which is a unitless entity, even though its derivation is tied to the cubic meter unit of volumetric capacity.Tonnage measurements are now governed by an IMO Convention (International Convention on Tonnage Measurement of Ships, 1969 (London-Rules)), which applies to all ships built after July 1982. In accordance with the Convention, the correct term to use now is GT, which is a function of the moulded volume of all enclosed spaces of the ship.
7	Net Ton	Net tonnage (NT) is based on a calculation of the volume of all cargo spaces of the ship. It indicates a vessels earning space and is a function of the moulded volume of all cargo spaces of the ship.
9	Suez Canal Net Tonnage	The Suez Canal Net Tonnage (SCNT) is derived with a number of modifications from the former net register tonnage of the Moorsom System and was established by the International Commission of Constantinople in its Protocol of 18 December 1873. It is still in use, as amended by the Rules of Navigation of the Suez Canal Authority, and is registered in the Suez Canal Tonnage Certificate.

16.86 Vessels Characteristics Value

Definition : The value of a particular characteristic such as a dimension or tonnage of a vessel.

Type : real

CamelCase : vesselsCharacteristicsValue

Alias :

Remarks : Indicates range limits in expressions characterizing vessels by dimensions and tonnages. The unit of measure, characteristic, and comparison operator (greater, less, etc.) are encoded separately.

16.87 Action or Activity

Definition : The action or activity of a vessel.

Type : S100_CodeList

CamelCase : actionOrActivity

Alias :

Remarks :

Code	Label	Definition
1	Navigating With a Pilot	Carrying a qualified pilot as part of the vessel navigation team.
2	Entering Port	Navigating a vessel into a port.
3	Leaving Port	Navigating a vessel out of a port.
4	Berthing	A signal station for the control of vessels when berthing.
5	Slipping	Detaching a vessel from a wharf or jetty.
6	Anchoring	Attaching a vessel to the seabed by means of an anchor and cable.
7	Weighing Anchor	Detaching a vessel from the seabed by recovering an anchor and cable.
8	Transiting	Navigating a vessel along a route or through a narrow gap, such as under a bridge or through a lock.
9	Overtaking	Navigating a vessel past another traveling broadly in the same direction.
10	Reporting	Providing details such as the name, location or intentions of a vessel.
11	Working Cargo	Loading or unloading cargo.
12	Landing	Placing crew or passengers on shore.
13	Diving	A signal or message warning of diving activity.
14	Fishing	Hunting or catching fish.
15	Discharging Overboard	Releasing anything into the sea; often ballast water; or spoil from dredging elsewhere.
16	Passing	Navigating a vessel past another travelling broadly in the opposite direction.
17	Ballast Water Exchange	Discharge and uptake of ballast water.
18	Hull Cleaning	The removal or treatment of biofouling (accumulation of aquatic organisms including microfouling and macrofouling) from a ship's submerged surfaces, including hull and niche areas, conducted either in-water or during dry-docking. The process includes both proactive cleaning (periodic removal of microfouling) and reactive cleaning (removal of micro- and macrofouling as corrective action).
19	Scientific Research	The conduct of observational, sampling, or experimental activities by authorised personnel to collect scientific or environmental data, which may involve the deployment of scientific instruments, collection of biological or geological samples, or in-water survey operations.

Code	Label	Definition
20	Tourism	Organised recreational visitation and leisure activities in marine areas, including sight-seeing, wildlife observation, glass-bottom vessel tours, and guided nature excursions conducted by commercial or permitted operators.
21	Education	Structured activities conducted for training, awareness, or interpretive purposes involving groups or individuals learning about the marine environment, including guided educational programs, school activities, and field instruction conducted within designated marine areas.
22	Infrastructure Maintenance	Inspection, repair, or upkeep of existing marine or coastal infrastructure such as wharves, piers, pipelines, moorings, subsea cables, navigational aids, or coastal protection structures, including minor works that do not expand the original footprint.

16.88 Category of Marine Protected Area

Definition : Classification of marine protected areas based on IUCN (International Union for Conservation of Nature and Natural Resources) categories.

Type : S100_CodeList

CamelCase : categoryOfMarineProtectedArea

Alias :

Remarks :

Code	Label	Definition
1	IUCN Category Ia	Strict Nature Reserve: Protected area managed mainly for science.
2	IUCN Category Ib	Wilderness Area: Protected area managed mainly for wilderness protection.
3	IUCN Category II	National Park: Protected area managed mainly for ecosystem protection and recreation.
4	IUCN Category III	Natural Monument: Protected area managed mainly for conservation of specific natural features.
5	IUCN Category IV	Habitat/Species Management Area: Protected area managed mainly for conservation through management intervention.
6	IUCN Category V	Protected Landscape/Seascape: Protected area managed mainly for landscape/ seascape conservation and recreation.
7	IUCN Category VI	Managed Resource Protected Area: Protected area managed mainly for the sustainable use of natural ecosystems.

16.89 Category of RxN

Definition : The principal subject matter of regulations, restrictions, recommendations or nautical information.

Type : S100_CodeList

CamelCase : categoryOfRxN

Alias :

Remarks :

Code	Label	Definition
1	Navigation	The process of directing the movement of a craft from one point to another.
2	Communication	Transmitting and/or receiving electronic communication signals.
3	Environmental Protection	Pertaining to environmental protection.
4	Wildlife Protection	Pertaining to wildlife protection.
5	Security	Pertaining to security.
6	Customs	The agency or establishment for collecting duties, tolls.
7	Cargo Operation	Pertaining to cargo operations.
8	Refuge	Pertaining to a place of safety or refuge.
9	Health	The authority with responsibility for checking the validity of the health declaration of a vessel and for declaring free pratique.
10	Natural Resources or Exploitation	Pertaining to natural resources or exploitation.
11	Port	Person or corporation, owners of, or entrusted with or invested with the power of managing a port. May be called a Harbour Board, Port Trust, Port Commission, Harbour Commission, Marine Department.
12	Finance	An authority with responsibility for the control and movement of money.
13	Agriculture	The science, art, or practice of cultivating the soil, producing crops, and raising livestock and in varying degrees the preparation and marketing of the resulting products.

16.90 Category of Vessel

Definition : Classification of vessels by function or use.

Type : S100_CodeList

CamelCase : categoryOfVessel

Alias :

Remarks :

Code	Label	Definition
1	General Cargo Vessel	A vessel which is designed for carrying general cargo, e.g. boxes, sacks.
2	Container Carrier	A vessel designed to carry ISO containers.
3	Tanker	A vessel which is designed for carrying liquid goods, for example oil or water.
4	Bulk Carrier	A vessel which is designed for carrying bulk goods, e.g. coal, ore or grain.
5	Passenger Vessel	A day trip or cabin vessel constructed and equipped to carry more than 12 passengers.
6	Roll-On Roll-Off	A vessel designed to allow road vehicles to be driven on and off; often a ferry.
7	Refrigerated Cargo Vessel	A vessel designed to carry refrigerated cargo.
8	Fishing Vessel	A vessel that is used and equipped for the fishing of living aquatic resources.

Code	Label	Definition
9	Service	A vessel which provides a service such as a tug, anchor handler, survey or supply vessel.
10	Warship	A vessel designed for the conduct of military operations.
11	Towed or Pushed Composite Unit	Either a tug and tow, or any combination of a tug providing propulsion to barges or vessels secured ahead or alongside.
12	Tug and Tow	A combination of tug(s) and non-powered tow(s).
13	Light Recreational	A pleasure boat or watercraft, or an excursion vessel used for short cruises such as whale watching.
14	Semi-Submersible Offshore Installation	An installation which is designed to float at all times and which is normally anchored in position when deployed in the offshore gas and oil industry.
15	Jack-Up Exploration or Project Installation	An exploration or project installation with legs which can be raised and lowered. The legs are raised when the installation is re-positioned. When stationary the legs are lowered to the sea floor and the working platform is raised clear of the sea surface.
16	Livestock Carrier	A vessel designed to carry large quantities of live animals.
17	Sport Fishing	A vessel used in fishing for pleasure or competition.

Page intentionally left blank

17 Complex Attributes

17.1 Bearing Information

Definition : A bearing is the direction one object is from another object.

CamelCase : bearingInformation

Alias :

Remarks :

- At least one of the sub-attributes must be populated.

Sub-attributes :

Sub-Attribute	Type	Multiplicity
cardinalDirection	enumeration	0,1
distance	real	0,1
information	Complex	0,*
orientation	Complex	0,1

17.2 Contact Address

Definition : Direction or superscription of a letter, package, etc., specifying the name of the place to which it is directed, and optionally a contact person or organisation who should receive it.

CamelCase : contactAddress

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
deliveryPoint	text	0,* (ordered)
cityName	text	0,1
administrativeDivision	text	0,1
countryName	text	0,1
postalCode	text	0,1

17.3 Designation

Definition : An official name, title or description. This can be an identifier or an identifier which is an instance of a named designation scheme.

CamelCase : designation

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
designationScheme	text	0,1
designationIdentifier	text	0,1
jurisdiction	enumeration	0,1
text	text	0,1

17.4 Feature Name

Definition : Provides the name of an entity, defines the national language of the name, and provides the option to display the name at various system display settings.

CamelCase : featureName

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
language	text	1,1
name	text	1,1
nameUsage	enumeration	0,1

17.5 Fixed Date Range

Definition : An active period of a single fixed event or occurrence, as the date range between discrete start and end dates.

CamelCase : fixedDateRange

Alias :

Remarks : Dates must be encoded in the format YYYYMMDD; using 4 digits for the calendar year (YYYY) and, optionally, 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD). When no specific month and/or day is required/known, the values are replaced with dashes (-). The date range of a recurring event or occurrence must be encoded using periodicDateRange.

Sub-attributes :

Sub-Attribute	Type	Multiplicity
dateStart	S100_TruncatedDate	0,1
dateEnd	S100_TruncatedDate	0,1

17.6 Frequency Pair

Definition : A pair of frequencies for transmitting and receiving radio signals. The shore station transmits and receives on the frequencies indicated.

CamelCase : frequencyPair

Alias : FRQPAR

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
frequencyShoreStationReceives	integer	0,1
frequencyShoreStationTransmits	integer	1,1

17.7 Graphic

Definition : Pictorial information such as a photograph, sketch or other graphic, optionally accompanied by descriptive information about the graphic and the location relative to its subject from which it was made.

CamelCase : graphic

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
pictorialRepresentation	text	1,*
pictureCaption	text	0,1
sourceDate	date	0,1
pictureInformation	text	0,1
bearingInformation	Complex	0,1

17.8 Horizontal Position Uncertainty

Definition : The best estimate of the accuracy of a position.

CamelCase : horizontalPositionUncertainty

Alias : POSACC

Remarks : The expected input is the maximum of the two-dimensional error. The error is assumed to be positive and negative.

Sub-attributes :

Sub-Attribute	Type	Multiplicity
uncertaintyFixed	real	1,1
uncertaintyVariableFactor	real	0,1

17.9 Information

Definition : Textual information about the feature. The information may be provided as a string of text or as a file name of a single external text file that contains the text.

CamelCase : information

Alias : INFORM

Remarks : At least one of the sub-attributes file reference or text must be populated.The sub-attribute file reference is generally used for long text strings or those that require formatting, however, there is

no restriction on the type of text (except for lexical level) that can be held in files referenced by sub-attribute file reference.

Sub-attributes :

Sub-Attribute	Type	Multiplicity
fileLocator	text	0,1
fileReference	text	0,1
headline	text	0,* (ordered)
language	text	0,1
text	text	0,1

17.10 Online Resource

Definition : Information about online sources from which a resource or data can be obtained.

CamelCase : onlineResource

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
linkage	URI	1,1
protocol	text	0,1
applicationProfile	text	0,1
nameOfResource	text	0,1
onlineResourceDescription	text	0,1
protocolRequest	text	0,1
onlineFunction	enumeration	0,1

17.11 Orientation

Definition : (1) The angular distance measured from true north to the major axis of the feature. (2) In ECDIS, the mode in which information on the ECDIS is being presented. Typical modes include: north-up — as shown on a nautical chart, north is at the top of the display; Ships head-up — based on the actual heading of the ship, (e.g. Ships gyrocompass); course-up display — based on the course or route being taken.

CamelCase : orientation

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
orientationUncertainty	real	0,1
orientationValue	real	1,1

17.12 Periodic Date Range

Definition : The active period of a recurring event or occurrence.

CamelCase : periodicDateRange

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
dateStart	S100_TruncatedDate	1,1
dateEnd	S100_TruncatedDate	1,1

17.13 RxN Code

Definition : A summary of the impact of the most common types of regulation, restriction, recommendation and nautical information on a vessel.

CamelCase : rxNCode

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
categoryOfRxN	S100_CodeList	0,1
actionOrActivity	S100_CodeList	0,1
headline	text	0,1

17.14 Schedule by Day of Week

Definition : The nature and timings of a daily schedule by days of the week.

CamelCase : scheduleByDayOfWeek

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
categoryOfSchedule	enumeration	0,1
text	text	0,1
timeIntervalsByDayOfWeek	Complex	1,*

17.15 Spatial Accuracy

Definition : Provides an indication of the vertical and horizontal positional uncertainty of bathymetric data, optionally within a specified date range.

CamelCase : spatialAccuracy

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
fixedDateRange	Complex	0,1
horizontalPositionUncertainty	Complex	0,1

17.16 Source Indication

Definition : Information about the source document, publication, or reference from which object data or textual material included or referenced in a dataset are derived.

CamelCase : sourceIndication

Alias :

Remarks : Content of featureName is source authority name.

Sub-attributes :

Sub-Attribute	Type	Multiplicity
categoryOfAuthority	enumeration	0,1
countryName	text	0,1
source	text	0,1
sourceType	enumeration	0,1
reportedDate	S100_TruncatedDate	0,1
featureName	Complex	0,*

17.17 Survey Date Range

Definition : The complex attribute describes the period of the hydrographic survey, as the time between its sub-attributes.

CamelCase : surveyDateRange

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
dateStart	S100_TruncatedDate	0,1
dateEnd	S100_TruncatedDate	1,1

17.18 Telecommunications

Definition : A means or channel of communicating at a distance by electrical or electromagnetic means such as telegraphy, telephony, or broadcasting.

CamelCase : telecommunications

Alias :

Remarks : If no value is populated for the sub-attribute telecommunication service, this means the service is by voice communication. If no value is populated for the sub-attribute telecommunication carrier, this means the service is by land line communication.

Sub-attributes :

Sub-Attribute	Type	Multiplicity
categoryOfCommunicationPreference	enumeration	0,1
telecommunicationIdentifier	text	1,1
telecommunicationCarrier	text	0,1
contactInstructions	text	0,1
telecommunicationService	enumeration	0,*

17.19 Text Content

Definition : Textual material, or a pointer to a resource providing textual material. May be accompanied by basic information about its source and relationship to the source.

CamelCase : textContent

Alias : TXTCON

Remarks : Exactly one of sub-attributes onlineResource or information must be completed in one instance of textContent. Product specifications may restrict the use or content of onlineResource for security. For example, a product specification may forbid populating onlineResource. Product specification authors must consider whether applications using the data product may be prevented from accessing off-system resources by security policies.

Sub-attributes :

Sub-Attribute	Type	Multiplicity
categoryOfText	enumeration	0,1
information	Complex	0,*
onlineResource	Complex	0,1
sourceIndication	Complex	0,*

17.20 Time Intervals by Day of Week

Definition : The regular weekly operation times of a service or schedule.

CamelCase : timeIntervalsByDayOfWeek

Alias :

Remarks :

Sub-attributes :

Sub-Attribute	Type	Multiplicity
dayOfWeek	enumeration	0,7 (ordered)
dayOfWeekIsRange	boolean	0,1
timeOfDayStart	time	0,* (ordered)

Sub-Attribute	Type	Multiplicity
timeOfDayEnd	time	0,* (ordered)

17.21 Vessel Measurements Specification

Definition : Combinations of values of measurable characteristics or dimensions of vessels, used to specify size and tonnage ranges.

CamelCase : vesselMeasurementsSpecification

Alias :

Remarks : Combines (i) specifications of vessels' measurable characteristics (length, beam, tonnages, etc.), (ii) limit values for the specified characteristics (with units), (iii) arithmetical comparison operators (greater than, etc.), and (iv) logical operators (AND/OR) to define a subset of vessels characterized by the specified ranges. For example, the combination (draught, 10.5, metres, greaterThan) describes "vessels with draught greater than 10.5 metres".

Sub-attributes :

Sub-Attribute	Type	Multiplicity
comparisonOperator	enumeration	1,1
vesselsCharacteristics	enumeration	1,1
vesselsCharacteristicsValue	real	1,1
vesselsCharacteristicsUnit	enumeration	1,1